and the two data values 62 and 67 can be represented as

Stem	Leaf
6	2,7

Example 2.2.c. Table 2.5 gives the monthly and yearly average daily minimum temperatures in 35 U.S. cities.

The annual average daily minimum temperatures from Table 2.5 are represented in the following stem and leaf plot.

7 0.0
6 9.0
5 1.0, 1.3, 2.0, 5.5, 7.1, 7.4, 7.6, 8.5, 9.3
4 0.0, 1.0, 2.4, 3.6, 3.7, 4.8, 5.0, 5.2, 6.0, 6.7, 8.1, 9.0, 9.2
3 3.1, 4.1, 5.3, 5.8, 6.2, 9.0, 9.5, 9.5
2 9.0, 9.8 ■

2.3 Summarizing data sets

Modern-day experiments often deal with huge sets of data. For instance, in an attempt to learn about the health consequences of certain common practices, in 1951 the medical statisticians R. Doll and A. B. Hill sent questionnaires to all doctors in the United Kingdom and received approximately 40,000 replies. Their questions dealt with age, eating habits, and smoking habits. The respondents were then tracked for the ensuing 10 years and the causes of death for those who died were monitored. To obtain a feel for such a large amount of data, it is useful to be able to summarize it by some suitably chosen measures. In this section we present some summarizing *statistics*, where a statistic is a numerical quantity whose value is determined by the data.

2.3.1 Sample mean, sample median, and sample mode

In this section we introduce some statistics that are used for describing the center of a set of data values. To begin, suppose that we have a data set consisting of the n numerical values $x_1, x_2, \ldots, x_n$. The sample mean is the arithmetic average of these values.

Definition. The *sample mean*, designated by $\bar{x}$, is defined by

$$\bar{x} = \sum_{i=1}^{n} x_i/n$$

The computation of the sample mean can often be simplified by noting that if for constants a and b

$$y_i = ax_i + b, \qquad i = 1, \ldots, n$$

Table 2.5 Normal Daily Minimum Temperature — Selected Cities
In Fahrenheit degrees. Airport data except as noted. Based on standard 30-year period, 1961 through 1990

State	Station	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept	Oct.	Nov.	Dec.	Annual avg.
AL	Mobile	40.0	42.7	50.1	57.1	64.4	70.7	73.2	72.9	68.7	57.3	49.1	43.1	57.4
AK	Juneau	19.0	22.7	26.7	32.1	38.9	45.0	48.1	47.3	42.9	37.2	27.2	22.6	34.1
AZ	Phoenix	41.2	44.7	48.8	55.3	63.9	72.9	81.0	79.2	72.8	60.8	48.9	41.8	59.3
AR	Little Rock	29.1	33.2	42.2	50.7	59.0	67.4	71.5	69.8	63.5	50.9	41.5	33.1	51.0
CA	Los Angeles	47.8	49.3	50.5	52.8	56.3	59.5	62.8	64.2	63.2	59.2	52.8	47.9	55.5
	Sacramento	37.7	41.4	43.2	45.5	50.3	55.3	58.1	58.0	55.7	50.4	43.4	37.8	48.1
	San Diego	48.9	50.7	52.8	55.6	59.1	61.9	65.7	67.3	65.6	60.9	53.9	48.8	57.6
	San Francisco	41.8	45.0	45.8	47.2	49.7	52.6	53.9	55.0	55.2	51.8	47.1	42.7	49.0
CO	Denver	16.1	20.2	25.8	34.5	43.6	52.4	58.6	56.9	47.6	36.4	25.4	17.4	36.2
CT	Hartford	15.8	18.6	28.1	37.5	47.6	56.9	62.2	60.4	51.8	40.7	32.8	21.3	39.5
DE	Wilmington	22.4	24.8	33.1	41.8	52.2	61.6	67.1	65.9	58.2	45.7	37.0	27.6	44.8
DC	Washington	26.8	29.1	37.7	46.4	56.6	66.5	71.4	70.0	62.5	50.3	41.1	31.7	49.2
FL	Jacksonville	40.5	43.3	49.2	54.9	62.1	69.1	71.9	71.8	69.0	59.3	50.2	43.4	57.1
	Miami	59.2	60.4	64.2	67.8	72.1	75.1	76.2	76.7	75.9	72.1	66.7	61.5	69.0
GA	Atlanta	31.5	34.5	42.5	50.2	58.7	66.2	69.5	69.0	63.5	51.9	42.8	35.0	51.3
HI	Honolulu	65.6	65.4	67.2	68.7	70.3	72.2	73.5	74.2	73.5	72.3	70.3	67.0	70.0
ID	Boise	21.6	27.5	31.9	36.7	43.9	52.1	57.7	56.8	48.2	39.0	31.1	22.5	39.1
IL	Chicago	12.9	17.2	28.5	38.6	47.7	57.5	62.6	61.6	53.9	42.2	31.6	19.1	39.5
	Peoria	13.2	17.7	29.8	40.8	50.9	60.7	65.4	63.1	55.2	43.1	32.5	19.3	41.0
IN	Indianapolis	17.2	20.9	31.9	41.5	51.7	61.0	65.2	62.8	55.6	43.5	34.1	23.2	42.4
IA	Des Moines	10.7	15.6	27.6	40.0	51.5	61.2	66.5	63.6	54.5	42.7	29.9	16.1	40.0
KS	Wichita	19.2	23.7	33.6	44.5	54.3	64.6	69.9	67.9	59.2	46.6	33.9	23.0	45.0
KY	Louisville	23.2	26.5	36.2	45.4	54.7	62.9	67.3	65.8	58.7	45.8	37.3	28.6	46.0
LA	New Orleans	41.8	44.4	51.6	58.4	65.2	70.8	73.1	72.8	69.5	58.7	51.0	44.8	58.5
ME	Portland	11.4	13.5	24.5	34.1	43.4	52.1	58.3	57.1	48.9	38.3	30.4	17.8	35.8
MD	Baltimore	23.4	25.9	34.1	42.5	52.6	61.8	66.8	65.7	58.4	45.9	37.1	28.2	45.2
MA	Boston	21.6	23.0	31.3	40.2	49.8	59.1	65.1	64.0	56.8	46.9	38.3	26.7	43.6
MI	Detroit	15.6	17.6	27.0	36.8	47.1	56.3	61.3	59.6	52.5	40.9	32.2	21.4	39.0
	Sault Ste. Marie	4.6	4.8	15.3	28.4	38.4	45.5	51.3	51.3	44.3	36.2	25.9	11.8	29.8
MN	Duluth	−2.2	2.8	15.7	28.9	39.6	48.5	55.1	53.3	44.5	35.1	21.5	4.9	29.0
	Minneapolis-St. Paul	2.8	9.2	22.7	36.2	47.6	57.6	63.1	60.3	50.3	38.8	25.2	10.2	35.3
MS	Jackson	32.7	35.7	44.1	51.9	60.0	67.1	70.5	69.7	63.7	50.3	42.3	36.1	52.0
MO	Kansas City	16.7	21.8	32.6	43.8	53.9	63.1	68.2	65.7	56.9	45.7	33.6	21.9	43.7
	St. Louis	20.8	25.1	35.5	46.4	56.0	65.7	70.4	67.9	60.5	48.3	37.7	26.0	46.7
MT	Great Falls	11.6	17.2	22.8	31.9	40.9	48.6	53.2	52.2	43.5	35.8	24.3	14.6	33.1

Source: U.S. National Oceanic and Atmospheric Administration, Climatography of the United States, No. 81.

then the sample mean of the data set $y_1, \ldots, y_n$ is

$$\bar{y} = \sum_{i=1}^{n} (ax_i + b)/n = \sum_{i=1}^{n} ax_i/n + \sum_{i=1}^{n} b/n = a\bar{x} + b$$

Example 2.3.a. The winning scores in the U.S. Masters golf tournament in the years from 2004 to 2013 were as follows:

280, 278, 272, 276, 281, 279, 276, 281, 289, 280

Find the sample mean of these scores.

Solution. Rather than directly adding these values, it is easier to first subtract 280 from each one to obtain the new values $y_i = x_i - 280$:

$$0, -2, -8, -4, 1, -1, -4, 1, 9, 0$$

Because the arithmetic average of the transformed data set is

$$\bar{y} = -8/10$$

it follows that

$$\bar{x} = \bar{y} + 280 = 279.2 \quad \blacksquare$$

Sometimes we want to determine the sample mean of a data set that is presented in a frequency table listing the k distinct values $v_1, \ldots, v_k$ having corresponding frequencies $f_1, \ldots, f_k$. Since such a data set consists of $n = \sum_{i=1}^{k} f_i$ observations, with the value v_i appearing f_i times, for each $i = 1, \ldots, k$, it follows that the sample mean of these n data values is

$$\bar{x} = \sum_{i=1}^{k} v_i f_i / n$$

By writing the preceding as

$$\bar{x} = \frac{f_1}{n} v_1 + \frac{f_2}{n} v_2 + \cdots + \frac{f_k}{n} v_k$$

we see that the sample mean is a *weighted average* of the distinct values, where the weight given to the value v_i is equal to the proportion of the n data values that are equal to $v_i, i = 1, \ldots, k$.

Example 2.3.b. The following is a frequency table giving the ages of members of a symphony orchestra for young adults.

Age	Frequency
15	2
16	5
17	11
18	9
19	14
20	13

Find the sample mean of the ages of the 54 members of the symphony.

Solution.

$$\bar{x} = (15 \cdot 2 + 16 \cdot 5 + 17 \cdot 11 + 18 \cdot 9 + 19 \cdot 14 + 20 \cdot 13)/54 \approx 18.24 \quad \blacksquare$$

Another statistic used to indicate the center of a data set is the *sample median*; loosely speaking, it is the middle value when the data set is arranged in increasing order.

Definition. Order the values of a data set of size n from smallest to largest. If n is odd, the *sample median* is the value in position $(n + 1)/2$; if n is even, it is the average of the values in positions $n/2$ and $n/2 + 1$.

Thus the sample median of a set of three values is the second smallest; of a set of four values, it is the average of the second and third smallest.

Example 2.3.c. Find the sample median for the data described in Example 2.3.b.

Solution. Since there are 54 data values, it follows that when the data are put in increasing order, the sample median is the average of the values in positions 27 and 28. Thus, the sample median is 18.5. $\blacksquare$

The sample mean and sample median are both useful statistics for describing the central tendency of a data set. The sample mean makes use of all the data values and is affected by extreme values that are much larger or smaller than the others; the sample median makes use of only one or two of the middle values and is thus not affected by extreme values. Which of them is more useful depends on what one is trying to learn from the data. For instance, if a city government has a flat rate income tax and is trying to estimate its total revenue from the tax, then the sample mean of its residents' income would be a more useful statistic. On the other hand, if the city was thinking about constructing middle-income housing, and wanted to determine the proportion of its population able to afford it, then the sample median would probably be more useful.

Example 2.3.d. In a study reported in Hoel, D. G., "A representation of mortality data by competing risks," *Biometrics*, **28**, pp. 475–488, 1972, a group of 5-week-old mice were each given a radiation dose of 300 rad. The mice were then divided into two groups; the first group was kept in a germ-free environment, and the second in conventional laboratory conditions. The numbers of days until death were then observed. The data for those whose death was due to thymic lymphoma are given in the following stem and leaf plots (whose stems are in units of hundreds of days); the first plot is for mice living in the germ-free conditions and the second for mice living under ordinary laboratory conditions.

Germ-Free Mice

1	58, 92, 93, 94, 95
2	02, 12, 15, 29, 30, 37, 40, 44, 47, 59
3	01, 01, 21, 37
4	15, 34, 44, 85, 96
5	29, 37
6	24
7	07
8	00

Conventional Mice

1	59, 89, 91, 98
2	35, 45, 50, 56, 61, 65, 66, 80
3	43, 56, 83
4	03, 14, 28, 32

Determine the sample means and the sample medians for the two sets of mice.

Solution. It is clear from the stem and leaf plots that the sample mean for the set of mice put in the germ-free setting is larger than the sample mean for the set of mice in the usual laboratory setting; indeed, a calculation gives that the former sample mean is 344.07, whereas the latter one is 292.32. On the other hand, since there are 29 data values for the germ-free mice, the sample median is the 15th largest data value, namely, 259; similarly, the sample median for the other set of mice is the 10th largest data value, namely, 265. Thus, whereas the sample mean is quite a bit larger for the first data set, the sample medians are approximately equal. The reason for this is that whereas the sample mean for the first set is greatly affected by the five data values greater than 500, these values have a much smaller effect on the sample median. Indeed, the sample median would remain unchanged if these values were replaced by any other five values greater than or equal to 259. It appears from the stem and leaf plots that the germ-free conditions probably improved the life span of the five longest living rats, but it is unclear what, if any, effect it had on the life spans of the other rats. ■

Another statistic that has been used to indicate the central tendency of a data set is the *sample mode*, defined to be the value that occurs with the greatest frequency. If no single value occurs most frequently, then all the values that occur at the highest frequency are called *modal values*.

Example 2.3.e. The following frequency table gives the values obtained in 40 rolls of a die.

Value	Frequency
1	9
2	8
3	5
4	5
5	6
6	7

Find **(a)** the sample mean, **(b)** the sample median, and **(c)** the sample mode.

Solution. **(a)** The sample mean is

$$\bar{x} = (9 + 16 + 15 + 20 + 30 + 42)/40 = 3.05$$

(b) The sample median is the average of the 20th and 21st smallest values, and is thus equal to 3. **(c)** The sample mode is 1, the value that occurred most frequently. ■

2.3.2 Sample variance and sample standard deviation

Whereas we have presented statistics that describe the central tendencies of a data set, we are also interested in ones that describe the spread or variability of the data values. A statistic that could be used for this purpose would be one that measures the average value of the squares of the distances between the data values and the sample mean. This is accomplished by the sample variance, which for technical reasons divides the sum of the squares of the differences by $n - 1$ rather than n, where n is the size of the data set.

Definition. The *sample variance*, call it s^2, of the data set $x_1, \ldots, x_n$ is defined by

$$s^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2/(n-1)$$

Example 2.3.f. Find the sample variances of the data sets **A** and **B** given below.

A: 3, 4, 6, 7, 10 **B:** −20, 5, 15, 24

Solution. As the sample mean for data set **A** is $\bar{x} = (3 + 4 + 6 + 7 + 10)/5 = 6$, it follows that its sample variance is

$$s^2 = [(-3)^2 + (-2)^2 + 0^2 + 1^2 + 4^2]/4 = 7.5$$

The sample mean for data set **B** is also 6; its sample variance is

$$s^2 = [(-26)^2 + (-1)^2 + 9^2 + (18)^2]/3 \approx 360.67$$

Thus, although both data sets have the same sample mean, there is a much greater variability in the values of the **B** set than in the **A** set. ■

The following algebraic identity is often useful for computing the sample variance:

An algebraic identity

$$\sum_{i=1}^{n}(x_i-\bar{x})^2=\sum_{i=1}^{n}x_i^2-n\bar{x}^2$$

The identity is proven as follows:

$$\begin{aligned}\sum_{i=1}^{n}(x_i-\bar{x})^2&=\sum_{i=1}^{n}\left(x_i^2-2x_i\bar{x}+\bar{x}^2\right)\\&=\sum_{i=1}^{n}x_i^2-2\bar{x}\sum_{i=1}^{n}x_i+\sum_{i=1}^{n}\bar{x}^2\\&=\sum_{i=1}^{n}x_i^2-2n\bar{x}^2+n\bar{x}^2\\&=\sum_{i=1}^{n}x_i^2-n\bar{x}^2\end{aligned}$$

The computation of the sample variance can also be eased by noting that if

$$y_i=a+bx_i,\qquad i=1,\ldots,n$$

then $\bar{y}=a+b\bar{x}$, and so

$$\sum_{i=1}^{n}(y_i-\bar{y})^2=b^2\sum_{i=1}^{n}(x_i-\bar{x})^2$$

That is, if s_y^2 and s_x^2 are the respective sample variances, then

$$s_y^2=b^2s_x^2$$

In other words, adding a constant to each data value does not change the sample variance; whereas multiplying each data value by a constant results in a new sample variance that is equal to the old one multiplied by the square of the constant. ■

Example 2.3.g. The following data give the worldwide number of fatal airline accidents of commercially scheduled air transports in the years from 1997 to 2005.

Year	1997	1998	1999	2000	2001	2002	2003	2004	2005
Accidents	25	20	21	18	13	13	7	9	18

Source: National Safety Council.

Find the sample variance of the number of accidents in these years.

Solution. Let us start by subtracting 18 from each value, to obtain the new data set:

$$7, 2, 3, 0, -5, -5, -11, -9, 0$$

Calling the transformed data $y_1, \ldots, y_9$, we have

$$\bar{y} = \sum_{i=1}^{9} y_i/9 = -2, \qquad \sum_{i=1}^{n} y_i^2 = 49 + 4 + 9 + 25 + 25 + 121 + 81 = 314$$

Hence, since the sample variance of the transformed data is equal to that of the original data, upon using the algebraic identity we obtain

$$s^2 = \frac{314 - 9(4)}{8} = 34.75 \quad \blacksquare$$

Program 2.3 on the text disk can be used to obtain the sample variance for large data sets.

The positive square root of the sample variance is called the *sample standard deviation*.

Definition. The quantity s, defined by

$$s = \sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2/(n-1)}$$

is called the *sample standard deviation*.

The sample standard deviation is measured in the same units as the data.

2.3.3 Sample percentiles and box plots

Loosely speaking, the sample $100p$ percentile of a data set is that value such that $100p$ percent of the data values are less than or equal to it, $0 \leq p \leq 1$. More formally, we have the following definition.

Definition. The *sample 100*p *percentile* is that data value such that at least $100p$ percent of the data are less than or equal to it and at least $100(1-p)$ percent are greater than or equal to it. If two data values satisfy this condition, then the sample $100p$ percentile is the arithmetic average of these two values.

To determine the sample $100p$ percentile of a data set of size n, we need to determine the data values such that

1. At least np of the values are less than or equal to it.
2. At least $n(1-p)$ of the values are greater than or equal to it.

To accomplish this, first arrange the data in increasing order. Then, note that if np is not an integer, then the only data value that satisfies the preceding conditions is the one whose position when the data are ordered from smallest to largest is the smallest integer exceeding np. For instance, if $n = 22,\ p = .8$, then we require a data value such that at least 17.6 of the values are less than or equal to it, and at least 4.4 of them are greater than or equal to it. Clearly, only the 18th smallest value satisfies both conditions and this is the sample 80 percentile. On the other hand, if np is an integer, then it is easy to check that both the values in positions np and $np+1$ satisfy the preceding conditions, and so the sample $100p$ percentile is the average of these values. For instance, if we wanted the 90 percentile of a data set of size 20, then both the (18)th and (19)th smallest values would be such that at least 90 percent of the data values are less than or equal to them, and at least 10 percent of the data values are greater than or equal to them. Thus, the 90 percentile is the average of these two values.

Example 2.3.h. Table 2.6 lists the populations of the 25 most populous U.S. cities for the year 2006. For this data set, find **(a)** the sample 10 percentile and **(b)** the sample 80 percentile.

Solution. **(a)** Because the sample size is 25 and $25(.10) = 2.5$, the sample 10 percentile is the third smallest value, equal to 590, 763.

(b) Because $25(.80) = 20$, the sample 80 percentile is the average of the twentieth and the twenty-first smallest values. Hence, the sample 80 percentile is

$$\frac{1{,}512{,}986 + 1{,}448{,}394}{2} = 1{,}480{,}690 \quad \blacksquare$$

The sample 50 percentile is, of course, just the sample median. Along with the sample 25 and 75 percentiles, it makes up the sample quartiles.

Definition. The sample 25 percentile is called the *first quartile*; the sample 50 percentile is called the sample median or the *second quartile*; the sample 75 percentile is called the *third quartile*.

The quartiles break up a data set into four parts, with roughly 25 percent of the data being less than the first quartile, 25 percent being between the first and second quartile, 25 percent being between the second and third quartile, and 25 percent being greater than the third quartile.

Table 2.6 Population of 25 Largest U.S. Cities, July 2006.

Rank	City	Population
1	New York, NY	8,250,567
2	Los Angeles, CA	3,849,378
3	Chicago, IL	2,833,321
4	Houston, TX	2,144,491
5	Phoenix, AR	1,512,986
6	Philadelphia, PA	1,448,394
7	San Antonio, TX	1,296,682
8	San Diego, CA	1,256,951
9	Dallas, TX	1,232,940
10	San Jose, CA	929, 936
11	Detroit, MI	918, 849
12	Jacksonville, FL	794, 555
13	Indianapolis, IN	785, 597
14	San Francisco, CA	744, 041
15	Columbus, OH	733, 203
16	Austin, TX	709, 893
17	Memphis, TN	670, 902
18	Fort Worth, TX	653, 320
19	Baltimore, MD	640, 961
20	Charlotte, NC	630, 478
21	El Paso, TX	609, 415
22	Milwaukee, WI	602, 782
23	Boston, MA	590, 763
24	Seattle, WA	582, 454
25	Washington, DC	581, 530

Example 2.3.i. Noise is measured in decibels, denoted as dB. One decibel is about the level of the weakest sound that can be heard in a quiet surrounding by someone with good hearing; a whisper measures about 30 dB; a human voice in normal conversation is about 70 dB; a loud radio is about 100 dB. Ear discomfort usually occurs at a noise level of about 120 dB.

The following data give noise levels measured at 36 different times directly outside of Grand Central Station in Manhattan.

82, 89, 94, 110, 74, 122, 112, 95, 100, 78, 65, 60, 90, 83, 87, 75, 114, 85
69, 94, 124, 115, 107, 88, 97, 74, 72, 68, 83, 91, 90, 102, 77, 125, 108, 65

Determine the quartiles.

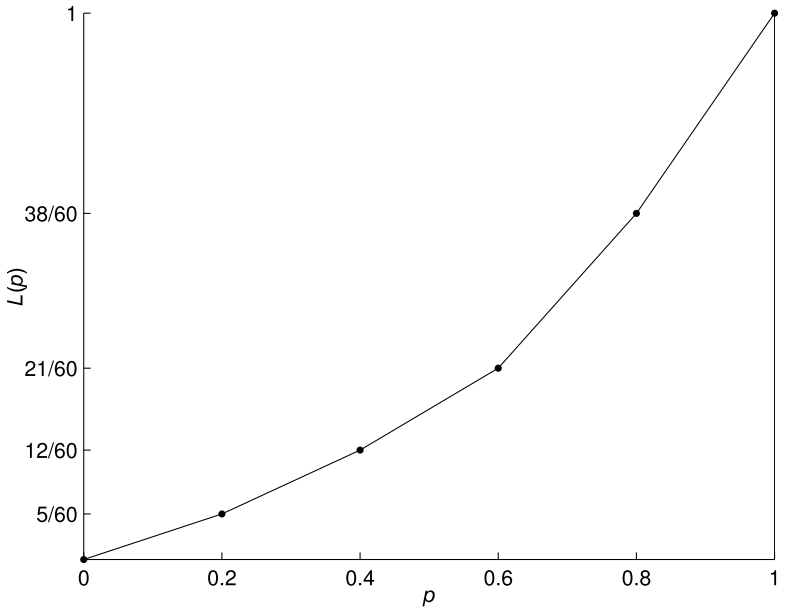


FIGURE 2.16
The Lorenz curve for Example 2.7.a.

$$L(.4) = 12/60,$$
$$L(.6) = 21/60,$$
$$L(.8) = 38/60,$$
$$L(1) = 60/60.$$

Connecting with straight lines between these points gives the Lorenz curve (Figure 2.16). ■

Lorenz curves are also used to plot the wealths of members of a population, with $L(p)$ now representing the proportion of all wealth owned by the least wealthy $100p$ percent of the population. For instance, if the population consists of 1000 people, then $L(.22)$ is the proportion of the populations's wealth that is owned by its 220 poorest members.

Now, the average of the numbers in a set always increases when a new value that is larger than any of the previous ones is added to the set. Consequently, if the incomes, in increasing value, of the members of the population are $x_1 \leq x_2 \leq x_3 \cdots \leq x_n$, then for any $j = 1, \ldots, n$

$$\frac{x_1 + \ldots + x_j}{j} \leq \frac{x_1 + \ldots + x_{j+1}}{j+1} \leq \ldots \leq \frac{x_1 + \ldots + x_n}{n},$$

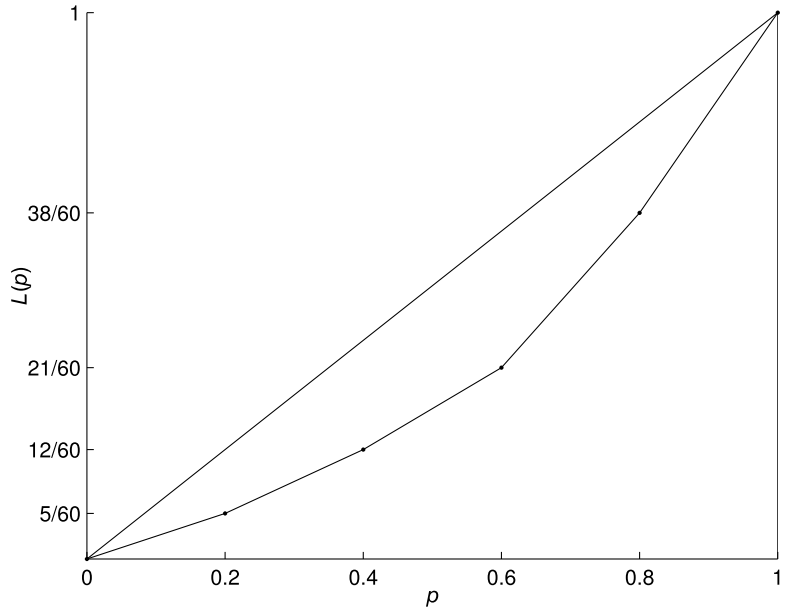


FIGURE 2.17
The Lorenz curve and the straight line.

implying that

$$\frac{x_1 + \ldots + x_j}{x_1 + \ldots + x_n} \leq \frac{j}{n}.$$

Thus,

$$L(j/n) \leq j/n.$$

In addition, it can be verified that $L(j/n) = j/n$, $j = 1, \ldots, n$, only when all n incomes are equal. Consequently, unless all incomes are equal, the Lorenz curve is always below the straight line from $(0, 0)$ to $(1, 1)$. (Figure 2.17 indicates this for the data of Example 2.7.a.)

Thus, the Lorenz curve is equal to the straight line from $(0, 0)$ to $(1, 1)$ when all incomes are equal and dips below the line when incomes are not all equal. The more unequal the incomes the greater the "hump", equal to the region between the straight line and the Lorenz curve (the shaded region in Figure 2.18).

A measure of the hump and thus of the inequality of the incomes is given by the *Gini index*, which is the ratio of the area of the hump divided by the area under the straight line. Because the area of a triangle is one half its base times its height, the area under the straight line from $(0, 0)$ to $(1, 1)$ is equal to $1/2$.

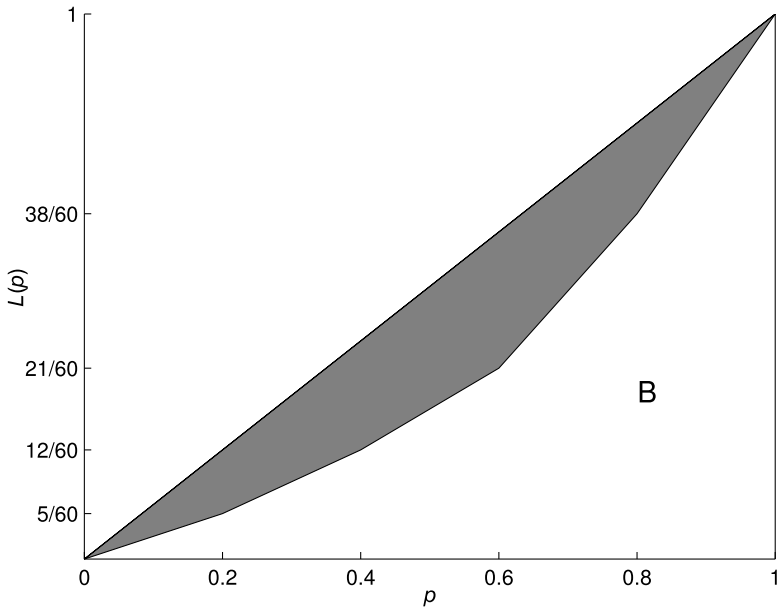


FIGURE 2.18
The hump of the Lorenz curve.

Hence, the Gini index, call it G, is given by

$$G = \frac{\text{area of the hump of the Lorenz curve}}{1/2}.$$

Letting B be the area beneath the Lorenz curve, and using that the area of the hump is equal to the area under the straight line minus the area under the Lorenz curve, we see that the value of the Gini index is

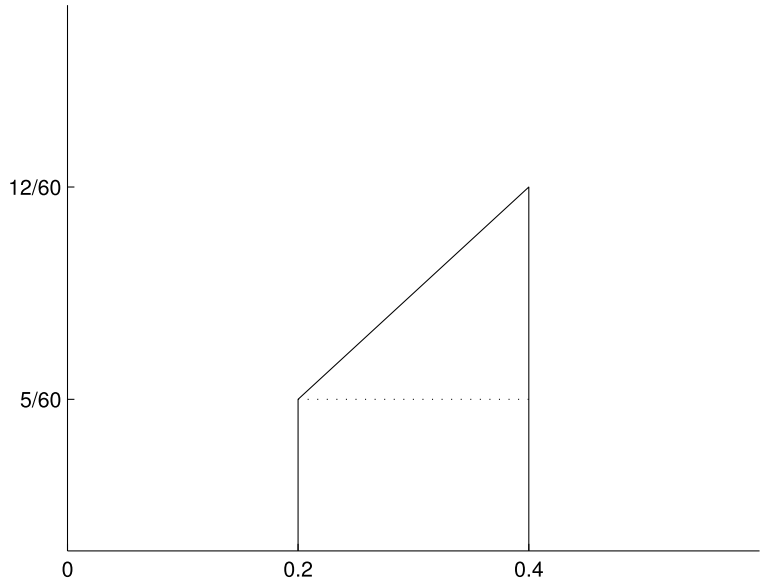
$$G = \frac{1/2 - B}{1/2} = 1 - 2B.$$

Example 2.7.b. Compute the Gini index for the data of Example 2.7.a.

Solution. To compute B, the area under the Lorenz curve, we write $B = B_1 + B_2 + B_3 + B_4 + B_5$, where B_1 is the area of the region under the Lorenz curve from 0 to .2; B_2 is the area of the region between .2 and .4; B_3 is the area of the region between .4 and .6; B_4 is the area of the region between .6 and .8; and B_5 is the area of the region between .8 and 1. Now, B_1 is the area of a triangle whose base is .2 and whose height is 5/60, showing that

$$B_1 = (1/2)(.2)(5/60) = 5/600.$$

Now, B_2, B_3, B_4, and B_5 are the areas of regions composed of a triangle on top of a rectangle. For instance, the following is the region B_2:



Now all of the triangles in these regions have base .2 and respective heights $5/60, 7/60, 9/60, 17/60, 22/60$. Consequently, the sum of the areas of all these triangles is

$$\text{Area of triangles} = \frac{1}{2}(.2)(5+7+9+17+22)/60 = .1.$$

The four rectangles all have base length .2 and respective heights 5/60, 12/60, 21/60, 38/60, yielding

$$\text{Area of rectangles} = .2(5+12+21+38)/60 \approx .25333.$$

Adding the area of the rectangles and the area of the triangles gives $B = .1 + .25333 = .35333$, and thus

$$G = 1 - .70666 = .29334.$$

In general, suppose the incomes are, in increasing order, $x_1 \le x_2 \le x_3 \cdots \le x_n$. Let

$$s_j = x_1 + \ldots + x_j, \quad j = 1, \ldots, n.$$

Then all the triangles will have base $1/n$ and will have respective heights $x_1/s_n, x_2/s_n, \ldots, x_n/s_n$. Consequently, the sum of the area of all the triangles is

$$\text{Area of triangles} = \frac{1}{2n}(x_1 + \ldots + x_n)/s_n = \frac{1}{2n}.$$

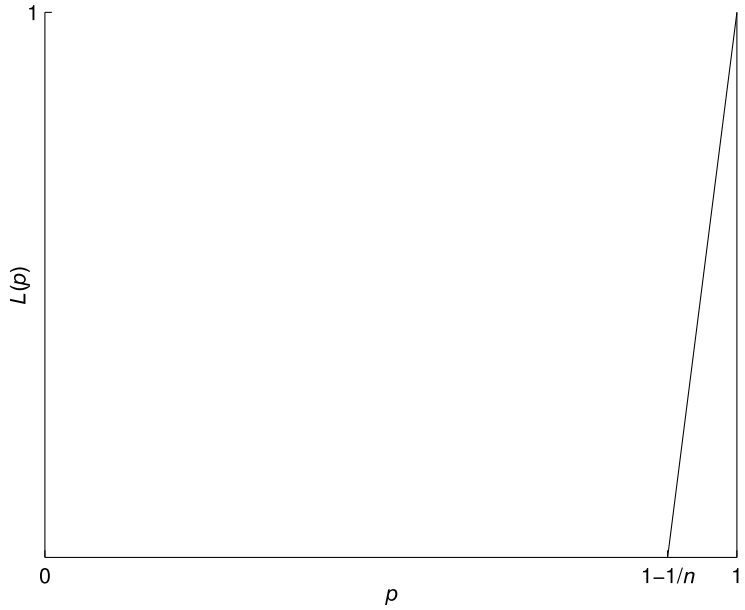


FIGURE 2.19
Largest possible Gini index value.

The rectangles will all have length $1/n$ and will have respective heights $s_1/s_n, s_2/s_n, \ldots, s_{n-1}/s_n$. Hence,

$$\text{Area of rectangles} = \frac{s_1 + s_2 + \ldots + s_{n-1}}{ns_n},$$

giving

$$B = \frac{1}{2n} + \frac{s_1 + s_2 + \ldots + s_{n-1}}{ns_n}. \quad \blacksquare$$

The Gini index is 0 when all incomes are equal (and so the area between the straight line and the Lorenz curve is 0). On the opposite extreme, the largest possible value of the Gini index occurs when only one of the n members of the population has a positive income. In this case, the area under the Lorenz curve is the area of a triangle whose base is of length $1/n$ and whose height is 1 (see Figure 2.19). Thus, $B = \frac{1}{2n}$, and $G = 1 - 1/n$.

2.8 Using R

Suppose you want the sample mean and sample variance of the values $x_1, \ldots, x_n$. Type
$x = c(x_1, \ldots, x_n)$

and hit return. [Interpret $x = c(x_1, \ldots, x_n)$ as the sequence of values $x_1, \ldots, x_n$.]
On the next line type
mean(x)
and when you hit return, the sample mean is displayed. On the next line, type
var(x)
and when you hit return, the variance is displayed.

For instance, suppose you wanted the sample mean and sample variance of the values 4, 6, 12, 9, 21, 14. Then

```
> x=c(4,6,12,9,21,14)
> mean(x)
[1] 11
> var(x)
[1] 37.6
```

By the way, you do not type > at the beginning of a line. R always starts a line with this symbol, which is a prompt asking what to do next. (The [1] in front of the answers returned is to indicate that you only asked for a single item.)

Other commands are sum(x), which returns the sum of the values; median(x), which returns the sample median; and sd(x), which returns the sample standard deviation (equal to the square root of var(x)). Also, if $x = c(x_1, \ldots, x_n)$ and $y = c(y_1, \ldots, y_n)$, then $x + y = c(x_1 + y_1, \ldots, x_n + y_n)$, $x - y = c(x_1 - y_1, \ldots, x_n - y_n)$, $x * y = c(x_1 y_1, \ldots, x_n y_n)$, and if all $y_i \neq 0$, $x/y = c(x_1/y_1, \ldots, x_n/y_n)$. Also, $x^2 = x * x = c(x_1^2, \ldots, x_n^2)$. If a is a real number, then $a + x = c(a + x_1, \ldots, a + x_n)$, $a * x = c(ax_1, \ldots, ax_n)$ and, for $a \neq 0$, $x/a = c(x_1/a, \ldots, x_n/a)$. In addition, if all $x_i \neq 0$, then $a/x = c(a/x_1, \ldots, a/x_n)$.

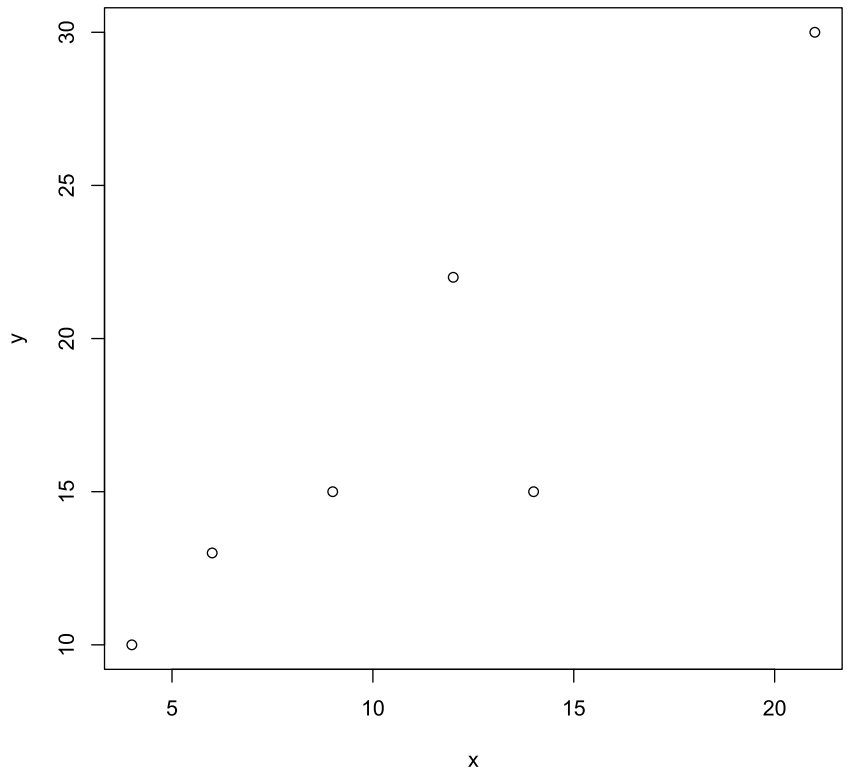
If you have data pairs $(x_1, y_1), \ldots, (x_n, y_n)$, then to obtain the sample correlation coefficient, do the following:

```
> x = c(x_1, ..., x_n)
> y = c(y_1, ..., y_n)
> cor(x, y)
```

To obtain a scatter diagram of the data, type
> plot(x,y)
and hit return.

For instance, suppose we have the data pairs (4, 10), (6, 13), (12, 22), (9, 15), (21, 30), (14, 15), then we obtain the sample correlation coefficient and scatter diagram as follows:

```
> x = c(4, 6, 12, 9, 21, 14)
> y = c(10, 13, 22, 15, 30, 15)
> cor(x, y)
[1] 0.9041494
> plot(x,y)
```



R can also be used for computation. For instance, suppose we wanted to compute $\frac{18\sqrt{177}}{677}$. This is accomplished by

$>$ 18 $*$ sqrt(177)/677
[1] 0.3537288

Suppose now that we want to select an element of the vector $x = c(x_1, \ldots, x_n)$, say x_i, its value in position i. This is accomplished by the R command $> x[i]$. If we wanted the new vector, consisting of the elements of the vector x, going from positions i to j, use the R command $x[i : j]$.

For instance,

$> x = c(3, 18, 9, 7, 22, 5, 17)$
$> x[3]$
[1] 9
$> x[2 : 5]$
[1]18 9 7 22

process will have useful lifetimes that are independent with a common unknown distribution F. Physical reasons sometimes suggest the parametric form of the distribution F; for instance, it may lead us to believe that F is a normal distribution, or that F is an exponential distribution. In such cases, we are confronted with a parametrical statistical problem in which we would want to use the observed data to estimate the parameters of F. For instance, if F were assumed to be a normal distribution, then we would want to estimate its mean and variance; if F were assumed to be exponential, we would want to estimate its mean. In other situations, there might not be any physical justification for supposing that F has any particular form; in this case the problem of making inferences about F would constitute a nonparametric inference problem. ■

In this chapter, we will be concerned with the probability distributions of certain statistics that arise from a sample, where a *statistic* is a random variable whose value is determined by the sample data. Two important statistics that we will discuss are the sample mean and the sample variance. In Section 6.2, we consider the sample mean and derive its expectation and variance. We note that when the sample size is at least moderately large, the distribution of the sample mean is approximately normal. This follows from the central limit theorem, one of the most important theoretical results in probability, which is discussed in Section 6.3. In Section 6.4, we introduce the sample variance and determine its expected value. In Section 6.5, we suppose that the population distribution is normal and present the joint distribution of the sample mean and the sample variance. In Section 6.6, we suppose that we are sampling from a finite population of elements and explain what it means for the sample to be a "random sample." When the population size is large in relation to the sample size, we often treat it as if it were of infinite size; this is illustrated and its consequences are discussed.

6.2 The sample mean

Consider a population of elements, each of which has a numerical value attached to it. For instance, the population might consist of the adults of a specified community and the value attached to each adult might be his or her annual income, or height, or age, and so on. We often suppose that the value associated with any member of the population can be regarded as being the value of a random variable having expectation μ and variance σ^2. The quantities μ and σ^2 are called the *population mean* and the *population variance*, respectively. Let $X_1, X_2, \ldots, X_n$ be a sample of values from this population. The sample mean is defined by

$$\overline{X} = \frac{X_1 + \cdots + X_n}{n}$$

Since the value of the sample mean $\overline{X}$ is determined by the values of the random variables in the sample, it follows that $\overline{X}$ is also a random variable. Its

expected value and variance are obtained as follows:

$$\begin{aligned} E[\overline{X}] &= E\left[\frac{X_1 + \cdots + X_n}{n}\right] \\ &= \frac{1}{n}(E[X_1] + \cdots + E[X_n]) \\ &= \mu \end{aligned}$$

and

$$\begin{aligned} \text{Var}(\overline{X}) &= \text{Var}\left(\frac{X_1 + \cdots + X_n}{n}\right) \\ &= \frac{1}{n^2}[\text{Var}(X_1) + \cdots + \text{Var}(X_n)] \quad \text{by independence} \\ &= \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

where μ and σ^2 are the population mean and variance, respectively. Hence, the expected value of the sample mean is the population mean μ whereas its variance is $1/n$ times the population variance. As a result, we can conclude that $\overline{X}$ is also centered about the population mean μ, but its spread becomes more and more reduced as the sample size increases. Figure 6.1 plots the probability density function of the sample mean from a standard normal population for a variety of sample sizes.

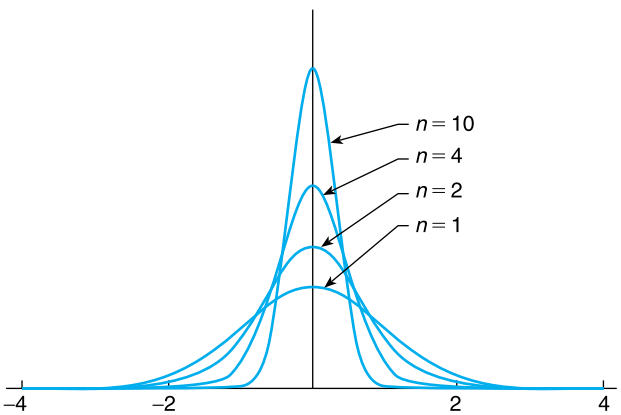


FIGURE 6.1
Densities of sample means from a standard normal population.

6.3 The central limit theorem

In this section, we will consider one of the most remarkable results in probability — namely, the *central limit theorem*. Loosely speaking, this theorem asserts that the sum of a large number of independent random variables has a distribution that is approximately normal. Hence, it not only provides a simple method for computing approximate probabilities for sums of independent random variables, but it also helps explain the remarkable fact that the empirical frequencies of so many natural populations exhibit a bell-shaped (that is, a normal) curve.

In its simplest form, the central limit theorem is as follows:

Theorem 6.3.1 (The central limit theorem). Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables each having mean μ and variance σ^2. Then for n large, the distribution of

$$X_1 + \cdots + X_n$$

is approximately normal with mean $n\mu$ and variance $n\sigma^2$.

It follows from the central limit theorem that

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

is approximately a standard normal random variable; thus, for n large,

$$P\left\{\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} < x\right\} \approx P\{Z < x\}$$

where Z is a standard normal random variable.

Example 6.3.a. An insurance company has 25,000 automobile policy holders. If the yearly claim of a policy holder is a random variable with mean 320 and standard deviation 540, approximate the probability that the total yearly claim exceeds 8.3 million.

Solution. Let X denote the total yearly claim. Number the policy holders, and let X_i denote the yearly claim of policy holder i. With $n = 25{,}000$, we have from the central limit theorem that $X = \sum_{i=1}^{n} X_i$ will have approximately a normal distribution with mean $320 \times 25{,}000 = 8 \times 10^6$ and standard deviation $540\sqrt{25{,}000} = 8.5381 \times 10^4$. Therefore,

$$P\{X > 8.3 \times 10^6\} = P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{8.3 \times 10^6 - 8 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$= P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{.3 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$\approx P\{Z > 3.51\} \qquad \text{where } Z \text{ is a standard normal}$$
$$\approx .00023$$

Thus, there are only 2.3 chances out of 10,000 that the total yearly claim will exceed 8.3 million.

Of course, we could have used R to calculate the desired probability:

```
> 1 − pnorm(8.3 ∗ (10)^6, 320 ∗ 25000, 540 ∗ sqrt(25000))
[1] 0.0002210042
```

■

Example 6.3.b. Civil engineers believe that W, the amount of weight (in units of 1000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1000 pounds) of a car is a random variable with mean 3 and standard deviation .3. How many cars would have to be on the bridge span for the probability of structural damage to exceed .1?

Solution. Let P_n denote the probability of structural damage when there are n cars on the bridge. That is,

$$\begin{aligned} P_n &= P\{X_1 + \cdots + X_n \geq W\} \\ &= P\{X_1 + \cdots + X_n - W \geq 0\} \end{aligned}$$

where X_i is the weight of the ith car, $i = 1, \ldots, n$. Now it follows from the central limit theorem that $\sum_{i=1}^{n} X_i$ is approximately normal with mean $3n$ and variance $.09n$. Hence, since W is independent of the $X_i, i = 1, \ldots, n$, and is also normal, it follows that $\sum_{i=1}^{n} X_i - W$ is approximately normal, with mean and variance given by

$$E\left[\sum_{1}^{n} X_i - W\right] = 3n - 400$$

$$\text{Var}\left(\sum_{1}^{n} X_i - W\right) = \text{Var}\left(\sum_{1}^{n} X_i\right) + \text{Var}(W) = .09n + 1{,}600$$

Thus,

$$\begin{aligned} P_n &= P\left\{\frac{X_1 + \cdots + X_n - W - (3n - 400)}{\sqrt{.09n + 1{,}600}} \geq \frac{-(3n - 400)}{\sqrt{.09n + 1{,}600}}\right\} \\ &\approx P\left\{Z \geq \frac{400 - 3n}{\sqrt{.09n + 1{,}600}}\right\} \end{aligned}$$

where Z is a standard normal random variable. Now $P\{Z \geq 1.28\} \approx .1$, and so if the number of cars n is such that

$$\frac{400 - 3n}{\sqrt{.09n + 1{,}600}} \leq 1.28$$

or

$$n \geq 117$$

then there is at least 1 chance in 10 that structural damage will occur. ■

One of the most important applications of the central limit theorem is in regard to binomial random variables. Since such a random variable X having parameters (n, p) represents the number of successes in n independent trials when each trial is a success with probability p, we can express it as

$$X = X_1 + \cdots + X_n$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise} \end{cases}$$

Because

$$E[X_i] = p, \qquad \text{Var}(X_i) = p(1 - p)$$

it follows from the central limit theorem that for n large

$$\frac{X - np}{\sqrt{np(1 - p)}}$$

will approximately be a standard normal random variable [see Figure 6.2, which graphically illustrates how the probability mass function of a binomial (n, p) random variable becomes more and more "normal" as n becomes larger and larger].

Example 6.3.c. The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year students attend this college.

Solution. Let X denote the number of students that attend; then assuming that each accepted applicant will independently attend, it follows that X is a binomial random variable with parameters $n = 450$ and $p = .3$. Since the binomial is a discrete and the normal a continuous distribution, it is best to compute

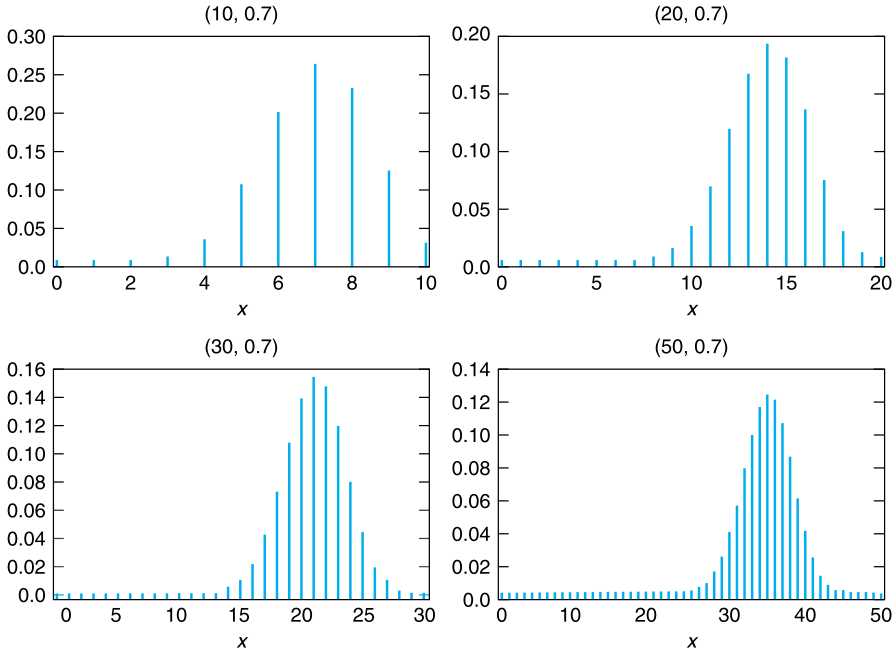


FIGURE 6.2
Binomial probability mass functions converging to the normal density.

$P\{X = i\}$ as $P\{i - .5 < X < i + .5\}$ when applying the normal approximation (this is called the continuity correction). This yields the approximation

$$P\{X > 150.5\} = P\left\{\frac{X - (450)(.3)}{\sqrt{450(.3)(.7)}} \geq \frac{150.5 - (450)(.3)}{\sqrt{450(.3)(.7)}}\right\}$$
$$\approx P\{Z > 1.59\} = .06$$

Hence, only 6 percent of the time do more than 150 of the first 450 accepted actually attend. ■

It should be noted that we now have two possible approximations to binomial probabilities: The Poisson approximation, which yields a good approximation when n is large and p small, and the normal approximation, which can be shown to be quite good when $np(1 - p)$ is large. [The normal approximation will, in general, be quite good for values of n satisfying $np(1 - p) \geq 10$.]

6.3.1 Approximate distribution of the sample mean

Let $X_1, \ldots, X_n$ be a sample from a population having mean μ and variance σ^2. The central limit theorem can be used to approximate the distribution of the

sample mean

$$\overline{X} = \sum_{i=1}^{n} X_i/n$$

Since a constant multiple of a normal random variable is also normal, it follows from the central limit theorem that $\overline{X}$ will be approximately normal when the sample size n is large. Since the sample mean has expected value μ and standard deviation $\sigma/\sqrt{n}$, it then follows that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

has approximately a standard normal distribution.

Example 6.3.d. The weights of a population of workers have mean 167 and standard deviation 27.

(a) If a sample of 36 workers is chosen, approximate the probability that the sample mean of their weights lies between 163 and 170.
(b) Repeat part (a) when the sample is of size 144.

Solution. Let Z be a standard normal random variable.

(a) It follows from the central limit theorem that $\overline{X}$ is approximately normal with mean 167 and standard deviation $27/\sqrt{36} = 4.5$. Therefore, with Z being a standard normal random variable,

$$\begin{aligned} P\{163 < \overline{X} < 170\} &= P\left\{\frac{163-167}{4.5} < \frac{\overline{X}-167}{4.5} < \frac{170-167}{4.5}\right\} \\ &\approx P\{-.8889 < Z < .8889\} \\ &= P\{Z < .8889\} - P\{Z < -.8889\} \\ &= 2P\{Z < .8889\} - 1 \\ &\approx 2 * \text{ pnorm}(.8889) - 1 \\ &= 0.6259432 \end{aligned}$$

(b) For a sample of size 144, the sample mean will be approximately normal with mean 167 and standard deviation $27/\sqrt{144} = 2.25$. Therefore,

$$\begin{aligned} P\{163 < \overline{X} < 170\} &= P\left\{\frac{163-167}{2.25} < \frac{\overline{X}-167}{2.25} < \frac{170-167}{2.25}\right\} \\ &= P\left\{-1.7778 < \frac{\overline{X}-167}{4.5} < 1.7778\right\} \end{aligned}$$

$$\approx 2P\{Z < 1.7778\} - 1$$
$$= 2 * \text{ pnorm}(1.7778) - 1$$
$$= 0.9245633$$

Thus increasing the sample size from 36 to 144 increases the probability from .6259 to .9246. ■

Example 6.3.e. An astronomer wants to measure the distance from her observatory to a distant star. However, due to atmospheric disturbances, any measurement will not yield the exact distance d. As a result, the astronomer has decided to make a series of measurements and then use their average value as an estimate of the actual distance. If the astronomer believes that the values of the successive measurements are independent random variables with a mean of d light-years and a standard deviation of 2 light-years, how many measurements need she make to be at least 95 percent certain that her estimate is accurate to within $\pm$ .5 light-years?

Solution. If the astronomer makes n measurements, then $\overline{X}$, the sample mean of these measurements, will be approximately a normal random variable with mean d and standard deviation $2/\sqrt{n}$. Thus, the probability that it will lie between $d \pm .5$ is obtained as follows:

$$P\{-.5 < \overline{X} - d < .5\} = P\left\{\frac{-.5}{2/\sqrt{n}} < \frac{\overline{X} - d}{2/\sqrt{n}} < \frac{.5}{2/\sqrt{n}}\right\}$$
$$\approx P\{-\sqrt{n}/4 < Z < \sqrt{n}/4\}$$
$$= 2P\{Z < \sqrt{n}/4\} - 1$$

where Z is a standard normal random variable.

Thus, the astronomer should make n measurements, where n is such that

$$2P\{Z < \sqrt{n}/4\} - 1 \geq .95$$

or, equivalently,

$$P\{Z < \sqrt{n}/4\} \geq .975$$

Since $P\{Z < 1.96\} = .975$, it follows that n should be chosen so that

$$\sqrt{n}/4 \geq 1.96$$

That is, at least 62 observations are necessary. ■

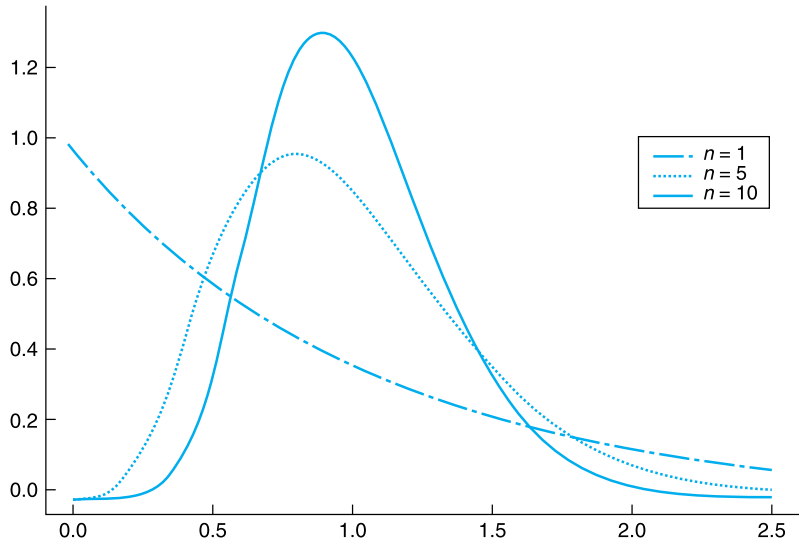


FIGURE 6.3
Densities of the average of n exponential random variables having mean 1.

6.3.2 How large a sample is needed?

The central limit theorem leaves open the question of how large the sample size n needs to be for the normal approximation to be valid, and indeed the answer depends on the population distribution of the sample data. For instance, if the underlying population distribution is normal, then the sample mean $\overline{X}$ will also be normal regardless of the sample size. A general rule of thumb is that one can be confident of the normal approximation whenever the sample size n is at least 30. That is, practically speaking, no matter how nonnormal the underlying population distribution is, the sample mean of a sample of size at least 30 will be approximately normal. In most cases, the normal approximation is valid for much smaller sample sizes. Indeed, a sample of size 5 will often suffice for the approximation to be valid. Figure 6.3 presents the distribution of the sample means from an exponential population distribution for samples of sizes $n = 1, 5, 10$.

6.4 The sample variance

Let $X_1, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Let $\overline{X}$ be the sample mean, and recall the following definition from Section 2.3.2.

Definition. The statistic S^2, defined by

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n - 1}$$

is called the *sample variance*. $S = \sqrt{S^2}$ is called the *sample standard deviation*.

To compute $E[S^2]$, we use an identity that was proven in Section 2.3.2: For any numbers $x_1, \ldots, x_n$

$$\sum_{i=1}^{n}(x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i^2 - n\overline{x}^2$$

where $\overline{x} = \sum_{i=1}^{n} x_i/n$. It follows from this identity that

$$(n-1)S^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2$$

Taking expectations of both sides of the preceding yields, upon using the fact that for any random variable W, $E[W^2] = \mathrm{Var}(W) + (E[W])^2$,

$$\begin{aligned}(n-1)E[S^2] &= E\left[\sum_{i=1}^{n} X_i^2\right] - nE[\overline{X}^2] \\ &= nE[X_1^2] - nE[\overline{X}^2] \\ &= n\mathrm{Var}(X_1) + n(E[X_1])^2 - n\mathrm{Var}(\overline{X}) - n(E[\overline{X}])^2 \\ &= n\sigma^2 + n\mu^2 - n(\sigma^2/n) - n\mu^2 \\ &= (n-1)\sigma^2\end{aligned}$$

or

$$E[S^2] = \sigma^2$$

That is, the expected value of the sample variance S^2 is equal to the population variance σ^2.

6.5 Sampling distributions from a normal population

Let $X_1, X_2, \ldots, X_n$ be a sample from a normal population having mean μ and variance σ^2. That is, they are independent and $X_i \sim \mathcal{N}(\mu, \sigma^2)$, $i = 1, \ldots, n$. Also let

$$\overline{X} = \sum_{i=1}^{n} X_i/n$$

and

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$$

denote the sample mean and sample variance, respectively. We would like to compute their distributions.

6.5.1 Distribution of the sample mean

Since the sum of independent normal random variables is normally distributed, it follows that $\overline{X}$ is normal with mean

$$E[\overline{X}] = \sum_{i=1}^{n} \frac{E[X_i]}{n} = \mu$$

and variance

$$\text{Var}(\overline{X}) = \frac{1}{n^2} \sum_{i=1}^{n} \text{Var}(X_i) = \sigma^2/n$$

That is, $\overline{X}$, the average of the sample, is normal with a mean equal to the population mean but with a variance reduced by a factor of $1/n$. It follows from this that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

is a standard normal random variable.

6.5.2 Joint distribution of $\overline{X}$ and S^2

In this section, we not only obtain the distribution of the sample variance S^2, but we also discover a fundamental fact about normal samples — namely, that $\overline{X}$ and S^2 are independent with $(n-1)S^2/\sigma^2$ having a chi-square distribution with $n-1$ degrees of freedom.

To start, for numbers $x_1, \ldots, x_n$, let $y_i = x_i - \mu$, $i = 1, \ldots, n$. Then as $\overline{y} = \overline{x} - \mu$, it follows from the identity

$$\sum_{i=1}^{n} (y_i - \overline{y})^2 = \sum_{i=1}^{n} y_i^2 - n\overline{y}^2$$

that

$$\sum_{i=1}^{n} (x_i - \overline{x})^2 = \sum_{i=1}^{n} (x_i - \mu)^2 - n(\overline{x} - \mu)^2$$

Now, if $X_1, \ldots, X_n$ is a sample from a normal population having mean μ and variance σ^2, then we obtain from the preceding identity that

$$\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2}=\frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2}+\frac{n(\overline{X}-\mu)^2}{\sigma^2}$$

or, equivalently,

$$\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2=\frac{\sum_{i=1}^{n}(X_i-\overline{X})^2}{\sigma^2}+\left[\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}\right]^2 \tag{6.5.1}$$

Because $(X_i-\mu)/\sigma, i=1,\ldots,n$ are independent standard normals, it follows that the left side of Equation (6.5.1) is a chi-square random variable with n degrees of freedom. Also, as shown in Section 6.5.1, $\sqrt{n}(\overline{X}-\mu)/\sigma$ is a standard normal random variable and so its square is a chi-square random variable with 1 degree of freedom. Thus Equation (6.5.1) equates a chi-square random variable having n degrees of freedom to the sum of two random variables, one of which is chi-square with 1 degree of freedom. But it has been established that the sum of two independent chi-square random variables is also chi-square with a degree of freedom equal to the sum of the two degrees of freedom. Thus, it would seem that there is a reasonable possibility that the two terms on the right side of Equation (6.5.1) are independent, with $\sum_{i=1}^{n}(X_i-\overline{X})^2/\sigma^2$ having a chi-square distribution with $n-1$ degrees of freedom. Since this result can indeed be established, we have the following fundamental result.

Theorem 6.5.1. If $X_1, \ldots, X_n$ is a sample from a normal population having mean μ and variance σ^2, then $\overline{X}$ and S^2 are independent random variables, with $\overline{X}$ being normal with mean μ and variance σ^2/n and $(n-1)S^2/\sigma^2$ being chi-square with $n-1$ degrees of freedom.

Theorem 6.5.1 not only provides the distributions of $\overline{X}$ and S^2 for a normal population but also establishes the important fact that they are independent. In fact, it turns out that this independence of $\overline{X}$ and S^2 is a unique property of the normal distribution. Its importance will become evident in the following chapters.

Example 6.5.a. The time it takes a central processing unit to process a certain type of job is normally distributed with mean 20 seconds and standard deviation 3 seconds. If a sample of 15 such jobs is observed, what is the probability that the sample variance will exceed 12?

Solution. Since the sample is of size $n=15$ and $\sigma^2=9$, write

$$P\{S^2>12\}=P\left\{\frac{14S^2}{9}>\frac{14}{9}\cdot 12\right\}$$

Using R this gives the solution

```
> 1 − pchisq(56/3, 14)
[1] 0.1780811
```
■

The following corollary of Theorem 6.5.1 will be quite useful in the following chapters.

Corollary 6.5.2. Let $X_1, \ldots, X_n$ be a sample from a normal population with mean μ. If $\overline{X}$ denotes the sample mean and S the sample standard deviation, then

$$\sqrt{n}\frac{(\overline{X}-\mu)}{S} \sim t_{n-1}$$

That is, $\sqrt{n}(\overline{X}-\mu)/S$ has a t-distribution with $n-1$ degrees of freedom.

Proof. Recall that a t-random variable with n degrees of freedom is defined as the distribution of

$$\frac{Z}{\sqrt{\chi_n^2/n}}$$

where Z is a standard normal random variable that is independent of χ_n^2, a chi-square random variable with n degrees of freedom. Because Theorem 6.5.1 gives that $\sqrt{n}(\overline{X}-\mu)/\sigma$ is a standard normal that is independent of $(n-1)S^2/\sigma^2$, which is chi-square with $n-1$ degrees of freedom, we can conclude that

$$\frac{\sqrt{n}(\overline{X}-\mu)/\sigma}{\sqrt{S^2/\sigma^2}} = \sqrt{n}\frac{(\overline{X}-\mu)}{S}$$

is a t-random variable with $n-1$ degrees of freedom. ■

6.6 Sampling from a finite population

Consider a population of N elements, and suppose that p is the proportion of the population that has a certain characteristic of interest; that is, Np elements have this characteristic, and $N(1-p)$ do not. A sample of size n from this population is said to be a *random sample* if it is chosen in such a manner that each of the $\binom{N}{n}$ population subsets of size n is equally likely to be the sample. For instance, if the population consists of the three elements a, b, c, then a random sample of size 2 is one that is chosen so that each of the subsets $\{a, b\}$, $\{a, c\}$, and $\{b, c\}$ is equally likely to be the sample. A random subset can be chosen

sequentially by letting its first element be equally likely to be any of the N elements of the population, then letting its second element be equally likely to be any of the remaining $N-1$ elements of the population, and so on.

Suppose now that a random sample of size n has been chosen from a population of size N. For $i = 1, \ldots, n$, let

$$X_i = \begin{cases} 1 & \text{if the } i\text{th member of the sample has the characteristic} \\ 0 & \text{otherwise} \end{cases}$$

Consider now the sum of the X_i; that is, consider

$$X = X_1 + X_2 + \cdots + X_n$$

Because the term X_i contributes 1 to the sum if the ith member of the sample has the characteristic and 0 otherwise, it follows that X is equal to the number of members of the sample that possess the characteristic. In addition, the sample mean

$$\overline{X} = X/n = \sum_{i=1}^{n} X_i/n$$

is equal to the proportion of the members of the sample that possess the characteristic.

Let us now consider the probabilities associated with the statistics X and $\overline{X}$. To begin, note that since each of the N members of the population is equally likely to be the ith member of the sample, it follows that

$$P\{X_i = 1\} = \frac{Np}{N} = p$$

Also,

$$P\{X_i = 0\} = 1 - P\{X_i = 1\} = 1 - p$$

That is, each X_i is equal to either 1 or 0 with respective probabilities p and $1-p$.

It should be noted that the random variables $X_1, X_2, \ldots, X_n$ are not independent. For instance, since the second selection is equally likely to be any of the N members of the population, of which Np have the characteristic, it follows that the probability that the second selection has the characteristic is $Np/N = p$. That is, without any knowledge of the outcome of the first selection,

$$P\{X_2 = 1\} = p$$

However, the conditional probability that $X_2 = 1$, given that the first selection has the characteristic, is

$$P\{X_2 = 1|X_1 = 1\} = \frac{Np - 1}{N - 1}$$

which is seen by noting that if the first selection has the characteristic, then the second selection is equally likely to be any of the remaining $N - 1$ elements, of which $Np - 1$ have the characteristic. Similarly, the probability that the second selection has the characteristic given that the first one does not is

$$P\{X_2 = 1|X_1 = 0\} = \frac{Np}{N - 1}$$

Thus, knowing whether or not the first element of the random sample has the characteristic changes the probability for the next element. However, when the population size N is large in relation to the sample size n, this change will be very slight. For instance, if $N = 1000$, $p = .4$, then

$$P\{X_2 = 1|X_1 = 1\} = \frac{399}{999} = .3994$$

which is very close to the unconditional probability that $X_2 = 1$; namely,

$$P\{X_2 = 1\} = .4$$

Similarly, the probability that the second element of the sample has the characteristic given that the first does not is

$$P\{X_2 = 1|X_1 = 0\} = \frac{400}{999} = .4004$$

which is again very close to .4.

Indeed, it can be shown that when the population size N is large with respect to the sample size n, then $X_1, X_2, \ldots, X_n$ are approximately independent. Now if we think of each X_i as representing the result of a trial that is a success if X_i equals 1 and a failure otherwise, it follows that $X = \sum_{i=1}^{n} X_i$ can be thought of as representing the total number of successes in n trials. Hence, if the X_i were independent, then X would be a binomial random variable with parameters n and p. In other words, when the population size N is large in relation to the sample size n, then the distribution of the number of members of the sample that possess the characteristic is approximately that of a binomial random variable with parameters n and p.

Remark

Of course, X is a hypergeometric random variable (Section 5.4); and so the preceding shows that a hypergeometric can be approximated by a binomial random variable when the number chosen is small in relation to the total number of elements.

For the remainder of this text, we will suppose that the underlying population is large in relation to the sample size and we will take the distribution of X to be binomial.

By using the formulas given in Section 5.4 for the mean and standard deviation of a binomial random variable, we see that

$$E[X] = np \quad \text{and} \quad SD(X) = \sqrt{np(1-p)}$$

Moreover $\overline{X} = X/n$, the proportion of the sample that has the characteristic, has mean and variance given by

$$E[\overline{X}] = E[X]/n = p$$

and

$$\text{Var}(\overline{X}) = \text{Var}(X)/n^2 = p(1-p)/n$$

Example 6.6.a. Suppose that 45 percent of the population favors a certain candidate in an upcoming election. If a random sample of size 200 is chosen, find

(a) the expected value and standard deviation of the number of members of the sample that favor the candidate;
(b) the probability that more than half the members of the sample favor the candidate.

Solution. (a) The expected value and standard deviation of the proportion that favor the candidate are

$$E[X] = 200(.45) = 90, \quad SD(X) = \sqrt{200(.45)(1-.45)} = 7.0356$$

(b) Because X is binomial with parameters 200 and .45, use R to obtain

$$P(X \geq 101) =$$

```
> 1 - pbinom(100, 200, .45)
[1] 0.06807525
```

■

Even when each element of the population has more than two possible values, it still remains true that if the population size is large in relation to the sample size, then the sample data can be regarded as being independent random variables from the population distribution.

Example 6.6.b. According to the U.S. Department of Agriculture's *World Livestock Situation*, the country with the greatest per capita consumption of pork is Denmark. In 2013, the amount of pork consumed by a person residing in Denmark had a mean value of 147 pounds with a standard deviation of 62 pounds. If a random sample of 25 Danes is chosen, approximate the probability that

the average amount of pork consumed by the members of this group in 2013 exceeded 150 pounds.

Solution. If we let X_i be the amount consumed by the ith member of the sample, $i = 1, \ldots, 25$, then the desired probability is

$$P\left\{\frac{X_1 + \cdots + X_{25}}{25} > 150\right\} = P\{\overline{X} > 150\}$$

where $\overline{X}$ is the sample mean of the 25 sample values. Since we can regard the X_i as being independent random variables with mean 147 and standard deviation 62, it follows from the central limit theorem that their sample mean will be approximately normal with mean 147 and standard deviation 62/5. Thus, with Z being a standard normal random variable, we have

$$P\{\overline{X} > 150\} = P\left\{\frac{\overline{X} - 147}{12.4} > \frac{150 - 147}{12.4}\right\}$$

$$P(\bar{X} > 150) \approx P\left(Z > \frac{150 - 147}{12.4}\right)$$

$$> 1 - \text{ pnorm}(3/12.4)$$

$$[1]\ 0.4044151 \quad \blacksquare$$

Problems

1. Suppose that X_1, X_2, X_3 are independent with the common probability mass function

   $$P\{X_i = 0\} = .2, \quad P\{X_i = 1\} = .3, \quad P\{X_i = 3\} = .5, \quad i = 1, 2, 3$$

   a. Plot the probability mass function of $\overline{X}_2 = \dfrac{X_1 + X_2}{2}$.
   b. Determine $E[\overline{X}_2]$ and $\text{Var}(\overline{X}_2)$.
   c. Plot the probability mass function of $\overline{X}_3 = \dfrac{X_1 + X_2 + X_3}{3}$.
   d. Determine $E[\overline{X}_3]$ and $\text{Var}(\overline{X}_3)$.
2. If 10 fair dice are rolled, approximate the probability that the sum of the values obtained (which ranges from 10 to 60) is between 30 and 40 inclusive.
3. Approximate the probability that the sum of 16 independent uniform (0, 1) random variables exceeds 10.
4. A roulette wheel has 38 slots, numbered 0, 00, and 1 through 36. If you bet 1 on a specified number, you either win 35 if the roulette ball lands on that number or lose 1 if it does not. If you continually make such bets, approximate the probability that
   a. you are winning after 34 bets;

b. you are winning after 1000 bets;
c. you are winning after 100,000 bets.

Assume that each roll of the roulette ball is equally likely to land on any of the 38 numbers.

5. A highway department has enough salt to handle a total of 80 inches of snowfall. Suppose the daily amount of snow has a mean of 1.5 inches and a standard deviation of .3 inch.
   a. Approximate the probability that the salt on hand will suffice for the next 50 days.
   b. What assumption did you make in solving part (a)?
   c. Do you think this assumption is justified? Explain briefly.
6. Fifty numbers are rounded off to the nearest integer and then summed. If the individual roundoff errors are uniformly distributed between $-.5$ and .5, what is the approximate probability that the resultant sum differs from the exact sum by more than 3?
7. A six-sided die, in which each side is equally likely to appear, is repeatedly rolled until the total of all rolls exceeds 400. Approximate the probability that this will require more than 140 rolls.
8. The amount of time that a certain type of battery functions is a random variable with mean 5 weeks and standard deviation 1.5 weeks. Upon failure, it is immediately replaced by a new battery. Approximate the probability that 13 or more batteries will be needed in a year.
9. The lifetime of a certain electrical part is a random variable with mean 100 hours and standard deviation 20 hours. If 16 such parts are tested, find the probability that the sample mean is
   a. less than 104;
   b. between 98 and 104 hours.
10. A tobacco company claims that the amount of nicotine in its cigarettes is a random variable with mean 2.2 mg and standard deviation .3 mg. However, the sample mean nicotine content of 100 randomly chosen cigarettes was 3.1 mg. What is the approximate probability that the sample mean would have been as high or higher than 3.1 if the company's claims were true?
11. The lifetime (in hours) of a type of electric bulb has expected value 500 and standard deviation 80. Approximate the probability that the sample mean of n such bulbs is greater than 525 when
   a. $n = 4$;
   b. $n = 16$;
   c. $n = 36$;
   d. $n = 64$.
12. An instructor knows from past experience that student exam scores have mean 77 and standard deviation 15. At present the instructor is teaching two separate classes — one of size 25 and the other of size 64.
   a. Approximate the probability that the average test score in the class of size 25 lies between 72 and 82.

b. Repeat part (a) for a class of size 64.
c. What is the approximate probability that the average test score in the class of size 25 is higher than that of the class of size 64?
d. Suppose the average scores in the two classes are 76 and 83. Which class, the one of size 25 or the one of size 64, do you think was more likely to have averaged 83?

13. If X is binomial with parameters $n = 150$, $p = .6$, compute the exact value of $P\{X \leq 80\}$ and compare with its normal approximation both (a) making use of and (b) not making use of the continuity correction.

14. Teams 1, 2, 3, 4 are all scheduled to play with each of the other teams 10 times. Whenever team i plays team j, team i is the winner with probability $P_{i,j}$, and team j is the winner with probability $P_{j,i} = 1 - P_{i,j}$. If

$$P_{1,2} = .6, \quad P_{1,3} = .7, \quad P_{1,4} = .75,$$
$$P_{2,3} = .6, \quad P_{2,4} = .70, \quad P_{3,4} = .5,$$

a. approximate the probability that team 1 wins at least 20 games.
Suppose we want to approximate the probability that team 2 wins at least as many games as does team 1. To do so, let X be the number of games that team 2 wins against team 1, let Y be the total number of games that team 2 wins against teams 3 and 4, and let Z be the total number of games that team 1 wins against teams 3 and 4.
b. Are X, Y, Z independent?
c. Express the event that team 2 wins at least as many games as does team 1 in terms of the random variables X, Y, Z.
d. Approximate the probability that team 2 wins at least as many games as team 1.

15. A club basketball team will play a 60-game season. Thirty-two of these games are against class A teams and 28 are against class B teams. The outcomes of all the games are independent. The team will win each game against a class A opponent with probability .5, and it will win each game against a class B opponent with probability .7. Let X denote its total number of victories in the season.
a. Is X a binomial random variable?
b. Let X_A and X_B denote, respectively, the number of victories against class A and class B teams. What are the distributions of X_A and X_B?
c. What is the relationship between X_A, X_B, and X?
d. Approximate the probability that the team wins 40 or more games.

16. Argue, based on the central limit theorem, that a Poisson random variable having mean λ will approximately have a normal distribution with mean and variance both equal to λ when λ is large. If X is Poisson with mean 100, compute the exact probability that X is less than or equal to 116 and compare it with its normal approximation both when a continuity correction is utilized and when it is not. The convergence of the Poisson to the normal is indicated in Figure 6.4.

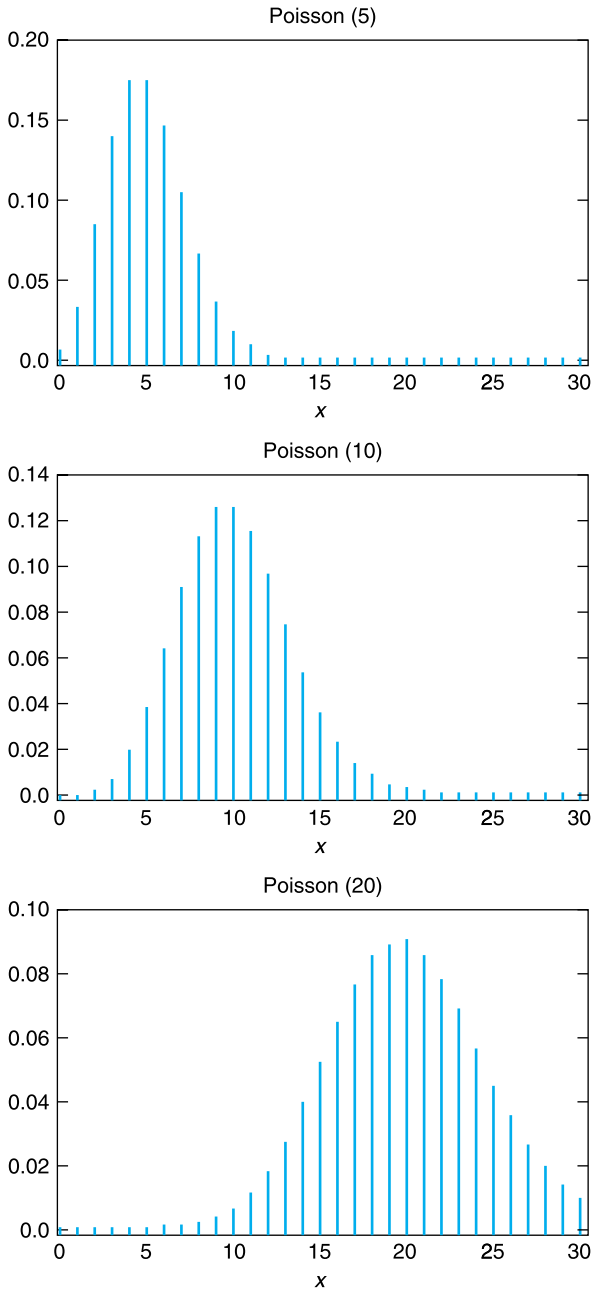


FIGURE 6.4
Poisson probability mass functions.

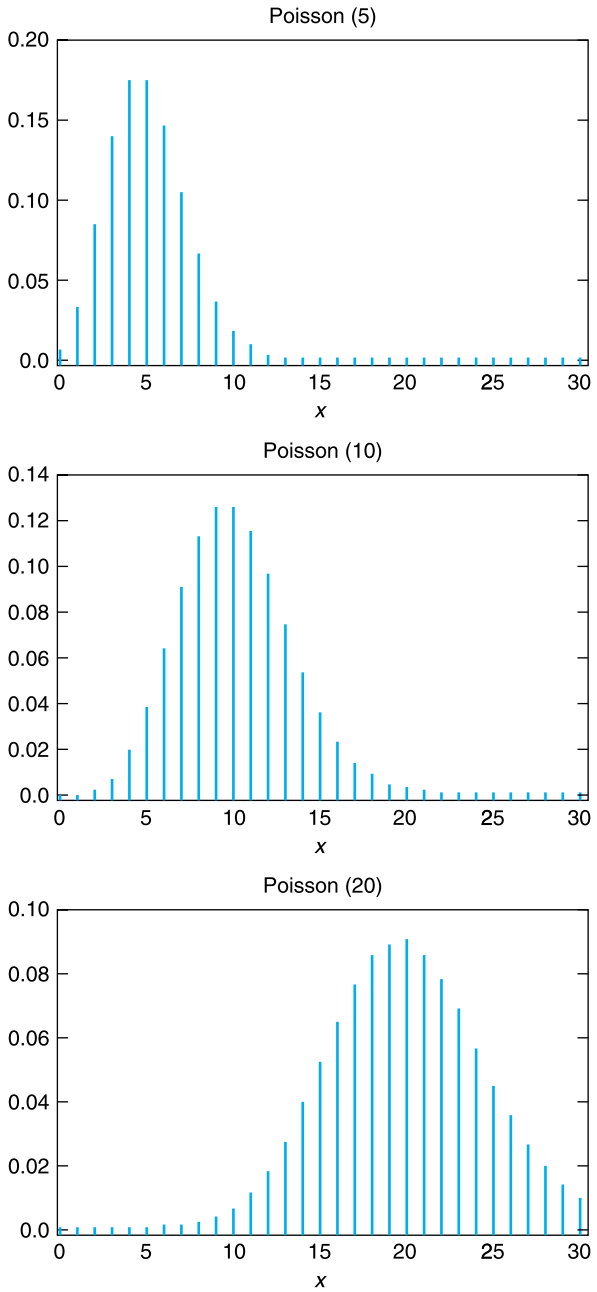


FIGURE 6.4
Poisson probability mass functions.

17. Use the text disk to compute $P\{X \leq 10\}$ when X is a binomial random variable with parameters $n = 100$, $p = .1$. Now compare this with its **(a)** Poisson and **(b)** normal approximation. In using the normal approximation, write the desired probability as $P\{X < 10.5\}$ so as to utilize the continuity correction.
18. The temperature at which a thermostat goes off is normally distributed with variance σ^2. If the thermostat is to be tested five times, find
    a. $P\{S^2/\sigma^2 \leq 1.8\}$
    b. $P\{.85 \leq S^2/\sigma^2 \leq 1.15\}$

    where S^2 is the sample variance of the five data values.
19. In Problem 18, how large a sample would be necessary to ensure that the probability in part (a) is at least .95?
20. Consider two independent samples — the first of size 10 from a normal population having variance 4 and the second of size 5 from a normal population having variance 2. Compute the probability that the sample variance from the second sample exceeds the one from the first. (*Hint*: Relate it to the F-distribution.)
21. Twelve percent of the population is left-handed. Find the probability that there are between 10 and 14 left-handers in a random sample of 100 members of this population. That is, find $P\{10 \leq X \leq 14\}$, where X is the number of left-handers in the sample.
22. Fifty-two percent of the residents of a certain city are in favor of teaching evolution in high school. Find or approximate the probability that at least 50 percent of a random sample of size n is in favor of teaching evolution, when
    a. $n = 10$;
    b. $n = 100$;
    c. $n = 1000$;
    d. $n = 10{,}000$.
23. The following table gives the percentages of individuals of a given city, categorized by gender, that follow certain negative health practices. Suppose a random sample of 300 men is chosen. Approximate the probability that
    a. at least 150 of them rarely eat breakfast;
    b. fewer than 100 of them smoke.

	Sleeps 6 Hours or Less per Night	Smoker	Rarely Eats Breakfast	Is 20 Percent or More Overweight
Men	22.7	28.4	45.4	29.6
Women	21.4	22.8	42.0	25.6

Source: U.S. National Center for Health Statistics, Health Promotion and Disease Prevention.

24. (Use the table from Problem 23.) Suppose a random sample of 300 women is chosen. Approximate the probability that

a. at least 60 of them are overweight by 20 percent or more;
b. fewer than 50 of them sleep 6 hours or less nightly.

25. (Use the table from Problem 23.) Suppose random samples of 300 women and of 300 men are chosen. Approximate the probability that more women than men rarely eat breakfast.
26. The following table uses data concerning the percentages of teenage male and female full-time workers whose annual salaries fall in different salary groupings. Suppose random samples of 1000 men and 1000 women were chosen. Use the table to approximate the probability that
    a. at least half of the women earned less than $20,000;
    b. more than half of the men earned $20,000 or more;
    c. more than half of the women and more than half of the men earned $20,000 or more;
    d. 250 or fewer of the women earned at least $25,000;
    e. at least 200 of the men earned $50,000 or more;
    f. more women than men earned between $20,000 and $24,999.

Earnings Range	Percentage of Women	Percentage of Men
$4999 or less	2.8	1.8
$5000 to $9999	10.4	4.7
$10,000 to $19,999	41.0	23.1
$20,000 to $24,999	16.5	13.4
$25,000 to $49,999	26.3	42.1
$50,000 and over	3.0	14.9

Source: *U.S. Department of Commerce, Bureau of the Census.*

27. Today, roughly 10.5 percent of the labor force belong to a union. If five workers are randomly chosen, what is the probability that none of them belong to a union? Compare your answer to what it would have been in 1983 when 20.1 percent of the workforce belonged to a union.
28. The sample mean and sample standard deviation of all San Francisco student scores on the most recent Scholastic Aptitude Test examination in mathematics were 517 and 120. Approximate the probability that a random sample of 144 students would have an average score exceeding
    a. 507;
    b. 517;
    c. 537;
    d. 550.
29. The average salary of newly graduated students with bachelor's degrees in chemical engineering is $53,600, with a standard deviation of $3200. Approximate the probability that the average salary of a sample of 12 recently graduated chemical engineers exceeds $55,000.

30. A certain component is critical to the operation of an electrical system and must be replaced immediately upon failure. If the mean lifetime of this type of component is 100 hours and its standard deviation is 30 hours, how many of the components must be in stock so that the probability that the system is in continual operation for the next 2000 hours is at least .95?

able. This is the *Bayesian* approach, which supposes that prior to observing the data, information about θ is always available to the decision maker, and that this information can be expressed in terms of a probability distribution on θ. In such a situation, we show how to compute the *Bayes estimator*, which is the estimator whose expected squared distance from θ is minimal.

7.2 Maximum likelihood estimators

Any statistic used to estimate the value of an unknown parameter θ is called an *estimator* of θ. The observed value of the estimator is called the *estimate*. For instance, as we shall see, the usual estimator of the mean of a normal population, based on a sample $X_1, \ldots, X_n$ from that population, is the sample mean $\overline{X} = \sum_i X_i/n$. If a sample of size 3 yields the data $X_1 = 2$, $X_2 = 3$, $X_3 = 4$, then the estimate of the population mean, resulting from the estimator $\overline{X}$, is the value 3.

Suppose that the random variables $X_1, \ldots, X_n$, whose joint distribution is assumed given except for an unknown parameter θ, are to be observed. The problem of interest is to use the observed values to estimate θ. For example, the X_i's might be independent, exponential random variables each having the same unknown mean θ. In this case, the joint density function of the random variables would be given by

$$
\begin{aligned}
&f(x_1, x_2, \ldots, x_n) \\
&\quad = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n) \\
&\quad = \frac{1}{\theta} e^{-x_1/\theta} \frac{1}{\theta} e^{-x_2/\theta} \cdots \frac{1}{\theta} e^{-x_n/\theta}, \qquad 0 < x_i < \infty, i = 1, \ldots, n \\
&\quad = \frac{1}{\theta^n} \exp\left\{ -\sum_1^n x_i/\theta \right\}, \qquad 0 < x_i < \infty, i = 1, \ldots, n
\end{aligned}
$$

and the objective would be to estimate θ from the observed data $X_1, X_2, \ldots, X_n$.

A particular type of estimator, known as the *maximum likelihood* estimator, is widely used in statistics. It is obtained by reasoning as follows. Let $f(x_1, \ldots, x_n|\theta)$ denote the joint probability mass function of the random variables $X_1, X_2, \ldots, X_n$ when they are discrete, and let it be their joint probability density function when they are jointly continuous random variables. Because θ is assumed unknown, we also write f as a function of θ. Now since $f(x_1, \ldots, x_n|\theta)$ represents the likelihood that the values $x_1, x_2, \ldots, x_n$ will be observed when θ is the true value of the parameter, it would seem that a reasonable estimate of θ would be that value yielding the largest likelihood of the observed values. In other words, the maximum likelihood estimate $\hat{\theta}$ is defined to be that value of θ maximizing $f(x_1, \ldots, x_n|\theta)$ where $x_1, \ldots, x_n$ are the ob-

served values. The function $f(x_1, \ldots, x_n|\theta)$ is often referred to as the *likelihood* function of θ.

In determining the maximizing value of θ, it is often useful to use the fact that $f(x_1, \ldots, x_n|\theta)$ and $\log[f(x_1, \ldots, x_n|\theta)]$ have their maximum at the same value of θ. Hence, we may also obtain $\hat{\theta}$ by maximizing $\log[f(x_1, \ldots, x_n|\theta)]$.

Example 7.2.a (Maximum Likelihood Estimator of a Bernoulli Parameter). Suppose that n independent trials, each of which is a success with probability p, are performed. What is the maximum likelihood estimator of p?

Solution. The data consist of the values of $X_1, \ldots, X_n$ where

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success} \\ 0 & \text{otherwise} \end{cases}$$

Now

$$P\{X_i = 1\} = p = 1 - P\{X_i = 0\}$$

which can be succinctly expressed as

$$P\{X_i = x\} = p^x(1-p)^{1-x}, \quad x = 0, 1$$

Hence, by the assumed independence of the trials, the likelihood (that is, the joint probability mass function) of the data is given by

$$\begin{aligned} f(x_1, \ldots, x_n|p) &= P\{X_1 = x_1, \ldots, X_n = x_n|p\} \\ &= p^{x_1}(1-p)^{1-x_1} \cdots p^{x_n}(1-p)^{1-x_n} \\ &= p^{\sum_1^n x_i}(1-p)^{n-\sum_1^n x_i}, \quad x_i = 0, 1, \quad i = 1, \ldots, n \end{aligned}$$

To determine the value of p that maximizes the likelihood, first take logs to obtain

$$\log f(x_1, \ldots, x_n|p) = \sum_1^n x_i \log p + \left(n - \sum_1^n x_i\right) \log(1-p)$$

Differentiation yields

$$\frac{d}{dp} \log f(x_1, \ldots, x_n|p) = \frac{\sum_1^n x_i}{p} - \frac{\left(n - \sum_1^n x_i\right)}{1-p}$$

Upon equating to zero and solving, we obtain that the maximum likelihood estimate $\hat{p}$ satisfies

$$\frac{\sum_1^n x_i}{\hat{p}} = \frac{n - \sum_1^n x_i}{1 - \hat{p}}$$

or

$$\hat{p} = \frac{\sum_{i=1}^n x_i}{n}$$

Hence, the maximum likelihood estimator of the unknown mean of a Bernoulli distribution is given by

$$d(X_1, \ldots, X_n) = \frac{\sum_{i=1}^n X_i}{n}$$

Since $\sum_{i=1}^n X_i$ is the number of successful trials, we see that the maximum likelihood estimator of p is equal to the proportion of the observed trials that result in successes. For an illustration, suppose that each RAM (random access memory) chip produced by a certain manufacturer is, independently, of acceptable quality with probability p. Then if out of a sample of 1000 tested 921 are acceptable, it follows that the maximum likelihood estimate of p is .921. ■

Example 7.2.b. Two proofreaders were given the same manuscript to read. If proofreader 1 found n_1 errors, and proofreader 2 found n_2 errors, with $n_{1,2}$ of these errors being found by both proofreaders, estimate N, the total number of errors that are in the manuscript.

Solution. Before we can estimate N we need to make some assumptions about the underlying probability model. So let us assume that the results of the proofreaders are independent, and that each error in the manuscript is independently found by proofreader i with probability p_i, $i = 1, 2$.

To estimate N, we will start by deriving an estimator of p_1. To do so, note that each of the n_2 errors found by reader 2 will, independently, be found by proofreader 1 with probability p_i. Because proofreader 1 found $n_{1,2}$ of those n_2 errors, a reasonable estimate of p_1 is given by

$$\hat{p}_1 = \frac{n_{1,2}}{n_2}$$

However, because proofreader 1 found n_1 of the N errors in the manuscript, it is reasonable to suppose that p_1 is also approximately equal to $\frac{n_1}{N}$. Equating

this to $\hat{p}_1$ gives that

$$\frac{n_{1,2}}{n_2} \approx \frac{n_1}{N}$$

or

$$N \approx \frac{n_1 n_2}{n_{1,2}}$$

Because the preceding estimate is symmetric in n_1 and n_2, it follows that it is the same no matter which proofreader is designated as proofreader 1.

An interesting application of the preceding occurred when two teams of researchers recently announced that they had decoded the human genetic code sequence. As part of their work both teams estimated that the human genome consisted of approximately 33,000 genes. Because both teams independently arrived at the same number, many scientists found this number believable. However, most scientists were quite surprised by this relatively small number of genes; by comparison it is only about twice as many as a fruit fly has. However, a closer inspection of the findings indicated that the two groups only agreed on the existence of about 17,000 genes. (That is, 17,000 genes were found by both teams.) Thus, based on our preceding estimator, we would estimate that the actual number of genes, rather than being 33,000, is

$$\frac{n_1 n_2}{n_{1,2}} = \frac{33{,}000 \times 33{,}000}{17{,}000} \approx 64{,}000$$

(Because there is some controversy about whether some of the genes claimed to be found are actually genes, 64,000 should probably be taken as an upper bound on the actual number of genes.)

The estimation approach used when there are two proofreaders does not work when there are m proofreaders, when $m > 2$. Because, if for each i, we let $\hat{p}_i$ be the fraction of the errors found by at least one of the other proofreaders j, $(j \neq i)$, that are also found by i, and then set that equal to $\frac{n_i}{N}$, then the estimate of N, namely $\frac{n_i}{\hat{p}_i}$, would differ for different values of i. Moreover, with this approach it is possible that we may have that $\hat{p}_i > \hat{p}_j$ even if proofreader i finds fewer errors than does proofreader j. For instance, for $m = 3$, suppose proofreaders 1 and 2 find exactly the same set of 10 errors whereas proofreader 3 finds 20 errors with only 1 of them in common with the set of errors found by the others. Then, because proofreader 1 (and 2) found 10 of the 29 errors found by at least one of the other proofreaders, $\hat{p}_i = 10/29$, $i = 1, 2$. On the other hand, because proofreader 3 only found 1 of the 10 errors found by the others, $\hat{p}_3 = 1/10$. Therefore, although proofreader 3 found twice the number of errors as did proofreader 1, the estimate of p_3 is less than that of p_1. To obtain more reasonable estimates, we could take the preceding values of $\hat{p}_i$, $i = 1, \ldots, m$, as preliminary estimates of the p_i. Now, let n_f be the number of errors that

are found by at least one proofreader. Because n_f/N is the fraction of errors that are found by at least one proofreader, this should approximately equal $1 - \prod_{i=1}^{m}(1-p_i)$, the probability that an error is found by at least one proofreader. Therefore, we have

$$\frac{n_f}{N} \approx 1 - \prod_{i=1}^{m}(1-p_i)$$

suggesting that $N \approx \hat{N}$, where

$$\hat{N} = \frac{n_f}{1 - \prod_{i=1}^{m}(1-\hat{p}_i)} \tag{7.2.1}$$

With this estimate of N, we can then reset our estimates of the p_i by using

$$\hat{p}_i = \frac{n_i}{\hat{N}}, \quad i = 1, \ldots, m \tag{7.2.2}$$

We can then reestimate N by using the new value (Equation (7.2.1)). (The estimation need not stop here; each time we obtain a new estimate $\hat{N}$ of N we can use Equation (7.2.2) to obtain new estimates of the p_i, which can then be used to obtain a new estimate of N, and so on.) ■

Example 7.2.c (Maximum Likelihood Estimator of a Poisson Parameter). Suppose $X_1, \ldots, X_n$ are independent Poisson random variables each having mean λ. Determine the maximum likelihood estimator of λ.

Solution. The likelihood function is given by

$$\begin{aligned} f(x_1, \ldots, x_n|\lambda) &= \frac{e^{-\lambda}\lambda^{x_1}}{x_1!} \cdots \frac{e^{-\lambda}\lambda^{x_n}}{x_n!} \\ &= \frac{e^{-n\lambda}\lambda^{\sum_1^n x_i}}{x_1! \ldots x_n!} \end{aligned}$$

Thus,

$$\log\, f(x_1, \ldots, x_n|\lambda) = -n\lambda + \sum_{1}^{n} x_i \log\lambda - \log c$$

where $c = \prod_{i=1}^{n} x_i!$ does not depend on λ. Differentiation yields

$$\frac{d}{d\lambda} \log\, f(x_1, \ldots, x_n|\lambda) = -n + \frac{\sum_{1}^{n} x_i}{\lambda}$$

By equating to zero, we obtain that the maximum likelihood estimate $\hat{\lambda}$ equals

$$\hat{\lambda} = \frac{\sum_1^n x_i}{n}$$

and so the maximum likelihood estimator is given by

$$d(X_1, \ldots, X_n) = \frac{\sum_{i=1}^n X_i}{n}$$

For example, suppose that the number of people who enter a certain retail establishment in any day is a Poisson random variable having an unknown mean λ, which must be estimated. If after 20 days a total of 857 people have entered the establishment, then the maximum likelihood estimate of λ is $857/20 = 42.85$. That is, we estimate that on average, 42.85 customers will enter the establishment on a given day. ■

Example 7.2.d. The number of traffic accidents in Berkeley, California, in 10 randomly chosen nonrainy days in 1998 is as follows:

4, 0, 6, 5, 2, 1, 2, 0, 4, 3

Use these data to estimate the proportion of nonrainy days that had 2 or fewer accidents that year.

Solution. Since there are a large number of drivers, each of whom has a small probability of being involved in an accident in a given day, it seems reasonable to assume that the daily number of traffic accidents is a Poisson random variable. Since

$$\overline{X} = \frac{1}{10}\sum_{i=1}^{10} X_i = 2.7$$

it follows that the maximum likelihood estimate of the Poisson mean is 2.7. Since the long-run proportion of nonrainy days that have 2 or fewer accidents is equal to $P\{X \leq 2\}$, where X is the random number of accidents in a day, it follows that the desired estimate is

$$e^{-2.7}(1 + 2.7 + (2.7)^2/2) = .4936$$

That is, we estimate that a little less than half of the nonrainy days had 2 or fewer accidents. ■

Example 7.2.e (Maximum Likelihood Estimator in a Normal Population). Suppose $X_1, \ldots, X_n$ are independent, normal random variables each with unknown mean μ and unknown standard deviation σ. The joint density is given

by

$$f(x_1, \ldots, x_n|\mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[\frac{-(x_i - \mu)^2}{2\sigma^2}\right]$$

$$= \left(\frac{1}{2\pi}\right)^{n/2} \frac{1}{\sigma^n} \exp\left[\frac{-\sum_1^n (x_i - \mu)^2}{2\sigma^2}\right]$$

The logarithm of the likelihood is thus given by

$$\log\, f(x_1, \ldots, x_n|\mu, \sigma) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{\sum_1^n (x_i - \mu)^2}{2\sigma^2}$$

In order to find the value of μ and σ maximizing the foregoing, we compute

$$\frac{\partial}{\partial\mu} \log\, f(x_1, \ldots, x_n|\mu, \sigma) = \frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^2}$$

$$\frac{\partial}{\partial\sigma} \log\, f(x_1, \ldots, x_n|\mu, \sigma) = -\frac{n}{\sigma} + \frac{\sum_1^n (x_i - \mu)^2}{\sigma^3}$$

Equating these equations to zero yields that

$$\hat{\mu} = \sum_{i=1}^{n} x_i/n$$

and

$$\hat{\sigma} = \left[\sum_{i=1}^{n} (x_i - \hat{\mu})^2/n\right]^{1/2}$$

Hence, the maximum likelihood estimators of μ and σ are given, respectively, by

$$\overline{X} \quad \text{and} \quad \left[\sum_{i=1}^{n} (X_i - \overline{X})^2/n\right]^{1/2} \tag{7.2.3}$$

It should be noted that the maximum likelihood estimator of the standard deviation σ differs from the sample standard deviation

$$S = \left[\sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1) \right]^{1/2}$$

in that the denominator in Equation (7.2.3) is $\sqrt{n}$ rather than $\sqrt{n-1}$. However, for n of reasonable size, these two estimators of σ will be approximately equal. ■

Example 7.2.f. *Kolmogorov's law of fragmentation* states that the size of an individual particle in a large collection of particles resulting from the fragmentation of a mineral compound will have an approximate lognormal distribution, where a random variable X is said to have a *lognormal* distribution if $\log(X)$ has a normal distribution. The law, which was first noted empirically and then later given a theoretical basis by Kolmogorov, has been applied to a variety of engineering studies. For instance, it has been used in the analysis of the size of randomly chosen gold particles from a collection of gold sand. A less obvious application of the law has been to a study of the stress release in earthquake fault zones (see Lomnitz, C., "Global Tectonics and Earthquake Risk," *Developments in Geotectonics*, Elsevier, Amsterdam, 1979).

Suppose that a sample of 10 grains of metallic sand taken from a large sand pile have respective lengths (in millimeters):

2.2, 3.4, 1.6, 0.8, 2.7, 3.3, 1.6, 2.8, 2.5, 1.9

Estimate the percentage of sand grains in the entire pile whose length is between 2 and 3 mm.

Solution. Taking the natural logarithm of these 10 data values, the following transformed data set results

.7885, 1.2238, .4700, −.2231, .9933, 1.1939, .4700, 1.0296, .9163, .6419

Because the sample mean and sample standard deviation of these data are

$$\overline{x} = .7504, \quad s = .4351$$

it follows that the logarithm of the length of a randomly chosen grain has a normal distribution with mean approximately equal to .7504 and with standard deviation approximately equal to .4351. Hence, if X is the length of the grain, then

$$\begin{aligned} P\{2 < X < 3\} &= P\{\log(2) < \log(X) < \log(3)\} \\ &= P\left\{ \frac{\log(2) - .7504}{.4351} < \frac{\log(X) - .7504}{.4351} < \frac{\log(3) - .7504}{.4351} \right\} \end{aligned}$$

$$\begin{aligned}
&\approx \text{pnorm}((\log(3) - .7504)/.4351) \\
&\quad - \text{pnorm}((\log(2) - .7504)/.4351) \\
&= 0.3405766 \quad \blacksquare
\end{aligned}$$

The lognormal distribution is often assumed in situations where the random variable under interest can be regarded as the product of a large number of independent and identically distributed random variables. For instance, it is commonly used in finance as the distribution of the price of a security at some future time. To see why this might be reasonable, suppose that the current price of the security is s and that we are interested in $S(t)$, the price of the security after an additional time t. For a large value n, let $t_i = it/n$, and consider $S(t_1), \ldots, S(t_n)$, the prices of the security at the times $t_1, \ldots, t_n$. Now, a common assumption in finance is that the ratios $S(t_i)/S(t_{i-1})$ are approximately independent and identically distributed. Consequently, if we let $X_i = S(t_i)/S(t_{i-1})$, then writing

$$\begin{aligned}
S(t) = S(t_n) &= S(t_0) \cdot \frac{S(t_1)}{S(t_0)} \cdot \frac{S(t_2)}{S(t_1)} \cdots \frac{S(t_n)}{S(t_{n-1})} \\
&= s \prod_{i=1}^{n} X_i
\end{aligned}$$

we obtain, upon taking logarithms, that

$$\log(S(t)) = \log(s) + \sum_{i=1}^{n} \log(X_i)$$

Thus, by the central limit theorem $\log(S(t))$ will approximately have a normal distribution.

The lognormal distribution has also been shown to be a good fit for such random variables as length of patient stays in hospitals, and vehicle travel times.

In all of the foregoing examples, the maximum likelihood estimator of the population mean turned out to be the sample mean $\overline{X}$. To show that this is not always the situation, consider the following example.

Example 7.2.g (Estimating the Mean of a Uniform Distribution). Suppose $X_1, \ldots, X_n$ constitute a sample from a uniform distribution on $(0, \theta)$, where θ is unknown. Their joint density is thus

$$f(x_1, x_2, \ldots, x_n|\theta) = \begin{cases} \dfrac{1}{\theta^n} & 0 < x_i < \theta, \quad i = 1, \ldots, n \\ 0 & \text{otherwise} \end{cases}$$

This density is maximized by choosing θ as small as possible. Since θ must be at least as large as all of the observed values x_i, it follows that the smallest possible

choice of θ is equal to $\max(x_1, x_2, \ldots, x_n)$. Hence, the maximum likelihood estimator of θ is

$$\hat{\theta} = \max(X_1, X_2, \ldots, X_n)$$

It easily follows from the foregoing that the maximum likelihood estimator of $\theta/2$, the mean of the distribution, is $\max(X_1, X_2, \ldots, X_n)/2$. ■

7.2.1 Estimating life distributions[1]

Let X denote the age at death of a randomly chosen child born today. That is, $X = i$ if the newborn dies in its ith year, $i \geq 1$. To estimate the probability mass function of X, let λ_i denote the probability that a newborn who has survived his or her first $i - 1$ years dies in year i. That is,

$$\lambda_i = P\{X = i | X > i - 1\} = \frac{P\{X = i\}}{P\{X > i - 1\}}$$

Also, let

$$s_i = 1 - \lambda_i = \frac{P\{X > i\}}{P\{X > i - 1\}}$$

be the probability that a newborn who survives her first $i - 1$ years also survives year i. The quantity λ_i is called the *failure rate*, and s_i is called the *survival rate*, of an individual who is entering his or her ith year. Now,

$$\begin{aligned} s_1 s_2 \cdots s_i &= P\{X > 1\} \frac{P\{X > 2\} P\{X > 3\}}{P\{X > 1\} P\{X > 2\}} \cdots \frac{P\{X > i\}}{P\{X > i - 1\}} \\ &= P\{X > i\} \end{aligned}$$

Therefore,

$$P\{X = n\} = P\{X > n - 1\} \lambda_n = s_1 \cdots s_{n-1}(1 - s_n)$$

Consequently, we can estimate the probability mass function of X by estimating the quantities s_i, $i = 1, \ldots, n$. The value s_i can be estimated by looking at all individuals in the population who reached age i 1 year ago, and then letting the estimate $\hat{s}_i$ be the fraction of them who are alive today. We would then use $\hat{s}_1 \hat{s}_2 \cdots \hat{s}_{n-1}(1 - \hat{s}_n)$ as the estimate of $P\{X = n\}$. (Note that although we are using the most recent possible data to estimate the quantities s_i, our estimate of the probability mass function of the lifetime of a newborn assumes that the survival rate of the newborn when it reaches age i will be the same as last year's survival rate of someone of age i.)

[1] Optional section.

The use of the survival rate to estimate a life distribution is also of importance in health studies with partial information. For instance, consider a study in which a new drug is given to a random sample of 12 lung cancer patients. Suppose that after some time we have the following data on the number of months of survival after starting the new drug:

$$4, 7^*, 9, 11^*, 12, 3, 14^*, 1, 8, 7, 5, 3^*$$

where x means that the patient died in month x after starting the drug treatment, and x^* means that the patient has taken the drug for x months and is still alive.

Let X equal the number of months of survival after beginning the drug treatment, and let

$$s_i = P\{X > i | X > i - 1\} = \frac{P\{X > i\}}{P\{X > i - 1\}}$$

To estimate s_i, the probability that a patient who has survived the first $i - 1$ months will also survive month i, we should take the fraction of those patients who began their ith month of drug taking and survived the month. For instance, because 11 of the 12 patients survived month 1, $\hat{s}_1 = 11/12$. Because all 11 patients who began month 2 survived, $\hat{s}_2 = 11/11$. Because 10 of the 11 patients who began month 3 survived, $\hat{s}_3 = 10/11$. Because 8 of the 9 patients who began their fourth month of taking the drug (the 9 being all but the ones labeled 1, 3, and 3*) survived month 4, $\hat{s}_4 = 8/9$. Similar reasoning holds for the others, giving the following survival rate estimates:

$$\hat{s}_1 = 11/12$$
$$\hat{s}_2 = 11/11$$
$$\hat{s}_3 = 10/11$$
$$\hat{s}_4 = 8/9$$
$$\hat{s}_5 = 7/8$$
$$\hat{s}_6 = 7/7$$
$$\hat{s}_7 = 6/7$$
$$\hat{s}_8 = 4/5$$
$$\hat{s}_9 = 3/4$$
$$\hat{s}_{10} = 3/3$$
$$\hat{s}_{11} = 3/3$$
$$\hat{s}_{12} = 1/2$$
$$\hat{s}_{13} = 1/1$$
$$\hat{s}_{14} = 1/1$$

We can now use $\prod_{i=1}^{j} \hat{s}_i$ to estimate the probability that a drug taker survives at least j time periods, $j = 1, \ldots, 14$. For instance, our estimate of $P\{X > 6\}$ is 35/54.

7.3 Interval estimates

Suppose that $X_1, \ldots, X_n$ is a sample from a normal population having unknown mean μ and known variance σ^2. It has been shown that $\overline{X} = \sum_{i=1}^{n} X_i/n$ is the maximum likelihood estimator for μ. However, we don't expect that the sample mean $\overline{X}$ will exactly equal μ, but rather that it will "be close." Hence, rather than a point estimate, it is sometimes more valuable to be able to specify an interval for which we have a certain degree of confidence that μ lies within. To obtain such an interval estimator, we make use of the probability distribution of the point estimator. Let us see how it works for the preceding situation.

In the foregoing, since the point estimator $\overline{X}$ is normal with mean μ and variance σ^2/n, it follows that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma}$$

has a standard normal distribution. Therefore,

$$P\left\{-1.96 < \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma} < 1.96\right\} = .95$$

or, equivalently,

$$P\left\{-1.96\frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

Multiplying through by -1 yields the equivalent statement

$$P\left\{-1.96\frac{\sigma}{\sqrt{n}} < \mu - \overline{X} < 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

or, equivalently,

$$P\left\{\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

That is, 95 percent of the time the value of the sample average $\bar{X}$ will be such that the distance between it and the mean μ will be less than $1.96\sigma/\sqrt{n}$. If we now observe the sample and it turns out that $\overline{X} = \overline{x}$, then we say that "with 95 percent confidence"

$$\overline{x} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + 1.96\frac{\sigma}{\sqrt{n}} \tag{7.3.1}$$

That is, "with 95 percent confidence" we assert that the true mean lies within $1.96\sigma/\sqrt{n}$ of the observed sample mean. The interval

$$\left(\overline{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

is called a *95 percent confidence interval estimate* of μ.

Example 7.3.a. Suppose that when a signal having value μ is transmitted from location A the value received at location B is normally distributed with mean μ and variance 4. That is, if μ is sent, then the value received is $\mu + N$ where N, representing noise, is normal with mean 0 and variance 4. To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, let us construct a 95 percent confidence interval for μ.

Since

$$\overline{x} = \frac{81}{9} = 9$$

it follows, under the assumption that the values received are independent, that a 95 percent confidence interval for μ is

$$\left(9 - 1.96\frac{\sigma}{3},\ 9 + 1.96\frac{\sigma}{3}\right) = (7.69, 10.31)$$

Hence, we are "95 percent confident" that the true message value lies between 7.69 and 10.31. ■

The interval in Equation (7.3.1) is called a *two-sided confidence interval*. Sometimes, however, we are interested in determining a value so that we can assert with, say, 95 percent confidence, that μ is at least as large as that value.

To determine such a value, note that if Z is a standard normal random variable then

$$P\{Z < 1.645\} = .95$$

As a result,

$$P\left\{\sqrt{n}\frac{(\overline{X} - \mu)}{\sigma} < 1.645\right\} = .95$$

or

$$P\left\{\overline{X} - 1.645\frac{\sigma}{\sqrt{n}} < \mu\right\} = .95$$

Thus, a 95 percent *one-sided upper confidence interval* for μ is

$$\left(\overline{x} - 1.645\frac{\sigma}{\sqrt{n}}, \infty\right)$$

where $\overline{x}$ is the observed value of the sample mean.

A *one-sided lower confidence interval* is obtained similarly; when the observed value of the sample mean is $\overline{x}$, then the 95 percent one-sided lower confidence interval for μ is

$$\left(-\infty, \overline{x} + 1.645\frac{\sigma}{\sqrt{n}}\right)$$

Example 7.3.b. Determine the upper and lower 95 percent confidence interval estimates of μ in Example 7.3.a.

Solution. Since

$$1.645\frac{\sigma}{\sqrt{n}} = \frac{3.29}{3} = 1.097$$

the 95 percent upper confidence interval is

$$(9 - 1.097, \infty) = (7.903, \infty)$$

and the 95 percent lower confidence interval is

$$(-\infty, 9 + 1.097) = (-\infty, 10.097) \quad \blacksquare$$

We can also obtain confidence intervals of any specified level of confidence. To do so, recall that z_α is such that

$$P\{Z > z_\alpha\} = \alpha$$

when Z is a standard normal random variable. But this implies (see Figure 7.1) that for any α

$$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha$$

As a result, we see that

$$P\left\{-z_{\alpha/2} < \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma} < z_{\alpha/2}\right\} = 1 - \alpha$$

or

$$P\left\{-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

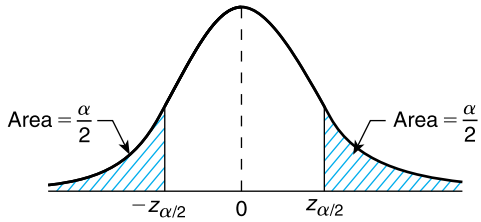


FIGURE 7.1
$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha$.

or

$$P\left\{-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu - \overline{X} < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

That is,

$$P\left\{\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

Hence, a $100(1 - \alpha)$ percent two-sided confidence interval for μ is

$$\left(\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

where $\overline{x}$ is the observed sample mean.

Similarly, knowing that $Z = \sqrt{n}\frac{(\overline{X} - \mu)}{\sigma}$ is a standard normal random variable, along with the identities

$$P\{Z > z_\alpha\} = \alpha$$

and

$$P\{Z < -z_\alpha\} = \alpha$$

results in one-sided confidence intervals of any desired level of confidence. Specifically, we obtain that

$$\left(\overline{x} - z_\alpha\frac{\sigma}{\sqrt{n}}, \ \infty\right)$$

and

$$\left(-\infty, \ \overline{x} + z_\alpha\frac{\sigma}{\sqrt{n}}\right)$$

are, respectively, $100(1 - \alpha)$ percent one-sided upper and $100(1 - \alpha)$ percent one-sided lower confidence intervals for μ.

Example 7.3.c. Use the data of Example 7.3.a to obtain a 99 percent confidence interval estimate of μ, along with 99 percent one-sided upper and lower intervals.

Solution. Since $z_{.005} = 2.58$, and

$$2.58\frac{\alpha}{\sqrt{n}} = \frac{5.16}{3} = 1.72$$

it follows that a 99 percent confidence interval for μ is

$$9 \pm 1.72$$

That is, the 99 percent confidence interval estimate is $(7.28, 10.72)$.

Also, since $z_{.01} = 2.33$, a 99 percent upper confidence interval is

$$(9 - 2.33(2/3),\ \infty) = (7.447,\ \infty)$$

Similarly, a 99 percent lower confidence interval is

$$(-\infty,\ 9 + 2.33(2/3)) = (-\infty,\ 10.553) \quad \blacksquare$$

Sometimes we are interested in a two-sided confidence interval of a certain level, say $1 - \alpha$, and the problem is to choose the sample size n so that the interval is of a certain size. For instance, suppose that we want to compute an interval of length .1 that we can assert, with 99 percent confidence, contains μ. How large need n be? To solve this, note that as $z_{.005} = 2.58$ it follows that the 99 percent confidence interval for μ from a sample of size n is

$$\left(\overline{x} - 2.58\frac{\sigma}{\sqrt{n}},\quad \overline{x} + 2.58\frac{\sigma}{\sqrt{n}}\right)$$

Hence, its length is

$$5.16\frac{\sigma}{\sqrt{n}}$$

Thus, to make the length of the interval equal to .1, we must choose

$$5.16\frac{\sigma}{\sqrt{n}} = .1$$

or

$$n = (51.6\sigma)^2$$

Remark

The interpretation of "a $100(1-\alpha)$ percent confidence interval" can be confusing. It should be noted that we are *not* asserting that the probability that $\mu \in (\overline{x} - 1.96\sigma/\sqrt{n},\ \overline{x} + 1.96\sigma/\sqrt{n})$ is .95, for there are no random variables involved in this assertion. What we are asserting is that the technique utilized to obtain this interval is such that 95 percent of the time that it is employed it will result in an interval in which μ lies. In other words, before the data are observed we can assert that with probability .95 the interval that will be obtained will contain μ, whereas after the data are obtained we can only assert that the resultant interval indeed contains μ "with confidence .95."

Example 7.3.d. From past experience it is known that the weights of salmon grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.3 pound. If we want to be 95 percent certain that our estimate of the present season's mean weight of a salmon is correct to within ± 0.1 pound, how large a sample is needed?

Solution. A 95 percent confidence interval estimate for the unknown mean μ, based on a sample of size n, is

$$\mu \in \left(\overline{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

Because the estimate $\overline{x}$ is within $1.96(\sigma/\sqrt{n}) = .588/\sqrt{n}$ of any point in the interval, it follows that we can be 95 percent certain that $\overline{x}$ is within 0.1 of μ provided that

$$\frac{.588}{\sqrt{n}} \leq 0.1$$

That is, provided that

$$\sqrt{n} \geq 5.88$$

or

$$n \geq 34.57$$

That is, a sample size of 35 or larger will suffice. ■

7.3.1 Confidence interval for a normal mean when the variance is unknown

Suppose now that $X_1, \ldots, X_n$ is a sample from a normal distribution with unknown mean μ and unknown variance σ^2, and that we wish to construct a

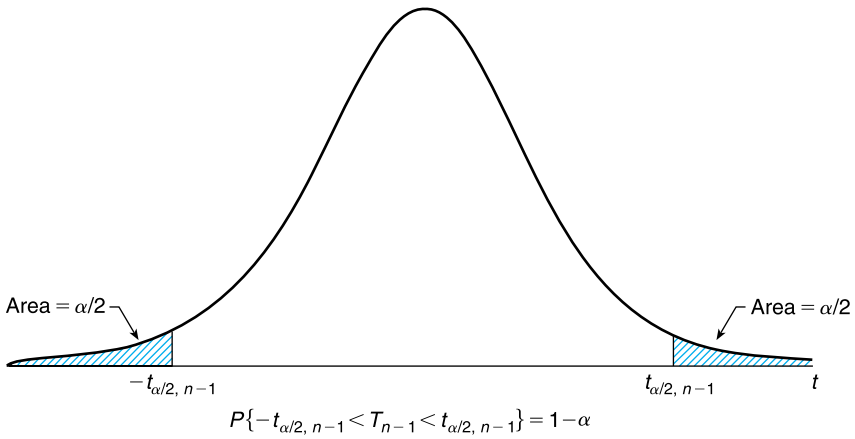


FIGURE 7.2
t-density function.

$100(1-\alpha)$ percent confidence interval for μ. Since σ is unknown, we can no longer base our interval on the fact that $\sqrt{n}(\overline{X}-\mu)/\sigma$ is a standard normal random variable. However, by letting $S^2 = \sum_{i=1}^{n}(X_i - \overline{X})^2/(n-1)$ denote the sample variance, then from Corollary 6.5.2 it follows that

$$\sqrt{n}\,\frac{(\overline{X}-\mu)}{S}$$

is a t-random variable with $n-1$ degrees of freedom. Hence, from the symmetry of the t-density function (see Figure 7.2), we have that for any $\alpha \in (0, 1/2)$,

$$P\left\{-t_{\alpha/2,n-1} < \sqrt{n}\,\frac{(\overline{X}-\mu)}{S} < t_{\alpha/2,n-1}\right\} = 1-\alpha$$

or, equivalently,

$$P\{-t_{\alpha/2,n-1}\,\frac{S}{\sqrt{n}} < \bar{X}-\mu < t_{\alpha/2,n-1}\,\frac{S}{\sqrt{n}}\} = 1-\alpha$$

Multiplying all sides of the preceding by -1 and then adding $\bar{X}$ yields that

$$P\left\{\overline{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}} < \mu < \overline{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right\} = 1-\alpha$$

Thus, if it is observed that $\overline{X} = \overline{x}$ and $S = s$, then we can say that "with $100(1-\alpha)$ percent confidence"

$$\mu \in \left(\overline{x} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}},\ \overline{x} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}\right)$$

Example 7.3.e. Let us again consider Example 7.3.a but let us now suppose that when the value μ is transmitted at location A then the value received at location B is normal with mean μ and variance σ^2 but with σ^2 being unknown. If 9 successive values are, as in Example 7.3.a, 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, compute a 95 percent confidence interval for μ.

Solution. A simple calculation yields that

$$\overline{x} = 9$$

and

$$s^2 = \frac{\sum x_i^2 - 9(\overline{x})^2}{8} = 9.5$$

or

$$s = 3.082$$

Hence, as $t_{.025,8} = 2.306$, a 95 percent confidence interval for μ is

$$\left[9 - 2.306\frac{(3.082)}{3},\ 9 + 2.306\frac{(3.082)}{3}\right] = (6.63, 11.37)$$

a larger interval than obtained in Example 7.3.a. The reason why the interval just obtained is larger than the one in Example 7.3.a is twofold. The primary reason is that we have a larger estimated variance than in Example 7.3.a. That is, in Example 7.3.a we assumed that σ^2 was known to equal 4, whereas in this example we assumed it to be unknown and our estimate of it turned out to be 9.5, which resulted in a larger confidence interval. In fact, the confidence interval would have been larger than in Example 7.3.a even if our estimate of σ^2 was again 4 because by having to estimate the variance we need to utilize the t-distribution, which has a greater variance and thus a larger spread than the standard normal (which can be used when σ^2 is assumed known). For instance, if it had turned out that $\overline{x} = 9$ and $s^2 = 4$, then our confidence interval would have been

$$(9 - 2.306 \cdot \tfrac{2}{3}, 9 + 2.306 \cdot \tfrac{2}{3}) = (7.46, 10.54)$$

which is larger than that obtained in Example 7.3.a. ■

Remarks

(a) The confidence interval for μ when σ is known is based on the fact that $\sqrt{n}(\overline{X} - \mu)/\sigma$ has a standard normal distribution. When σ is unknown, the foregoing approach is to estimate it by S and then use the fact that $\sqrt{n}(\overline{X} - \mu)/S$ has a t-distribution with $n - 1$ degrees of freedom.

(b) The length of a $100(1-2\alpha)$ percent confidence interval for μ is not always larger when the variance is unknown. For the length of such an interval is $2z_\alpha\sigma/\sqrt{n}$ when σ is known, whereas it is $2t_{\alpha,n-1}S/\sqrt{n}$ when σ is unknown; and it is certainly possible that the sample standard deviation S can turn out to be much smaller than σ. However, it can be shown that the mean length of the interval is longer when σ is unknown. That is, it can be shown that

$$t_{\alpha,n-1}E[S] \geq z_\alpha\sigma$$

Indeed, $E[S]$ is evaluated in Chapter 14 and it is shown, for instance, that

$$E[S] = \begin{cases} .94\,\sigma & \text{when } n = 5 \\ .97\,\sigma & \text{when } n = 9 \end{cases}$$

Since

$$z_{.025} = 1.96, \qquad t_{.025,4} = 2.78, \qquad t_{.025,8} = 2.31$$

the length of a 95 percent confidence interval from a sample of size 5 is $2 \times 1.96\sigma/\sqrt{5} = 1.75\,\sigma$ when σ is known, whereas its expected length is $2 \times 2.78 \times .94\sigma/\sqrt{5} = 2.34\,\sigma$ when σ is unknown — an increase of 33.7 percent. If the sample is of size 9, then the two values to compare are 1.31 σ and 1.49 σ — a gain of 13.7 percent. ■

A one-sided upper confidence interval can be obtained by noting that

$$P\left\{\sqrt{n}\frac{(\overline{X}-\mu)}{S} < t_{\alpha,n-1}\right\} = 1-\alpha$$

or

$$P\left\{\overline{X}-\mu < \frac{S}{\sqrt{n}}t_{\alpha,\,n-1}\right\} = 1-\alpha$$

or

$$P\left\{\mu > \overline{X} - \frac{S}{\sqrt{n}}t_{\alpha,\,n-1}\right\} = 1-\alpha$$

Hence, if it is observed that $\overline{X} = \overline{x}$, $S = s$, then we can assert "with $100(1-\alpha)$ percent confidence" that

$$\mu \in \left(\overline{x} - \frac{s}{\sqrt{n}}t_{\alpha,n-1},\ \infty\right)$$

Similarly, a $100(1-\alpha)$ lower confidence interval would be

$$\mu \in \left(-\infty,\ \overline{x} + \frac{s}{\sqrt{n}} t_{\alpha,\, n-1}\right)$$

We can use R to obtain confidence intervals. To obtain a $100(1-\alpha)$ percent confidence interval for the data values $x_1, ..., x_n$, input

$$> d = c(x_1, \ldots, x_n)$$
$$> l = \text{ mean}(d) - qt(1-\alpha/2, n-1) * \text{sqrt}(\text{var}(d)/n)$$
$$> u = \text{ mean}(d) + qt(1-\alpha/2, n-1) * \text{ sqrt}(\text{var}(d)/n)$$
$$> c(l, u)$$

and press return. For instance, suppose we want to use R to compute a 99 percent confidence interval for the mean of a normal data set when the variance is unknown. Let the data be 5, 9, 12, 8.6, 7.2, 13, 9.5. Then, write the following R commands:

$$> d = c(5, 9, 12, 8.6, 7.2, 13, 9.5)$$
$$> l = \text{ mean}(d) - qt(.995, 6) * \text{sqrt}(\text{var}(d)/7)$$
$$> u = \text{ mean}(d) + qt(.995, 6) * \text{sqrt}(\text{var}(d)/7)$$
$$> c(l, u)$$

Press return and R yields

[1] 5.37346331037313 12.9979652610554

We obtain upper and lower confidence intervals similarly.

Example 7.3.f. Determine a 95 percent confidence interval for the average resting pulse of the members of a health club if a random selection of 15 members of the club yielded the data 54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77. Also determine a 95 percent lower confidence interval for this mean.

Solution. We use R to obtain the two-sided 95 percent confidence interval, and the 95 percent lower confidence interval as follows:

$$>d = c(54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77)$$
$$>l = \text{ mean}(d) - qt(.975, 14) * \text{sqrt}(\text{var}(d)/15)$$
$$>u = \text{ mean}(d) + qt(.975, 14) * \text{sqrt}(\text{var}(d)/15)$$
$$>c(l, u)$$

[1] 60.8669366732691, 77.6663966600643

$$>ub = \text{ mean}(d) + qt(.95, 14) * \text{sqrt}(\text{var}(d)/15)$$

[1] 76.16457

Thus, the 95 percent two-sided confidence interval for μ is (60.8669366732691, 77.6663966600643), while the 95 percent lower confidence interval is $(-\infty, 76.16457)$. ■

Our derivations of the $100(1-\alpha)$ percent confidence intervals for the population mean μ have assumed that the population distribution is normal. However, even when this is not the case, if the sample size is reasonably large then the intervals obtained will still be approximate $100(1-\alpha)$ percent confidence intervals for μ. This is true because, by the central limit theorem, $\sqrt{n}(\overline{X}-\mu)/\sigma$ will have approximately a normal distribution, and $\sqrt{n}(\overline{X}-\mu)/S$ will have approximately a t-distribution.

Example 7.3.g. Simulation provides a powerful method for evaluating single and multidimensional integrals. For instance, let f be a function of an r-valued vector $(y_1, \ldots, y_r)$, and suppose that we want to estimate the quantity θ, defined by

$$\theta = \int_0^1 \int_0^1 \cdots \int_0^1 f(y_1, y_2, \ldots, y_r)\, dy_1 dy_2, \ldots, dy_r$$

To accomplish this, note that if $U_1, U_2, \ldots, U_r$ are independent uniform random variables on (0, 1), then

$$\theta = E[f(U_1, U_2, \ldots, U_r)]$$

Now, the values of independent uniform (0, 1) random variables can be approximated on a computer (by so-called *pseudo random numbers*); if we generate a vector of r of them, and evaluate f at this vector, then the value obtained, call it X_1, will be a random variable with mean θ. If we now repeat this process, then we obtain another value, call it X_2, which will have the same distribution as X_1. Continuing on, we can generate a sequence $X_1, X_2, \ldots, X_n$ of independent and identically distributed random variables with mean θ; we then use their observed values to estimate θ. This method of approximating integrals is called *Monte Carlo simulation*.

For instance, suppose we wanted to estimate the one-dimensional integral

$$\theta = \int_0^1 \sqrt{1-y^2}\, dy = E[\sqrt{1-U^2}]$$

where U is a uniform (0, 1) random variable. To do so, let $U_1, \ldots, U_{100}$ be independent uniform (0, 1) random variables, and set

$$X_i = \sqrt{1-U_i^2}, \qquad i = 1, \ldots, 100$$

In this way, we have generated a sample of 100 random variables having mean θ. Suppose that the computer-generated values of $U_1, \ldots, U_{100}$, resulting in

$X_1, \ldots, X_{100}$ having sample mean .786 and sample standard deviation .03. Consequently, since $t_{.025,99} = 1.985$, it follows that a 95 percent confidence interval for θ would be given by

$$.786 \pm 1.985(.003)$$

As a result, we could assert, with 95 percent confidence, that θ (which can be shown to equal $\pi/4$) is between .780 and .792. ■

7.3.2 Prediction intervals

Suppose that $X_1, \ldots, X_n, X_{n+1}$ is a sample from a normal distribution with unknown mean μ and unknown variance σ^2. Suppose further that the values of $X_1, \ldots, X_n$ are to be observed and that we want to use them to predict the value of X_{n+1}. To begin, note that if the mean μ were known, then it would be the natural predictor for X_{n+1}. As it is not known, it seems natural to use its current estimator after observing $X_1, \ldots, X_n$, namely the average of these observed values, as the predicted value of X_{n+1}. That is, we should use the observed value of $\bar{X}_n = \sum_{i=1}^n X_i/n$, as the predicted value of X_{n+1}.

Suppose now that we want to determine an interval in which we predict, with a certain degree of confidence, that X_{n+1} will lie. To obtain such a prediction interval, note that as $\bar{X}_n$ is normal with mean μ and variance σ^2/n, and is independent of X_{n+1} which is normal with mean μ and variance σ^2, it follows that $X_{n+1} - \bar{X}_n$ is normal with mean 0 and variance $sigma^2/n + \sigma^2$. Consequently,

$$\frac{X_{n+1} - \bar{X}_n}{\sigma\sqrt{1 + 1/n}} \qquad \text{is a standard normal random variable.}$$

Because this is independent of $S_n^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2/(n-1)$, it follows from the same argument used to establish Corollary 6.5.2, that replacing σ by its estimator S_n in the preceding expression will yield a t-random variable with $n-1$ degrees of freedom. That is,

$$\frac{X_{n+1} - \bar{X}_n}{S_n\sqrt{1 + 1/n}}$$

is a t-random variable with $n-1$ degrees of freedom. Hence, for any $\alpha \in (0, 1/2)$,

$$P\{-t_{\alpha/2,n-1} < \frac{X_{n+1} - \bar{X}_n}{S_n\sqrt{1 + 1/n}} < t_{\alpha/2,n-1}\} = 1 - \alpha$$

which is equivalent to

$$P\{\bar{X}_n - t_{\alpha/2,n-1}\, S_n\sqrt{1 + 1/n} < X_{n+1} < \bar{X}_n + t_{\alpha/2,n-1}\, S_n\sqrt{1 + 1/n}\}$$

Hence, if the observed values of $\bar{X}_n$ and S_n are, respectively, $\bar{x}_n$ and s_n, then we can predict, with $100(1-\alpha)$ percent confidence, that X_{n+1} will lie between $\bar{x}_n - t_{\alpha/2,n-1}s_n\sqrt{1+1/n}$ and $\bar{x}_n + t_{\alpha/2,n-1}s_n\sqrt{1+1/n}$. That is, with $100(1-\alpha)$ percent confidence, we can predict that

$$X_{n+1} \in \left(\bar{x}_n - t_{\alpha/2,n-1}s_n\sqrt{1+1/n}, \bar{x}_n + t_{\alpha/2,n-1}s_n\sqrt{1+1/n}\right)$$

Example 7.3.h. The following are the number of steps walked in each of the last 7 days

6822 5333 7420 7432 6252 7005 6752

Assuming that the daily number of steps can be thought of as being independent realizations from a normal distribution, give a prediction interval that, with 95 percent confidence, will contain the number of steps that will be walked tomorrow.

Solution. A simple calculation gives that the sample mean and sample variance of the 7 data values are

$$\bar{X}_7 = 6716.57 \qquad S_7 = 733.97$$

Because $t_{.025,6} = 2.447$, and $2.4447 \cdot 733.97\sqrt{1+1/7} = 1920.03$, we can predict, with 95 percent confidence, that tomorrow's number of steps will be between $6716.57 - 1920.03$ and $6716.57 + 1920.03$. That is, with 95 percent confidence, X_8 will lie in the interval (4796.54, 8636.60).

7.3.3 Confidence intervals for the variance of a normal distribution

If $X_1, \ldots, X_n$ is a sample from a normal distribution having unknown parameters μ and σ^2, then we can construct a confidence interval for σ^2 by using the fact that

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$

Hence,

$$P\left\{\chi^2_{1-\alpha/2,n-1} \le (n-1)\frac{S^2}{\sigma^2} \le \chi^2_{\alpha/2,n-1}\right\} = 1-\alpha$$

or, equivalently,

$$P\left\{\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right\} = 1-\alpha$$

or

$$P\{\mu - 1.645\sigma < \theta < \mu + 1.645\sigma\} = .90$$

it seems reasonable to take

$$1.645\sigma = a \quad \text{or} \quad \sigma = \frac{a}{1.645}$$

Thus, if your prior feelings can indeed be reasonably described by a normal distribution, then that distribution would have mean μ and standard deviation $\sigma = a/1.645$. As a test of whether this distribution indeed fits your prior feelings you might ask yourself such questions as whether you are 95 percent certain that θ will fall between $\mu - 1.96\sigma$ and $\mu + 1.96\sigma$, or whether you are 99 percent certain that θ will fall between $\mu - 2.58\sigma$ and $\mu + 2.58\sigma$, where these intervals are determined by the equalities

$$P\left\{-1.96 < \frac{\theta - \mu}{\sigma} < 1.96\right\} = .95$$
$$P\left\{-2.58 < \frac{\theta - \mu}{\sigma} < 2.58\right\} = .99$$

which hold when θ is normal with mean μ and variance σ^2.

Example 7.8.c. Consider the likelihood function $f(x_1, \ldots, x_n|\theta)$ and suppose that θ is uniformly distributed over some interval (a, b). The posterior density of θ given $X_1, \ldots, X_n$ equals

$$\begin{aligned} f(\theta|x_1, \ldots, x_n) &= \frac{f(x_1, \ldots, x_n|\theta)\, p(\theta)}{\int_a^b f(x_1, \ldots, x_n|\theta)\, p(\theta)\, d\theta} \\ &= \frac{f(x_1, \ldots, x_n|\theta)}{\int_a^b f(x_1, \ldots, x_n|\theta)\, d\theta} \qquad a < \theta < b \end{aligned}$$

Now the *mode* of a density $f(\theta)$ was defined to be that value of θ that maximizes $f(\theta)$. By the foregoing, it follows that the mode of the density $f(\theta|x_1, \ldots, x_n)$ is that value of θ maximizing $f(x_1, \ldots, x_n|\theta)$; that is, it is just the maximum likelihood estimate of θ [when it is constrained to be in (a, b)]. In other words, the maximum likelihood estimate equals the mode of the posterior distribution when a uniform prior distribution is assumed. ■

If, rather than a point estimate, we desire an interval in which θ lies with a specified probability — say $1 - \alpha$ — we can accomplish this by choosing values a and b such that

$$\int_a^b f(\theta|x_1, \ldots, x_n)\, d\theta = 1 - \alpha$$

Example 7.8.d. Suppose that if a signal of value s is sent from location A, then the signal value received at location B is normally distributed with mean s and variance 60. Suppose also that the value of a signal sent at location A is, *a priori*, known to be normally distributed with mean 50 and variance 100. If the value received at location B is equal to 40, determine an interval that will contain the actual value sent with probability .90.

Solution. It follows from Example 7.8.b that the conditional distribution of S, the signal value sent, given that 40 is the value received, is normal with mean and variance given by

$$E[S|\text{data}] = \frac{1/60}{1/60+1/100}40 + \frac{1/100}{1/60+1/100}50 = 43.75$$

$$\text{Var}(S|\text{data}) = \frac{1}{1/60+1/100} = 37.5$$

Hence, given that the value received is 40, $(S-43.75)/\sqrt{37.5}$ has a standard normal distribution and so

$$P\left\{-1.645 < \frac{S-43.75}{\sqrt{37.5}} < 1.645|\text{data}\right\} = .90$$

or

$$P\{43.75 - 1.645\sqrt{37.5} < S < 43.75 + 1.645\sqrt{37.5}|\text{data}\} = .95$$

That is, with *probability* .90, the true signal sent lies within the interval (33.68, 53.82). ■

Problems

1. Let $X_1, \ldots, X_n$ be a sample from the distribution whose density function is

$$f(x) = \begin{cases} e^{-(x-\theta)} & x \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

   Determine the maximum likelihood estimator of θ.
2. Determine the maximum likelihood estimator of θ when $X_1, \ldots, X_n$ is a sample with density function

$$f(x) = \tfrac{1}{2}e^{-|x-\theta|}, \qquad -\infty < x < \infty$$

3. Let $X_1, \ldots, X_n$ be a sample from a normal μ, σ^2 population. Determine the maximum likelihood estimator of σ^2 when μ is known. What is the expected value of this estimator?

4. Determine the maximum likelihood estimates of a and λ when $X_1, \ldots, X_n$ is a sample from the Pareto density function

$$f(x) = \begin{cases} \lambda a^{\lambda} x^{-(\lambda+1)}, & \text{if } x \geq a \\ 0, & \text{if } x < a \end{cases}$$

5. Suppose that $X_1, \ldots, X_n$ are normal with mean μ_1; $Y_1, \ldots, Y_n$ are normal with mean μ_2; and $W_1, \ldots, W_n$ are normal with mean $\mu_1 + \mu_2$. Assuming that all $3n$ random variables are independent with a common variance, find the maximum likelihood estimators of μ_1 and μ_2.
6. River floods are often measured by their discharges (in units of feet cubed per second). The value v is said to be the value of a 100-year flood if

$$P\{D \geq v\} = .01$$

where D is the discharge of the largest flood in a randomly chosen year. The following table gives the flood discharges of the largest floods of the Blackstone River in Woonsocket, Rhode Island, in each of the years from 1929 to 1965. Assuming that these discharges follow a lognormal distribution, estimate the value of a 100-year flood.

Table 7.4 Annual Floods of the Blackstone River (1929–1965).

Year	Flood Discharge (ft^3/s)
1929	4,570
1930	1,970
1931	8,220
1932	4,530
1933	5,780
1934	6,560
1935	7,500
1936	15,000
1937	6,340
1938	15,100
1939	3,840
1940	5,860
1941	4,480
1942	5,330
1943	5,310
1944	3,830
1945	3,410

continued on next page

Table 7.4 (*continued*)

Year	Flood Discharge (ft^3/s)
1946	3830
1947	3150
1948	5810
1949	2030
1950	3620
1951	4920
1952	4090
1953	5570
1954	9400
1955	32,900
1956	8710
1957	3850
1958	4970
1959	5398
1960	4780
1961	4020
1962	5790
1963	4510
1964	5520
1965	5300

7. Recall that X is said to have a lognormal distribution with parameters μ and σ^2 if $\log(X)$ is normal with mean μ and variance σ^2. Suppose X is such a lognormal random variable.
   a. Find $E[X]$.
   b. Find $\text{Var}(X)$.
   Hint: Make use of the formula for the moment generating function of a normal random variable.
   c. The following are, in minutes, travel times to work over a sequence of 10 days.

   42, 28, 53, 57, 67, 39, 35, 50, 44, 39

   Assuming an underlying lognormal distribution, use the data to estimate the mean travel time.
8. An electric scale gives a reading equal to the true weight plus a random error that is normally distributed with mean 0 and standard deviation $\sigma = .1$ mg. Suppose that the results of five successive weighings of the same object are as follows: 3.142, 3.163, 3.155, 3.150, 3.141.

a. Determine a 95 percent confidence interval estimate of the true weight.
b. Determine a 99 percent confidence interval estimate of the true weight.

9. The PCB concentration of a fish caught in Lake Michigan was measured by a technique that is known to result in an error of measurement that is normally distributed with a standard deviation of .08 ppm (parts per million). Suppose the results of 10 independent measurements of this fish are

11.2, 12.4, 10.8, 11.6, 12.5, 10.1, 11.0, 12.2, 12.4, 10.6

a. Give a 95 percent confidence interval for the PCB level of this fish.
b. Give a 95 percent lower confidence interval.
c. Give a 95 percent upper confidence interval.

10. The standard deviation of test scores on a certain achievement test is 11.3. If a random sample of 81 students had a sample mean score of 74.6, find a 90 percent confidence interval estimate for the average score of all students.

11. Let $X_1, \ldots, X_n, X_{n+1}$ be a sample from a normal population having an unknown mean μ and variance 1. Let $\bar{X}_n = \sum_{i=1}^{n} X_i/n$ be the average of the first n of them.
a. What is the distribution of $X_{n+1} - \bar{X}_n$?
b. If $\bar{X}_n = 4$, give an interval that, with 90 percent confidence, will contain the value of X_{n+1}.

12. If $X_1, \ldots, X_n$ is a sample from a normal population whose mean μ is unknown but whose variance σ^2 is known, show that $(-\infty, \overline{X} + z_\alpha \sigma/\sqrt{n})$ is a $100(1-\alpha)$ percent lower confidence interval for μ.

13. A sample of 20 cigarettes is tested to determine nicotine content and the average value observed was 1.2 mg. Compute a 99 percent two-sided confidence interval for the mean nicotine content of a cigarette if it is known that the standard deviation of a cigarette's nicotine content is $\sigma =$ .2 mg.

14. In Problem 13, suppose that the population variance is not known in advance of the experiment. If the sample variance is .04, compute a 99 percent two-sided confidence interval for the mean nicotine content.

15. In Problem 14, compute a value c for which we can assert "with 99 percent confidence" that c is larger than the mean nicotine content of a cigarette.

16. Suppose that when sampling from a normal population having an unknown mean μ and unknown variance σ^2, we wish to determine a sample size n so as to guarantee that the resulting $100(1-\alpha)$ percent confidence interval for μ will be of size no greater than A, for given values α and A. Explain how we can approximately do this by a double sampling scheme that first takes a subsample of size 30 and then chooses the total sample size by using the results of the first subsample.

17. The following data resulted from 24 independent measurements of the melting point of lead.

330°C	322°C	345°C
328.6°C	331°C	342°C
342.4°C	340.4°C	329.7°C
334°C	326.5°C	325.8°C
337.5°C	327.3°C	322.6°C
341°C	340°C	333°C
343.3°C	331°C	341°C
329.5°C	332.3°C	340°C

Assuming that the measurements can be regarded as constituting a normal sample whose mean is the true melting point of lead, determine a 95 percent two-sided confidence interval for this value. Also determine a 99 percent two-sided confidence interval.

18. The following are scores on IQ tests of a random sample of 18 students at a large eastern university.

130, 122, 119, 142, 136, 127, 120, 152, 141,
132, 127, 118, 150, 141, 133, 137, 129, 142

a. Construct a 95 percent confidence interval estimate of the average IQ score of all students at the university.
b. Construct a 95 percent lower confidence interval estimate.
c. Construct a 95 percent upper confidence interval estimate.

19. Suppose that a random sample of nine recently sold houses in a certain city resulted in a sample mean price of $222,000, with a sample standard deviation of $22,000. Give a 95 percent upper confidence interval for the mean price of all recently sold houses in this city.

20. A company self-insures its large fleet of cars against collisions. To determine its mean repair cost per collision, it has randomly chosen a sample of 16 accidents. If the average repair cost in these accidents is $2200 with a sample standard deviation of $800, find a 90 percent confidence interval estimate of the mean cost per collision.

21. A standardized test is given annually to all sixth-grade students in the state of Washington. To determine the average score of students in her district, a school supervisor selects a random sample of 100 students. If the sample mean of these students' scores is 320 and the sample standard deviation is 16, give a 95 percent confidence interval estimate of the average score of students in that supervisor's district.

22. Each of 20 science students independently measured the melting point of lead. The sample mean and sample standard deviation of these measurements were (in degrees centigrade) 330.2 and 15.4, respectively. Construct **(a)** a 95 percent and **(b)** a 99 percent confidence interval estimate of the true melting point of lead.

23. A random sample of 300 CitiBank VISA cardholder accounts indicated a sample mean debt of \$1220 with a sample standard deviation of \$840. Construct a 95 percent confidence interval estimate of the average debt of all cardholders.
24. In Problem 23, find the smallest value v that "with 90 percent confidence," exceeds the average debt per cardholder.
25. Verify the formula given in Table 7.1 for the $100(1-\alpha)$ percent lower confidence interval for μ when σ is unknown.
26. The following are the daily number of steps taken by a certain individual in 20 weekdays.

2,100	1,984	2,072	1,898
1,950	1,992	2,096	2,103
2,043	2,218	2,244	2,206
2,210	2,152	1,962	2,007
2,018	2,106	1,938	1,956

Assuming that the daily number of steps is normally distributed, construct **(a)** a 95 percent and **(b)** a 99 percent two-sided confidence interval for the mean number of steps. **(c)** Determine the largest value v that, "with 95 percent confidence," will be less than the mean range.
27. Studies were conducted in Los Angeles to determine the carbon monoxide concentration near freeways. The basic technique used was to capture air samples in special bags and to then determine the carbon monoxide concentration by using a spectrophotometer. The measurements in ppm (parts per million) over a sampled period during the year were 102.2, 98.4, 104.1, 101, 102.2, 100.4, 98.6, 88.2, 78.8, 83, 84.7, 94.8, 105.1, 106.2, 111.2, 108.3, 105.2, 103.2, 99, 98.8. Compute a 95 percent two-sided confidence interval for the mean carbon monoxide concentration.
28. A set of 10 determinations, by a method devised by the chemist Karl Fischer, of the percentage of water in a methanol solution yielded the following data.

.50, .55, .53, .56, .54,
.57, .52, .60, .55, .58

Assuming normality, use these data to give a 95 percent confidence interval for the actual percentage.
29. Suppose that $U_1, U_2, \ldots$ is a sequence of independent uniform (0,1) random variables, and define N by

$$N = \min\{n : U_1 + \cdots + U_n > 1\}$$

That is, N is the number of uniform (0, 1) random variables that need to be summed to exceed 1. Use random numbers to determine the value of 36 random variables having the same distribution as N, and then use

these data to obtain a 95 percent confidence interval estimate of $E[N]$. Based on this interval, guess the exact value of $E[N]$.

30. An important issue for a retailer is to decide when to reorder stock from a supplier. A common policy used to make the decision is of a type called s, S: The retailer orders at the end of a period if the on-hand stock is less than s, and orders enough to bring the stock up to S. The appropriate values of s and S depend on different cost parameters, such as inventory holding costs and the profit per item sold, as well as the distribution of the demand during a period. Consequently, it is important for the retailer to collect data relating to the parameters of the demand distribution. Suppose that the following data give the numbers of a certain type of item sold in each of 30 weeks.

    14, 8, 12, 9, 5, 22, 15, 12, 16, 7, 10, 9, 15, 15, 12,
    9, 11, 16, 8, 7, 15, 13, 9, 5, 18, 14, 10, 13, 7, 11

    Assuming that the numbers sold each week are independent random variables from a common distribution, use the data to obtain a 95 percent confidence interval for the mean number sold in a week.

31. A random sample of 16 professors at a large private university yielded a sample mean annual salary of \$90,450 with a sample standard deviation of \$9400. Determine a 95 percent confidence interval of the average salary of all professors at that university.

32. Let $X_1, \ldots, X_{n+1}$ be a sample from a population with mean μ and variance σ^2. As noted in the text, the natural predictor of X_{n+1} based on the data values $X_1, \ldots, X_n$ is $\bar{X}_n = \sum_{i=1}^{n} X_i/n$. Determine the mean square error of this predictor. That is, find $E[(X_{n+1} - \bar{X}_n)^2]$.

33. National Safety Council data show that the number of accidental deaths due to drowning in the United States in the years from 1990 to 1993 were (in units of one thousand) 5.2, 4.6, 4.3, 4.8. Use these data to give an interval that will, with 95 percent confidence, contain the number of such deaths in 1994.

34. The daily dissolved oxygen concentration for a water stream has been recorded over 30 days. If the sample average of the 30 values is 2.5 mg/liter and the sample standard deviation is 2.12 mg/liter, determine a value which, with 90 percent confidence, exceeds the mean daily concentration.

35. Verify the formulas given in Table 7.1 for the $100(1 - \alpha)$ percent lower and upper confidence intervals for σ^2.

36. The capacities (in ampere-hours) of 10 batteries were recorded as follows:

    140, 136, 150, 144, 148, 152, 138, 141, 143, 151

    a. Estimate the population variance σ^2.
    b. Compute a 99 percent two-sided confidence interval for σ^2.

CHAPTER 8

Hypothesis testing

8.1 Introduction

As in the previous chapter, let us suppose that a random sample from a population distribution, specified except for a vector of unknown parameters, is to be observed. However, rather than wishing to explicitly estimate the unknown parameters, let us now suppose that we are primarily concerned with using the resulting sample to test some particular hypothesis concerning them. As an illustration, suppose that a construction firm has just purchased a large supply of cables that have been guaranteed to have an average breaking strength of at least 7000 pounds per square inch (PSI). To verify this claim, the firm has decided to take a random sample of 10 of these cables to determine their breaking strengths. They will then use the result of this experiment to ascertain whether or not they accept the cable manufacturer's hypothesis that the population mean is at least 7000 PSI.

A statistical hypothesis is usually a statement about a set of parameters of a population distribution. It is called a hypothesis because it is not known whether or not it is true. A primary problem is to develop a procedure for determining whether or not the values of a random sample from this population are consistent with the hypothesis. For instance, consider a particular normally distributed population having an unknown mean value θ and known variance 1. The statement "θ is less than 1" is a statistical hypothesis that we could try to test by observing a random sample from this population. If the random sample is deemed to be consistent with the hypothesis under consideration, we say that the hypothesis has been "accepted"; otherwise we say that it has been "rejected."

Note that in accepting a given hypothesis we are not actually claiming that it is true but rather we are saying that the resulting data appear to be consistent with it. For instance, in the case of a normal $(\theta, 1)$ population, if a resulting sample of size 10 has an average value of 1.25, then although such a result cannot be regarded as being evidence in favor of the hypothesis "$\theta < 1$," it is not inconsistent with this hypothesis, which would thus be accepted. On the other hand, if the sample of size 10 has an average value of 3, then even though a sample value that large is possible when $\theta < 1$, it is so unlikely that it seems inconsistent with this hypothesis, which would thus be rejected.

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8.2 Significance levels

Consider a population having distribution F_θ, where θ is unknown, and suppose we want to test a specific hypothesis about θ. We shall denote this hypothesis by H_0 and call it the *null hypothesis*. For example, if F_θ is a normal distribution function with mean θ and variance equal to 1, then two possible null hypotheses about θ are

(a) $H_0 : \theta = 1$
(b) $H_0 : \theta \leq 1$

Thus the first of these hypotheses states that the population is normal with mean 1 and variance 1, whereas the second states that it is normal with variance 1 and a mean less than or equal to 1. Note that the null hypothesis in (a), when true, completely specifies the population distribution, whereas the null hypothesis in (b) does not. A hypothesis that, when true, completely specifies the population distribution is called a *simple* hypothesis; one that does not is called a *composite* hypothesis.

Suppose now that in order to test a specific null hypothesis H_0, a population sample of size n — say $X_1, \ldots, X_n$ — is to be observed. Based on these n values, we must decide whether or not to accept H_0. A test for H_0 can be specified by defining a region C in n-dimensional space with the proviso that the hypothesis is to be rejected if the random sample $X_1, \ldots, X_n$ turns out to lie in C and accepted otherwise. The region C is called the *critical region*. In other words, the statistical test determined by the critical region C is the one that

accepts H_0 if $(X_1, X_2, \ldots, X_n) \notin C$

and

rejects H_0 if $(X_1, \ldots, X_n) \in C$

For instance, a common test of the hypothesis that θ, the mean of a normal population with variance 1, is equal to 1 has a critical region given by

$$C = \{(X_1, \ldots, X_n) : |\bar{X} - 1| > 1.96/\sqrt{n}\} \tag{8.2.1}$$

Thus, this test calls for rejection of the null hypothesis that $\theta = 1$ when the sample average differs from 1 by more than 1.96 divided by the square root of the sample size.

It is important to note when developing a procedure for testing a given null hypothesis H_0 that, in any test, two different types of errors can result. The first of these, called a *type I error*, is said to result if the test incorrectly calls for rejecting H_0 when it is indeed correct. The second, called a *type II error*, results if the test calls for accepting H_0 when it is false.

Now, as was previously mentioned, the objective of a statistical test of H_0 is not to explicitly determine whether or not H_0 is true but rather to determine if its validity is consistent with the resultant data. Hence, with this objective it seems reasonable that H_0 should only be rejected if the resultant data are very unlikely when H_0 is true. The classical way of accomplishing this is to specify a value α and then require the test to have the property that whenever H_0 is true its probability of being rejected is never greater than α. The value α, called the *level of significance of the test*, is usually set in advance, with commonly chosen values being $\alpha = .1, .05, .005$. In other words, the classical approach to testing H_0 is to fix a significance level α and then require that the test have the property that the probability of a type I error occurring can never be greater than α.

Suppose now that we are interested in testing a certain hypothesis concerning θ, an unknown parameter of the population. Specifically, for a given set of parameter values w, suppose we are interested in testing

$$H_0 : \theta \in w$$

A common approach to developing a test of H_0, say at level of significance α, is to start by determining a point estimator of θ — say $d(\mathbf{X})$. The hypothesis is then rejected if $d(\mathbf{X})$ is "far away" from the region w. However, to determine how "far away" it need be to justify rejection of H_0, we need to determine the probability distribution of $d(\mathbf{X})$ when H_0 is true since this will usually enable us to determine the appropriate critical region so as to make the test have the required significance level α. For example, the test of the hypothesis that the mean of a normal $(\theta, 1)$ population is equal to 1, given by Equation (8.2.1), calls for rejection when the point estimate of θ — that is, the sample average — is farther than $1.96/\sqrt{n}$ away from 1. As we will see in the next section, the value $1.96/\sqrt{n}$ was chosen to meet a level of significance of $\alpha = .05$.

8.3 Tests concerning the mean of a normal population

8.3.1 Case of known variance

Suppose that $X_1, \ldots, X_n$ is a sample of size n from a normal distribution having an unknown mean μ and a known variance σ^2 and suppose we are interested in testing the null hypothesis

$$H_0 : \mu = \mu_0$$

against the alternative hypothesis

$$H_1 : \mu \neq \mu_0$$

where μ_0 is some specified constant.

Since $\overline{X} = \sum_{i=1}^{n} X_i/n$ is a natural point estimator of μ, it seems reasonable to accept H_0 if $\overline{X}$ is not too far from μ_0. That is, the critical region of the test would be of the form

$$C = \{X_1, \ldots, X_n : |\overline{X} - \mu_0| > c\} \tag{8.3.1}$$

for some suitably chosen value c.

If we desire that the test has significance level α, then we must determine the critical value c in Equation (8.3.1) that will make the type I error equal to α. That is, c must be such that

$$P_{\mu_0}\{|\overline{X} - \mu_0| > c\} = \alpha \tag{8.3.2}$$

where we write P_{μ_0} to mean that the preceding probability is to be computed under the assumption that $\mu = \mu_0$. However, when $\mu = \mu_0$, $\overline{X}$ will be normally distributed with mean μ_0 and variance σ^2/n and so Z, defined by

$$Z \equiv \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

will have a standard normal distribution. Now Equation (8.3.2) is equivalent to

$$P\left\{|Z| > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

or, equivalently,

$$2P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

where Z is a standard normal random variable. However, we know that

$$P\{Z > z_{\alpha/2}\} = \alpha/2$$

and so

$$\frac{c\sqrt{n}}{\sigma} = z_{\alpha/2}$$

or

$$c = \frac{z_{\alpha/2}\sigma}{\sqrt{n}}$$

Thus, the significance level α test is to reject H_0 if $|\overline{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$ and accept otherwise; or, equivalently, to

$$\begin{aligned} &\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| > z_{\alpha/2} \\ &\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| \leq z_{\alpha/2} \end{aligned} \tag{8.3.3}$$

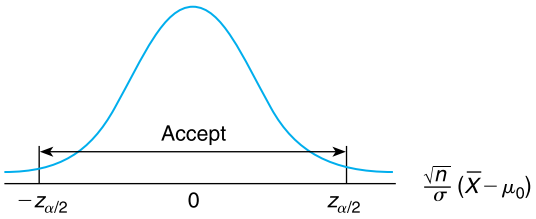


FIGURE 8.1

This can be pictorially represented as shown in Figure 8.1, where we have superimposed the standard normal density function [which is the density of the test statistic $\sqrt{n}(\overline{X} - \mu_0)/\sigma$ when H_0 is true].

Example 8.3.a. It is known that if a signal of value μ is sent from location A, then the value received at location B is normally distributed with mean μ and standard deviation 2. That is, the random noise added to the signal is an $N(0, 4)$ random variable. There is reason for the people at location B to suspect that the signal value $\mu = 8$ will be sent today. Test this hypothesis if the same signal value is independently sent five times and the average value received at location B is $\overline{X} = 9.5$.

Solution. Suppose we are testing at the 5 percent level of significance. To begin, we compute the test statistic

$$\frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| = \frac{\sqrt{5}}{2}(1.5) = 1.68$$

Since this value is less than $z_{.025} = 1.96$, the hypothesis is accepted. In other words, the data are not inconsistent with the null hypothesis in the sense that a sample average as far from the value 8 as observed would be expected, when the true mean is 8, over 5 percent of the time. Note, however, that if a less stringent significance level were chosen — say $\alpha = .1$ — then the null hypothesis would have been rejected. This follows since $z_{.05} = 1.645$, which is less than 1.68. Hence, if we would have chosen a test that had a 10 percent chance of rejecting H_0 when H_0 was true, then the null hypothesis would have been rejected.

The "correct" level of significance to use in a given situation depends on the individual circumstances involved in that situation. For instance, if rejecting a null hypothesis H_0 would result in large costs that would thus be lost if H_0 were indeed true, then we might elect to be quite conservative and so choose a significance level of .05 or .01. Also, if we initially feel strongly that H_0 was correct, then we would require very stringent data evidence to the contrary for us to reject H_0. (That is, we would set a very low significance level in this situation.) ■

The test given by Equation (8.3.3) can be described as follows: For any observed value of the test statistic $\sqrt{n}|\overline{X} - \mu_0|/\sigma$, call it v, the test calls for rejection of

the null hypothesis if the probability that the test statistic would be as large as v when H_0 is true is less than or equal to the significance level α. From this, it follows that we can determine whether or not to accept the null hypothesis by computing, first, the value of the test statistic and, second, the probability that a standard normal would (in absolute value) exceed that quantity. This probability — called the p-value of the test — gives the critical significance level in the sense that H_0 will be accepted if the significance level α is less than the p-value and rejected if it is greater than or equal.

In practice, the significance level is often not set in advance but rather the data are looked at to determine the resultant p-value. Sometimes, this critical significance level is clearly much larger than any we would want to use, and so the null hypothesis can be readily accepted. At other times the p-value is so small that it is clear that the hypothesis should be rejected.

Example 8.3.b. In Example 8.3.a, suppose that the average of the 5 values received is $\overline{X} = 8.5$. In this case,

$$\frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| = \frac{\sqrt{5}}{4} = .559$$

Since

$$\begin{aligned} P\{|Z| > .559\} &= 2P\{Z > .559\} \\ &= 2 \times .288 = .576 \end{aligned}$$

it follows that the p-value is .576 and thus the null hypothesis H_0 that the signal sent has value 8 would be accepted at any significance level $\alpha < .576$. Since we would clearly never want to test a null hypothesis using a significance level as large as .576, H_0 would be accepted.

On the other hand, if the average of the data values were 11.5, then the p-value of the test that the mean is equal to 8 would be

$$\begin{aligned} P\{|Z| > 1.75\sqrt{5}\} &= P\{|Z| > 3.913\} \\ &\approx .00005 \end{aligned}$$

For such a small p-value, the hypothesis that the value 8 was sent is rejected. ■

We have not yet talked about the probability of a type II error — that is, the probability of accepting the null hypothesis when the true mean μ is unequal to μ_0. This probability will depend on the value of μ, and so let us define $\beta(\mu)$ by

$$\beta(\mu) = P_\mu\{\text{acceptance of } H_0\}$$

$$= P_\mu \left\{ \left| \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \right| \le z_{\alpha/2} \right\}$$
$$= P_\mu \left\{ -z_{\alpha/2} \le \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2} \right\}$$

The function $\beta(\mu)$ is called the *operating characteristic* (or OC) *curve* and represents the probability that H_0 will be accepted when the true mean is μ.

To compute this probability, we use the fact that $\overline{X}$ is normal with mean μ and variance σ^2/n and so

$$Z \equiv \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim (0, 1)$$

Hence,

$$\begin{aligned}
\beta(\mu) &= P_\mu \left\{ -z_{\alpha/2} \le \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2} \right\} \\
&= P_\mu \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \le \frac{\overline{X} - \mu_0 - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\} \\
&= P_\mu \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \le Z - \frac{\mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\} \\
&= P \left\{ \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \le Z \le \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right\} \\
&= \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \right) \qquad (8.3.4)
\end{aligned}$$

where Φ is the standard normal distribution function.

For a fixed significance level α, the OC curve given by Equation (8.3.4) is symmetric about μ_0 and indeed will depend on μ only through $\sqrt{n}\,|\mu - \mu_0|/\sigma$. This curve with the abscissa changed from μ to $d = \sqrt{n}\,|\mu - \mu_0|/\sigma$ is presented in Figure 8.2 when $\alpha = .05$.

Example 8.3.c. For the problem presented in Example 8.3.a, let us determine the probability of accepting the null hypothesis that $\mu = 8$ when the actual value sent is 10. To do so, we compute

$$\frac{\sqrt{n}}{\sigma}(\mu_0 - \mu) = -\frac{\sqrt{5}}{2} \times 2 = -\sqrt{5}$$

As $z_{.025} = 1.96$, the desired probability is, from Equation (8.3.4),

$$\Phi(-\sqrt{5} + 1.96) - \Phi(-\sqrt{5} - 1.96)$$

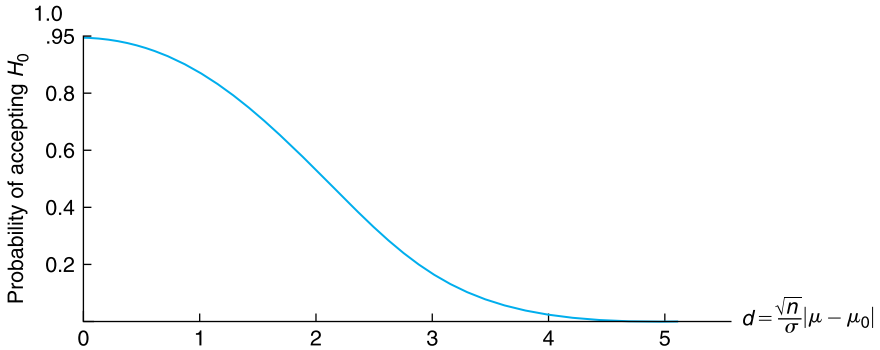


FIGURE 8.2
The OC curve for the two-sided normal test for significance level $\alpha = .05$.

$$= 1 - \Phi(\sqrt{5} - 1.96) - [1 - \Phi(\sqrt{5} + 1.96)]$$
$$= \Phi(4.196) - \Phi(.276)$$
$$= .392 \quad \blacksquare$$

Remark

The function $1 - \beta(\mu)$ is called the *power-function* of the test. Thus, for a given value μ, the power of the test is equal to the probability of rejection when μ is the true value. $\blacksquare$

The operating characteristic function is useful in determining how large the random sample need be to meet certain specifications concerning type II errors. For instance, suppose that we desire to determine the sample size n necessary to ensure that the probability of accepting $H_0 : \mu = \mu_0$ when the true mean is actually μ_1 is approximately β. That is, we want n to be such that

$$\beta(\mu_1) \approx \beta$$

But from Equation (8.3.4), this is equivalent to

$$\Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} + z_{\alpha/2}\right) - \Phi\left(\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2}\right) \approx \beta \tag{8.3.5}$$

Although the foregoing cannot be analytically solved for n, a solution can be obtained by using the standard normal distribution table. In addition, an approximation for n can be derived from Equation (8.3.5) as follows. To start, suppose that $\mu_1 > \mu_0$. Then, because this implies that

$$\frac{\sqrt{n}(\mu_0 - \mu_1)}{\sigma} - z_{\alpha/2} \leq -z_{\alpha/2}$$

it follows, since Φ is an increasing function, that

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\le\Phi(-z_{\alpha/2})=P\{Z\le -z_{\alpha/2}\}=P\{Z\ge z_{\alpha/2}\}=\alpha/2$$

Hence, we can take

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\approx 0$$

and so from Equation (8.3.5)

$$\beta\approx\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}+z_{\alpha/2}\right) \tag{8.3.6}$$

or, since

$$\beta=P\{Z>z_\beta\}=P\{Z<-z_\beta\}=\Phi(-z_\beta)$$

we obtain from Equation (8.3.6) that

$$-z_\beta\approx(\mu_0-\mu_1)\frac{\sqrt{n}}{\sigma}+z_{\alpha/2}$$

or

$$n\approx\frac{(z_{\alpha/2}+z_\beta)^2\sigma^2}{(\mu_1-\mu_0)^2} \tag{8.3.7}$$

In fact, the same approximation would result when $\mu_1<\mu_0$ (the details are left as an exercise) and so Equation (8.3.7) is in all cases a reasonable approximation to the sample size necessary to ensure that the type II error at the value $\mu=\mu_1$ is approximately equal to β.

Example 8.3.d. For the problem of Example 8.3.a, how many signals need be sent so that the .05 level test of $H_0:\mu=8$ has at least a 75 percent probability of rejection when $\mu=9.2$?

Solution. Since $z_{.025}=1.96$, $z_{.25}=.67$, the approximation from Equation (8.3.7) yields

$$n\approx\frac{(1.96+.67)^2}{(1.2)^2}4=19.21$$

Hence a sample of size 20 is needed. From Equation (8.3.4), we see that with $n=20$

$$\begin{aligned}\beta(9.2)&=\Phi\left(-\frac{1.2\sqrt{20}}{2}+1.96\right)-\Phi\left(-\frac{1.2\sqrt{20}}{2}-1.96\right)\\&=\Phi(-.723)-\Phi(-4.643)\end{aligned}$$

$$\approx 1 - \Phi(.723)$$
$$\approx .235$$

Therefore, if the message is sent 20 times, then there is a 76.5 percent chance that the null hypothesis $\mu = 8$ will be rejected when the true mean is 9.2. ■

8.3.1.1 One-sided tests

In testing the null hypothesis that $\mu = \mu_0$, we have chosen a test that calls for rejection when $\overline{X}$ is far from μ_0. That is, a very small value of $\overline{X}$ or a very large value appears to make it unlikely that μ (which $\overline{X}$ is estimating) could equal μ_0. However, what happens when the only alternative to μ being equal to μ_0 is for μ to be greater than μ_0? That is, what happens when the alternative hypothesis to $H_0 : \mu = \mu_0$ is $H_1 : \mu > \mu_0$? Clearly, in this latter case we would not want to reject H_0 when $\overline{X}$ is small (since a small $\overline{X}$ is more likely when H_0 is true than when H_1 is true). Thus, in testing

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu > \mu_0 \tag{8.3.8}$$

we should reject H_0 when $\overline{X}$, the point estimate of μ_0, is much greater than μ_0. That is, the critical region should be of the following form:

$$C = \{(X_1, \ldots, X_n) : \overline{X} - \mu_0 > c\}$$

Since the probability of rejection should equal α when H_0 is true (that is, when $\mu = \mu_0$), we require that c be such that

$$P_{\mu_0}\{\overline{X} - \mu_0 > c\} = \alpha \tag{8.3.9}$$

But since

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

has a standard normal distribution when H_0 is true, Equation (8.3.9) is equivalent to

$$P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

when Z is a standard normal. But since

$$P\{Z > z_\alpha\} = \alpha$$

we see that

$$c = \frac{z_\alpha \sigma}{\sqrt{n}}$$

Hence, the test of the Hypothesis 8.3.8 is to reject H_0 if $\overline{X} - \mu_0 > z_\alpha \sigma/\sqrt{n}$, and accept otherwise; or, equivalently, to

$$\begin{aligned} &\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \le z_\alpha \\ &\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) > z_\alpha \end{aligned} \tag{8.3.10}$$

This is called a *one-sided* critical region (since it calls for rejection only when $\overline{X}$ is large). Correspondingly, the hypothesis testing problem

$$H_0 : \mu = \mu_0$$
$$H_1 : \mu > \mu_0$$

is called a one-sided testing problem (in contrast to the *two-sided* problem that results when the alternative hypothesis is $H_1 : \mu \neq \mu_0$).

To compute the p-value in the one-sided test, Equation (8.3.10), we first use the data to determine the value of the statistic $\sqrt{n}(\overline{X} - \mu_0)/\sigma$. The p-value is then equal to the probability that a standard normal would be at least as large as this value.

Example 8.3.e. Suppose in Example 8.3.a that we know in advance that the signal value is at least as large as 8. What can be concluded in this case?

Solution. To see if the data are consistent with the hypothesis that the mean is 8, we test

$$H_0 : \mu = 8$$

against the one-sided alternative

$$H_1 : \mu > 8$$

The value of the test statistic is $\sqrt{n}(\overline{X} - \mu_0)/\sigma = \sqrt{5}(9.5 - 8)/2 = 1.68$, and the p-value is the probability that a standard normal would exceed 1.68, namely,

$$p\text{-value} = 1 - \Phi(1.68) = .0465$$

The test would call for rejection at all significance levels greater than or equal to .0465. It would, for instance, reject the null hypothesis at the $\alpha = .05$ level of significance. ■

The operating characteristic function of the one-sided test, Equation (8.3.10),

$$\beta(\mu) = P_\mu\{\text{accepting } H_0\}$$

can be obtained as follows:

$$\begin{aligned}\beta(\mu) &= P_\mu\left\{\overline{X} \le \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}\right\} \\ &= P\left\{\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \le \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha\right\} \\ &= P\left\{Z \le \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha\right\}, \quad Z \sim (0, 1)\end{aligned}$$

where the last equation follows since $\sqrt{n}(\overline{X} - \mu)/\sigma$ has a standard normal distribution. Hence we can write

$$\beta(\mu) = \Phi\left(\frac{\sqrt{n}(\mu_0 - \mu)}{\sigma} + z_\alpha\right)$$

Since Φ, being a distribution function, is increasing in its argument, it follows that $\beta(\mu)$ decreases in μ, which is intuitively pleasing since it certainly seems reasonable that the larger the true mean μ, the less likely it should be to conclude that $\mu \le \mu_0$. Also since $\Phi(z_\alpha) = 1 - \alpha$, it follows that

$$\beta(\mu_0) = 1 - \alpha$$

The test given by Equation (8.3.10), which was designed to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$, can also be used to test, at level of significance α, the one-sided hypothesis

$$H_0 : \mu \le \mu_0$$

versus

$$H_1 : \mu > \mu_0$$

To verify that it remains a level α test, we need to show that the probability of rejection is never greater than α when H_0 is true. That is, we must verify that

$$1 - \beta(\mu) \le \alpha \qquad \text{for all } \mu \le \mu_0$$

or

$$\beta(\mu) \ge 1 - \alpha \qquad \text{for all } \mu \le \mu_0$$

But it has previously been shown that for the test given by Equation (8.3.10), $\beta(\mu)$ decreases in μ and $\beta(\mu_0) = 1 - \alpha$. This gives that

$$\beta(\mu) \ge \beta(\mu_0) = 1 - \alpha \qquad \text{for all } \mu \le \mu_0$$

which shows that the test given by Equation (8.3.10) remains a level α test for $H_0 : \mu \le \mu_0$ against the alternative hypothesis $H_1 : \mu > \mu_0$.

Remark

We can also test the one-sided hypothesis

$$H_0 : \mu = \mu_0 \quad (\text{or } \mu \geq \mu_0) \qquad \text{versus} \qquad H_1 : \mu < \mu_0$$

at significance level α by

$$\text{accepting} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \geq -z_\alpha$$
$$\text{rejecting} \quad H_0 \quad \text{otherwise}$$

This test can alternatively be performed by first computing the value of the test statistic $\sqrt{n}(\overline{X} - \mu_0)/\sigma$. The p-value would then equal the probability that a standard normal would be less than this value, and the hypothesis would be rejected at any significance level greater than or equal to this p-value.

Example 8.3.f. All cigarettes presently on the market have an average nicotine content of at least 1.6 mg per cigarette. A firm that produces cigarettes claims that it has discovered a new way to cure tobacco leaves that will result in the average nicotine content of a cigarette being less than 1.6 mg. To test this claim, a sample of 20 of the firm's cigarettes were analyzed. If it is known that the standard deviation of a cigarette's nicotine content is .8 mg, what conclusions can be drawn, at the 5 percent level of significance, if the average nicotine content of the 20 cigarettes is 1.54?

Note: The above raises the question of how we would know in advance that the standard deviation is .8. One possibility is that the variation in a cigarette's nicotine content is due to variability in the amount of tobacco in each cigarette and not on the method of curing that is used. Hence, the standard deviation can be known from previous experience.

Solution. We must first decide on the appropriate null hypothesis. As was previously noted, our approach to testing is not symmetric with respect to the null and the alternative hypotheses since we consider only tests having the property that their probability of rejecting the null hypothesis when it is true will never exceed the significance level α. Thus, whereas rejection of the null hypothesis is a strong statement about the data not being consistent with this hypothesis, an analogous statement cannot be made when the null hypothesis is accepted. Hence, since in the preceding example we would like to endorse the producer's claims only when there is substantial evidence for it, we should take this claim as the alternative hypothesis. That is, we should test

$$H_0 : \mu \geq 1.6 \qquad \text{versus} \qquad H_1 : \mu < 1.6$$

Now, the value of the test statistic is

$$\sqrt{n}(\overline{X} - \mu_0)/\sigma = \sqrt{20}(1.54 - 1.6)/.8 = -.336$$

and so the p-value is given by

$$\begin{aligned} p\text{-value} &= P\{Z < -.336\}, \qquad Z \sim N(0,1) \\ &= .368 \end{aligned}$$

Since this value is greater than .05, the foregoing data do not enable us to reject, at the .05 percent level of significance, the hypothesis that the mean nicotine content exceeds 1.6 mg. In other words, the evidence, although supporting the cigarette producer's claim, is not strong enough to prove that claim. ■

Remarks

(a) There is a direct analogy between confidence interval estimation and hypothesis testing. For instance, for a normal population having mean μ and known variance σ^2, we have shown in Section 7.3 that a $100(1-\alpha)$ percent confidence interval for μ is given by

$$\mu \in \left(\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

where $\overline{x}$ is the observed sample mean. More formally, the preceding confidence interval statement is equivalent to

$$P\left\{\mu \in \left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)\right\} = 1-\alpha$$

Hence, if $\mu = \mu_0$, then the probability that μ_0 will fall in the interval

$$\left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

is $1-\alpha$, implying that a significance level α test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ is to reject H_0 when

$$\mu_0 \notin \left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

Similarly, since a $100(1-\alpha)$ percent one-sided confidence interval for μ is given by

$$\mu \in \left(\overline{X} - z_{\alpha}\frac{\sigma}{\sqrt{n}}, \infty\right)$$

it follows that an α-level significance test of $H_0 : \mu \leq \mu_0$ versus $H_1 : \mu > \mu_0$ is to reject H_0 when $\mu_0 \notin (\overline{X} - z_\alpha\sigma/\sqrt{n}, \infty)$ — that is, when $\mu_0 < \overline{X} - z_\alpha\sigma/\sqrt{n}$.

(b) A remark on robustness A test that performs well even when the underlying assumptions on which it is based are violated is said to be *robust*. For

Table 8.1 $X_1, \ldots, X_n$ Is a Sample from a (μ, σ^2) Population, σ^2 Is Known, $\overline{X} = \sum_{i=1}^{n} X_i/n$.

H_0	H_1	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $\|TS\| > z_{\alpha/2}$	$2P\{Z \geq \|t\|\}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS > z_\alpha$	$P\{Z \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/\sigma$	Reject if $TS < -z_\alpha$	$P\{Z \leq t\}$

Z is a standard normal random variable.

instance, the tests of Sections 8.3.1 and 8.3.1.1 were derived under the assumption that the underlying population distribution is normal with known variance σ^2. However, in deriving these tests, this assumption was used only to conclude that $\overline{X}$ also has a normal distribution. But, by the central limit theorem, it follows that for a reasonably large sample size, $\overline{X}$ will approximately have a normal distribution no matter what the underlying distribution. Thus we can conclude that these tests will be relatively robust for any population distribution with variance σ^2.

Table 8.1 summarizes the tests of this subsection.

8.3.2 Case of unknown variance: the t-test

Up to now we have supposed that the only unknown parameter of the normal population distribution is its mean. However, the more common situation is one where the mean μ and variance σ^2 are both unknown. Let us suppose this to be the case and again consider a test of the hypothesis that the mean is equal to some specified value μ_0. That is, consider a test of

$$H_0 : \mu = \mu_0$$

versus the alternative

$$H_1 : \mu \neq \mu_0$$

It should be noted that the null hypothesis is not a simple hypothesis since it does not specify the value of σ^2.

As before, it seems reasonable to reject H_0 when the sample mean $\overline{X}$ is far from μ_0. However, how far away it need be to justify rejection will depend on the variance σ^2. Recall that when the value of σ^2 was known, the test called for rejecting H_0 when $|\overline{X} - \mu_0|$ exceeded $z_{\alpha/2}\sigma/\sqrt{n}$ or, equivalently, when

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2}$$

Now when σ^2 is no longer known, it seems reasonable to estimate it by

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}$$

and then to reject H_0 when

$$\left| \frac{\overline{X} - \mu_0}{S/\sqrt{n}} \right|$$

is large.

To determine how large a value of the statistic

$$\left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right|$$

to require for rejection, in order that the resulting test have significance level α, we must determine the probability distribution of this statistic when H_0 is true. However, as shown in Section 6.5, the statistic T, defined by

$$T = \frac{\sqrt{n}(\overline{X} - \mu_0)}{S}$$

has, when $\mu = \mu_0$, a t-distribution with $n-1$ degrees of freedom. Hence,

$$P_{\mu_0}\left\{ -t_{\alpha/2,n-1} \le \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \le t_{\alpha/2,n-1} \right\} = 1 - \alpha \tag{8.3.11}$$

where $t_{\alpha/2,n-1}$ is the 100 $\alpha/2$ upper percentile value of the t-distribution with $n-1$ degrees of freedom. (That is, $P\{T_{n-1} \ge t_{\alpha/2,n-1}\} = P\{T_{n-1} \le -t_{\alpha/2,n-1}\} = \alpha/2$ when T_{n-1} has a t-distribution with $n-1$ degrees of freedom.) From Equation (8.3.11) we see that the appropriate significance level α test of

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \ne \mu_0$$

is, when σ^2 is unknown, to

$$\begin{aligned} &\text{accept} \quad H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right| \le t_{\alpha/2,\, n-1} \\ &\text{reject} \quad H_0 \quad \text{if} \quad \left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \right| > t_{\alpha/2,\, n-1} \end{aligned} \tag{8.3.12}$$

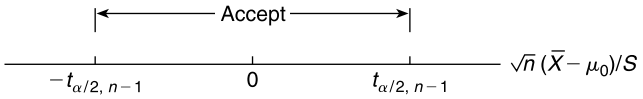


FIGURE 8.3
The two-sided t-test.

The test defined by Equation (8.3.12) is called a *two-sided t-test*. It is pictorially illustrated in Figure 8.3.

If we let t denote the observed value of the test statistic $T = \sqrt{n}(\overline{X} - \mu_0)/S$, then the p-value of the test is the probability that $|T|$ would exceed $|t|$ when H_0 is true. That is, the p-value is the probability that the absolute value of a t-random variable with $n - 1$ degrees of freedom would exceed $|t|$. The test then calls for rejection at all significance levels higher than the p-value and acceptance at all lower significance levels.

Example 8.3.g. Among a clinic's patients having blood cholesterol levels ranging in the medium to high range (at least 220 milliliters per deciliter of serum), volunteers were recruited to test a new drug designed to reduce blood cholesterol. A group of 50 volunteers was given the drug for 1 month and the changes in their blood cholesterol levels were noted. If the average change was a reduction of 14.8 with a sample standard deviation of 6.4, what conclusions can be drawn?

Solution. Let us start by testing the hypothesis that the change could be due solely to chance — that is, that the 50 changes constitute a normal sample with mean 0. Because the value of the t-statistic used to test the hypothesis that a normal mean is equal to 0 is

$$T = \sqrt{n}\,\overline{X}/S = \sqrt{50}\,14.8/6.4 = 16.352$$

it clear that we should reject the hypothesis that the changes were solely due to chance. Unfortunately, however, we are not justified at this point in concluding that the changes were due to the specific drug used and not to some other possibility. For instance, it is well known that any medication received by a patient (whether or not this medication is directly relevant to the patient's suffering) often leads to an improvement in the patient's condition — the so-called placebo effect. Also, another possibility that may need to be taken into account would be the weather conditions during the month of testing, for it is certainly conceivable that this affects blood cholesterol level. Indeed, it must be concluded that the foregoing was a very poorly designed experiment, for in order to test whether a specific treatment has an effect on a disease that may be affected by many things, we should try to design the experiment so as to neutralize all other possible causes. The accepted approach for accomplishing this is to divide the volunteers at random into two groups — one group to receive the drug and the other to receive a placebo (that is, a tablet that looks and tastes

like the actual drug but has no physiological effect). The volunteers should not be told whether they are in the actual or control group, and indeed it is best if even the clinicians do not have this information (the so-called double-blind test) so as not to allow their own biases to play a role. Since the two groups are chosen at random from among the volunteers, we can now hope that on average all factors affecting the two groups will be the same except that one received the actual drug and the other a placebo. Hence, any difference in performance between the groups can be attributed to the drug. ■

Example 8.3.h. A public health official claims that the mean home water use is 350 gallons a day. To verify this claim, a study of 20 randomly selected homes was instigated with the result that the average daily water uses of these 20 homes were as follows:

340	344	362	375
356	386	354	364
332	402	340	355
362	322	372	324
318	360	338	370

Do the data contradict the official's claim?

Solution. To determine if the data contradict the official's claim, we need to test

$$H_0 : \mu = 350 \qquad \text{versus} \qquad H_1 : \mu \neq 350$$

This can be accomplished by noting first that the sample mean and sample standard deviation of the preceding data set are

$$\overline{X} = 353.8, \qquad S = 21.8478$$

Thus, the value of the test statistic is

$$T = \frac{\sqrt{20}(3.8)}{21.8478} = .7778$$

Because this is less than $t_{.05,19} = 1.730$, the null hypothesis is accepted at the 10 percent level of significance. Indeed, the p-value of the test data is

$$p\text{-value} = P\{|T_{19}| > .7778\} = 2P\{T_{19} > .7778\} = .4462$$

indicating that the null hypothesis would be accepted at any reasonable significance level, and thus that the data are not inconsistent with the claim of the health official. ■

R can be used to perform the computations in a t-test. To test the null hypothesis that a mean μ is equal to μ_0 against the alternative that $\mu \neq \mu_0$ when the

data are $x_1, \ldots, x_n$, type

$>x = c(x_1, \ldots, x_n)$
$>$t.test(x, mu $= \mu_0$)

and hit return. Upon doing so, the output will include both a 95 percent confidence interval and the p-value. For instance, to use R to solve Example 8.3.h, do the following:

$>x = c(340, 356, 332, 362, 318, 344, 386, 402, 322, 360, 362, 354, 340, 372, 338,$
$375, 363, 355, 324, 370)$
$>$t.test(x, mu $= 350$)

After hitting return after the second line, you will receive the following output:

One Sample t-test
data: x
t = 0.77784, df = 19, p-value = 0.4462
alternative hypothesis: true mean is not equal to 350
95 percent confidence interval:
343.5749 364.0251
sample estimates:
mean of x
353.8

We can use a one-sided t-test to test the hypothesis

$H_0 : \mu = \mu_0 \qquad (\text{or } H_0 : \mu \leq \mu_0)$

against the one-sided alternative

$H_1 : \mu > \mu_0$

The significance level α test is to

$$\begin{aligned} &\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \leq t_{\alpha, n-1} \\ &\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} > t_{\alpha, n-1} \end{aligned} \tag{8.3.13}$$

If $\sqrt{n}(\overline{X} - \mu_0)/S = v$, then the p-value of the test is the probability that a t-random variable with $n - 1$ degrees of freedom would be at least as large as v.

The significance level α test of

$H_0 : \mu = \mu_0 \qquad (\text{or } H_0 : \mu \geq \mu_0)$

versus the alternative

$$H_1 : \mu < \mu_0$$

is to

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} \geq -t_{\alpha, n-1}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}(\overline{X} - \mu_0)}{S} < -t_{\alpha, n-1}$$

The p-value of this test is the probability that a t-random variable with $n - 1$ degrees of freedom would be less than or equal to the observed value of $\sqrt{n}(\overline{X} - \mu_0)/S$.

Example 8.3.i. The manufacturer of a new fiberglass tire claims that its average life will be at least 40,000 miles. To verify this claim a sample of 12 tires is tested, with their lifetimes (in 1000s of miles) being as follows:

Tire	1	2	3	4	5	6	7	8	9	10	11	12
Life	36.1	40.2	33.8	38.5	42	35.8	37	41	36.8	37.2	33	36

Test the manufacturer's claim at the 5 percent level of significance.

Solution. To determine whether the foregoing data are consistent with the hypothesis that the mean life is at least 40,000 miles, we will test

$$H_0 : \mu \geq 40{,}000 \qquad \text{versus} \qquad H_1 : \mu < 40{,}000$$

A computation gives that

$$\overline{X} = 37.2833, \qquad S = 2.7319$$

and so the value of the test statistic is

$$T = \frac{\sqrt{12}(37.2833 - 40)}{2.7319} = -3.4448$$

Since this is less than $-t_{.05,11} = -1.796$, the null hypothesis is rejected at the 5 percent level of significance. Indeed, the p-value of the test data is

$$p\text{-value} = P\{T_{11} < -3.4448\} = P\{T_{11} > 3.4448\} = .0027$$

indicating that the manufacturer's claim would be rejected at any significance level greater than .003. ■

The preceding could also have been solved by using R. R performs one-sided t-tests by specifying that the alternative hypothesis is that the mean is either less than or greater than μ_0. That is, to test

$H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$

when using data $x_1, \ldots, x_n$, just type

$>x = c(x_1, \ldots, x_n)$
$>$t.test(x, alternative =“greater”, mu = μ_0)

and hit return. Thus, for Example 8.3.i, we obtain

$>x = c(36.1, 40.2, 33.8, 38.5, 42, 35.8, 37, 41, 36.8, 37.2, 33, 36)$
$>$t.test(x, alternative = “less”, mu = 40)

One Sample t-test
data: x
t = -3.4448, df = 11, p-value = 0.002739
alternative hypothesis: true mean is less than 40
95 percent confidence interval:
-Inf 38.69963
sample estimates:
mean of x
37.28333

Example 8.3.j. In a single-server queueing system in which customers arrive according to a Poisson process, the long-run average queueing delay per customer depends on the service distribution through its mean and variance. Indeed, if μ is the mean service time, and σ^2 is the variance of a service time, then the average amount of time that a customer spends waiting in queue is given by

$$\frac{\lambda(\mu^2 + \sigma^2)}{2(1 - \lambda\mu)}$$

provided that $\lambda\mu < 1$, where λ is the arrival rate. (The average delay is infinite if $\lambda\mu \geq 1$.) As can be seen by this formula, the average delay is quite large when μ is only slightly smaller than $1/\lambda$, where, since λ is the arrival *rate*, $1/\lambda$ is the average time between arrivals.

Suppose that the owner of a service station will hire a second server if it can be shown that the average service time exceeds 8 minutes. The following data give the service times (in minutes) of 28 customers of this queueing system. Do they indicate that the mean service time is greater than 8 minutes?

8.6, 9.4, 5.0, 4.4, 3.7, 11.4, 10.0, 7.6, 14.4, 12.2, 11.0, 14.4, 9.3, 10.5,
10.3, 7.7, 8.3, 6.4, 9.2, 5.7, 7.9, 9.4, 9.0, 13.3, 11.6, 10.0, 9.5, 6.6

16. Suppose that team A and team B are to play a National Football League game and that team A is favored by f points. Let $S(A)$ and $S(B)$ denote the scores of teams A and B, and let $X = S(A) - S(B) - f$. That is, X is the amount by which team A beats the point spread. It has been claimed that the distribution of X is normal with mean 0 and standard deviation 14. Use data from randomly chosen football games to test this hypothesis.
17. A medical scientist believes that the average basal temperature of (outwardly) healthy individuals has increased over time and is now greater than 98.6 degrees Fahrenheit (37 degrees Celsius). To prove this, she has randomly selected 100 healthy individuals. If their mean temperature is 98.74 with a sample standard deviation of 1.1 degrees, does this prove her claim at the 5 percent level? What about at the 1 percent level?
18. Use the results of a Sunday's worth of NFL professional football games to test the hypothesis that the average number of points scored by winning teams is less than or equal to 28. Use the 5 percent level of significance.
19. Use the results of a Sunday's worth of major league baseball scores to test the hypothesis that the average number of runs scored by winning teams is at least 5.6. Use the 5 percent level of significance.
20. A car is advertised as having a gas mileage rating of at least 30 miles/gallon in highway driving. If the miles per gallon obtained in 10 independent experiments are 26, 24, 20, 25, 27, 25, 28, 30, 26, 33, should you believe the advertisement? What assumptions are you making?
21. A producer specifies that the mean lifetime of a certain type of battery is at least 240 hours. A sample of 18 such batteries yielded the following data.

237	242	232
242	248	230
244	243	254
262	234	220
225	236	232
218	228	240

Assuming that the life of the batteries is approximately normally distributed, do the data indicate that the specifications are not being met?
22. Use the data of Example 2.3.i of Chapter 2 to test the null hypothesis that the average noise level directly outside of Grand Central Station is less than or equal to 80 decibels.
23. An oil company claims that the sulfur content of its diesel fuel is at most .15 percent. To check this claim, the sulfur contents of 40 randomly chosen samples were determined; the resulting sample mean and sample standard deviation were .162 and .040. Using the 5 percent level of significance, can we conclude that the company's claims are invalid?
24. A company supplies plastic sheets for industrial use. A new type of plastic has been produced and the company would like to claim that the average

stress resistance of this new product is at least 30.0, where stress resistance is measured in pounds per square inch (psi) necessary to crack the sheet. The following random sample was drawn off the production line. Based on this sample, would the claim clearly be unjustified?

30.1	32.7	22.5	27.5
27.7	29.8	28.9	31.4
31.2	24.3	26.4	22.8
29.1	33.4	32.5	21.7

Assume normality and use the 5 percent level of significance.

25. It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of current gain is 200 with a sample standard deviation of 35, would the claim be rejected at the 5 percent level of significance if
    a. the sample size is 25;
    b. the sample size is 64?
26. A manufacturer of capacitors claims that the breakdown voltage of these capacitors has a mean value of at least 100 V. A test of 12 of these capacitors yielded the following breakdown voltages:

    96, 98, 105, 92, 111, 114, 99, 103, 95, 101, 106, 97

    Do these results prove the manufacturer's claim? Do they disprove them?
27. A sample of 10 fish were caught at lake A and their PCB concentrations were measured using a certain technique. The resulting data in parts per million were

    Lake A: 11.5, 10.8, 11.6, 9.4, 12.4, 11.4, 12.2, 11, 10.6, 10.8

    In addition, a sample of 8 fish were caught at lake B and their levels of PCB were measured by a different technique than that used at lake A. The resultant data were

    Lake B: 11.8, 12.6, 12.2, 12.5, 11.7, 12.1, 10.4, 12.6

    If it is known that the measuring technique used at lake A has a variance of .09 whereas the one used at lake B has a variance of .16, could you reject (at the 5 percent level of significance) a claim that the two lakes are equally contaminated?
28. A method for measuring the pH level of a solution yields a measurement value that is normally distributed with a mean equal to the actual pH of the solution and with a standard deviation equal to .05. An environmental pollution scientist claims that two different solutions come from the same source. If this were so, then the pH level of the solutions would

be equal. To test the plausibility of this claim, 10 independent measurements were made of the pH level for both solutions, with the following data resulting.

Measurements of Solution A	Measurements of Solution B
6.24	6.27
6.31	6.25
6.28	6.33
6.30	6.27
6.25	6.24
6.26	6.31
6.24	6.28
6.29	6.29
6.22	6.34
6.28	6.27

a. Do the data disprove the scientist's claim? Use the 5 percent level of significance.
b. What is the p-value?

29. The following are the values of independent samples from two different populations.

Sample 1	122, 114, 130, 165, 144, 133, 139, 142, 150
Sample 2	108, 125, 122, 140, 132, 120, 137, 128, 138

Let μ_1 and μ_2 be the respective means of the two populations. Find the p-value of the test of the null hypothesis

$$H_0 : \mu_1 \leq \mu_2$$

versus the alternative

$$H_1 : \mu_1 > \mu_2$$

when the population standard deviations are $\sigma_1 = 10$ and **(a)** $\sigma_2 = 5$; **(b)** $\sigma_2 = 10$; **(c)** $\sigma_2 = 20$.

30. The data below give the lifetimes in hundreds of hours of samples of two types of electronic tubes. Past lifetime data of such tubes have shown that they can often be modeled as arising from a lognormal distribution. That is, the logarithms of the data are normally distributed. Assuming that variance of the logarithms is equal for the two populations, test, at the 5 percent level of significance, the hypothesis that the two population distributions are identical.

Type 1	32, 84, 37, 42, 78, 62, 59, 74
Type 2	39, 111, 55, 106, 90, 87, 85

31. The viscosity of two different brands of car oil is measured and the following data resulted:

Brand 1	10.62, 10.58, 10.33, 10.72, 10.44, 10.74
Brand 2	10.50, 10.52, 10.58, 10.62, 10.55, 10.51, 10.53

Test the hypothesis that the mean viscosity of the two brands is equal, assuming that the populations have normal distributions with equal variances.

32. It is argued that the resistance of wire A is greater than the resistance of wire B. You make tests on each wire with the following results.

Wire A	Wire B
.140 ohm	.135 ohm
.138	.140
.143	.136
.142	.142
.144	.138
.137	.140

What conclusion can you draw at the 10 percent significance level? Explain what assumptions you are making.

In Problems 33 through 40, assume that the population distributions are normal and have equal variances.

33. Twenty-five men between the ages of 25 and 30, who were participating in a well-known heart study carried out in Framingham, Massachusetts, were randomly selected. Of these, 11 were smokers and 14 were not. The following data refer to readings of their systolic blood pressure.

Smokers	Nonsmokers
124	130
134	122
136	128
125	129
133	118
127	122
135	116
131	127
133	135
125	120
118	122
	120
	115
	123

Use these data to test the hypothesis that the mean blood pressures of smokers and nonsmokers are the same.

34. In a 1943 experiment (Whitlock and Bliss, "A Bioassay Technique for Anti-helminthics," *Journal of Parasitology*, **29**, pp. 48–58) 10 albino rats were used to study the effectiveness of carbon tetrachloride as a treatment for worms. Each rat received an injection of worm larvae. After 8 days, the rats were randomly divided into two groups of 5 each; each rat in the first group received a dose of .032 cc of carbon tetrachloride, whereas the dosage for each rat in the second group was .063 cc. Two days later the rats were killed, and the number of adult worms in each rat was determined. The numbers detected in the group receiving the .032 dosage were

    421, 462, 400, 378, 413

    whereas they were

    207, 17, 412, 74, 116

    for those receiving the .063 dosage. Do the data prove that the larger dosage is more effective than the smaller?

35. A professor claims that the average starting salary of industrial engineering graduating seniors is greater than that of civil engineering graduates. To study this claim, samples of 16 industrial engineers and 16 civil engineers, all of whom graduated in 2006, were chosen and sample members were queried about their starting salaries. If the industrial engineers had a sample mean salary of \$72,700 and a sample standard deviation of \$2400, and the civil engineers had a sample mean salary of \$71,400 and a sample standard deviation of \$2200, has the professor's claim been verified? Find the appropriate p-value.

36. In a certain experimental laboratory, a method A for producing gasoline from crude oil is being investigated. Before completing experimentation, a new method B is proposed. All other things being equal, it was decided to abandon A in favor of B only if the average yield of the latter was clearly greater. The yield of both processes is assumed to be normally distributed. However, there has been insufficient time to ascertain their true standard deviations, although there appears to be no reason why they cannot be assumed equal. Cost considerations impose size limits on the size of samples that can be obtained. If a 1 percent significance level is all that is allowed, what would be your recommendation based on the following random samples? The numbers represent percent yield of crude oil.

    A 23.2, 26.6, 24.4, 23.5, 22.6, 25.7, 25.5
    B 25.7, 27.7, 26.2, 27.9, 25.0, 21.4, 26.1

37. A study was instituted to learn how the diets of women changed during the winter and the summer. A random group of 12 women were observed during the month of July and the percentage of each woman's calories that came from fat was determined. Similar observations were made on a different randomly selected group of size 12 during the month of January. The results were as follows:

July	32.2, 27.4, 28.6, 32.4, 40.5, 26.2, 29.4, 25.8, 36.6, 30.3, 28.5, 32.0
January	30.5, 28.4, 40.2, 37.6, 36.5, 38.8, 34.7, 29.5, 29.7, 37.2, 41.5, 37.0

Test the hypothesis that the mean fat percentage intake is the same for both months. Use the **(a)** 5 percent level of significance and **(b)** 1 percent level of significance.

38. To learn about the feeding habits of bats, 22 bats were tagged and tracked by radio. Of these 22 bats, 12 were female and 10 were male. The distances flown (in meters) between feedings were noted for each of the 22 bats, and the following summary statistics were obtained.

Female Bats	Male Bats
$n = 12$	$m = 10$
$\overline{X} = 180$	$\overline{Y} = 136$
$S_x = 92$	$S_y = 86$

Test the hypothesis that the mean distance flown between feedings is the same for the populations of both male and of female bats. Use the 5 percent level of significance.

39. The following data summary was obtained from a comparison of the lead content of human hair removed from adult individuals that had died between 1880 and 1920 with the lead content of present-day adults. The data are in units of micrograms, equal to one-millionth of a gram.

	1880–1920	Today
Sample size:	30	100
Sample mean:	48.5	26.6
Sample standard deviation:	14.5	12.3

a. Do the above data establish, at the 1 percent level of significance, that the mean lead content of human hair is less today than it was in the years between 1880 and 1920? Clearly state what the null and alternative hypotheses are.

b. What is the p-value for the hypothesis test in part (a)?

40. Sample weights (in pounds) of newborn babies born in two adjacent counties in western Pennsylvania yielded the following data.

$$n = 53, \quad m = 44$$
$$\overline{X} = 6.8, \quad \overline{Y} = 7.2$$
$$S^2 = 5.2, \quad S^2 = 4.9$$

Consider a test of the hypothesis that the mean weight of newborns is the same in both counties. What is the resulting p-value?

41. To verify the hypothesis that blood lead levels tend to be higher for children whose parents work in a factory that uses lead in the manufacturing process, researchers examined lead levels in the blood of 33 children whose parents worked in a battery manufacturing factory (Morton, D., Saah, A., Silberg, S., Owens, W., Roberts, M., and Saah, M., "Lead Absorption in Children of Employees in a Lead-Related Industry," *American Journal of Epidemiology*, **115**, 549–555, 1982). Each of these children was then *matched* by another child who was of similar age, lived in a similar neighborhood, had a similar exposure to traffic, but whose parent did not work with lead. The blood levels of the 33 cases (sample 1) as well as those of the 33 controls (sample 2) were then used to test the hypothesis that the average blood levels of these groups are the same. If the resulting sample means and sample standard deviations were

$$\bar{x}_1 = .015, \quad s_1 = .004, \quad \bar{x}_2 = .006, \quad s_2 = .006$$

find the resulting p-value. Assume a common variance.

42. Ten pregnant women were given an injection of pitocin to induce labor. Their systolic blood pressures immediately before and after the injection were:

Patient	Before	After	Patient	Before	After
1	134	140	6	140	138
2	122	130	7	118	124
3	132	135	8	127	126
4	130	126	9	125	132
5	128	134	10	142	144

Do the data indicate that injection of this drug changes blood pressure?

43. A question of medical importance is whether jogging leads to a reduction in one's pulse rate. To test this hypothesis, 8 nonjogging volunteers agreed to begin a 1-month jogging program. After the month their pulse rates were determined and compared with their earlier values. If the data are as follows, can we conclude that jogging has had an effect on the

11.2 Goodness of fit tests when all parameters are specified

Suppose that n independent random variables — $Y_1, \ldots, Y_n$, each taking on one of the values $1, 2, \ldots, k$ — are to be observed and we are interested in testing the null hypothesis that $\{p_i, i = 1, \ldots, k\}$ is the probability mass function of the Y_j. That is, if Y represents any of the Y_j, then the null hypothesis is

$$H_0 : P\{Y = i\} = p_i, \qquad i = 1, \ldots, k$$

whereas the alternative hypothesis is

$$H_1 : P\{Y = i\} \neq p_i, \qquad \text{for some } i = 1, \ldots, k$$

To test the foregoing hypothesis, let $X_i, i = 1, \ldots, k$, denote the number of the Y_j's that equal i. Then as each Y_j will independently equal i with probability $P\{Y = i\}$, it follows that, under H_0, X_i is binomial with parameters n and p_i. Hence, when H_0 is true,

$$E[X_i] = np_i$$

and so $(X_i - np_i)^2$ will be an indication as to how likely it appears that p_i indeed equals the probability that $Y = i$. When this is large, say, in relationship to np_i, then it is an indication that H_0 is not correct. Indeed such reasoning leads us to consider the following test statistic:

$$T = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i} \tag{11.2.1}$$

and to reject the null hypothesis when T is large.

To determine the critical region, we need first specify a significance level α and then we must determine that critical value c such that

$$P_{H_0}\{T \geq c\} = \alpha$$

That is, we need to determine c so that the probability that the test statistic T is at least as large as c, when H_0 is true, is α. The test is then to reject the hypothesis, at the α level of significance, when $T \geq c$ and to accept when $T < c$.

It remains to determine c. The classical approach to doing so is to use the result that when n is large T will have, when H_0 is true, approximately (with the approximation becoming exact as n approaches infinity) a chi-square distribution with $k - 1$ degrees of freedom. Hence, for n large, c can be taken to equal $\chi^2_{\alpha,k-1}$; and so the approximate α level test is

reject H_0 if $T \geq \chi^2_{\alpha,k-1}$

accept H_0 otherwise

If the observed value of T is $T = t$, then the preceding test is equivalent to rejecting H_0 if the significance level α is at least as large as the p-value given by

$$\begin{aligned} p\text{-value} &= P_{H_0}\{T \geq t\} \\ &\approx P\{\chi^2_{k-1} \geq t\} \end{aligned}$$

where χ^2_{k-1} is a chi-square random variable with $k-1$ degrees of freedom.

An accepted rule of thumb as to how large n need be for the foregoing to be a good approximation is that it should be large enough so that $np_i \geq 1$ for each $i, i = 1, \ldots, k$, and also at least 80 percent of the values np_i should exceed 5.

Remarks

(a) A computationally simpler formula for T can be obtained by expanding the square in Equation (11.2.1) and using the results that $\sum_i p_i = 1$ and $\sum_i X_i = n$ (why is this true?):

$$\begin{aligned} T &= \sum_{i=1}^{k} \frac{X_i^2 - 2np_iX_i + n^2p_i^2}{np_i} \qquad (11.2.2) \\ &= \sum_i X_i^2/np_i - 2\sum_i X_i + n\sum_i p_i \\ &= \sum_i X_i^2/np_i - n \end{aligned}$$

(b) The intuitive reason why T, which depends on the k values $X_1, \ldots, X_k$, has only $k-1$ degrees of freedom is that 1 degree of freedom is lost because of the linear relationship $\sum_i X_i = n$.

(c) Whereas the proof that, asymptotically, T has a chi-square distribution is advanced, it can be easily shown when $k = 2$. In this case, since $X_1 + X_2 = n$, and $p_1 + p_2 = 1$, we see that

$$\begin{aligned} T &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2} \\ &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(n - X_1 - n[1-p_1])^2}{n(1-p_1)} \\ &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1-p_1)} \\ &= \frac{(X_1 - np_1)^2}{np_1(1-p_1)} \qquad \text{since} \qquad \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)} \end{aligned}$$

However, X_1 is a binomial random variable with mean np_1 and variance $np_1(1-p_1)$ and thus, by the normal approximation to the binomial, it follows

that $(X_1 - np_1)/\sqrt{np_1(1-p_1)}$ has, for large n, approximately a standard normal distribution, and so its square has approximately a chi-square distribution with 1 degree of freedom.

Example 11.2.a. In recent years, a correlation between mental and physical well-being has increasingly become accepted. An analysis of birthdays and death days of famous people could be used as further evidence in the study of this correlation. To use these data, we are supposing that being able to look forward to something betters a person's mental state, and that a famous person would probably look forward to his or her birthday because of the resulting attention, affection, and so on. If a famous person is in poor health and dying, then perhaps anticipating his birthday would "cheer him up and therefore improve his health and possibly decrease the chance that he will die shortly before his birthday." The data might therefore reveal that a famous person is less likely to die in the months before his or her birthday and more likely to die in the months afterward.

Solution. To test this, a sample of 1251 (deceased) Americans was randomly chosen from *Who Was Who in America*, and their birth and death days were noted. (The data are taken from D. Phillips, "Death Day and Birthday: An Unexpected Connection," in *Statistics: A Guide to the Unknown*, Holden-Day, 1972.) The data are summarized in Table 11.1.

If the death day does not depend on the birthday, then it would seem that each of the 1251 individuals would be equally likely to fall in any of the 12 categories. Thus, let us test the null hypothesis

$$H_0 = p_i = \frac{1}{12}, \qquad i = 1, \ldots, 12$$

Since $np_i = 1251/12 = 104.25$, the chi-square test statistic for this hypothesis is

$$T = \frac{(90)^2 + (100)^2 + (87)^2 + \cdots + (106)^2}{104.25} - 1251$$
$$= 17.192$$

The p-value is

$$\begin{aligned} p\text{-value} &\approx P(\chi^2_{11} \geq 17.192) \\ &> 1 - \text{pchisq}(17.192, 11) \\ &[1] \quad 0.1023232 \end{aligned}$$

The results of this test leave us somewhat up in the air about the hypothesis that an approaching birthday has no effect on an individual's remaining lifetime. For whereas the data are not quite strong enough (at least, at

Table 11.1 Number of Deaths Before, During, and After the Birth Month.

	6 Months Before	5 Months Before	4 Months Before	3 Months Before	2 Months Before	1 Month Before	The Month	1 Month After	2 Months After	3 Months After	4 Months After	5 Months After
Number of deaths	90	100	87	96	101	86	119	118	121	114	113	106

$n = 1251$.
$n/12 = 104.25$.

the 10 percent level of significance) to reject this hypothesis, they are certainly suggestive of its possible falsity. This raises the possibility that perhaps we should not have allowed as many as 12 data categories, and that we might have obtained a more powerful test by allowing for a fewer number of possible outcomes. For instance, let us determine what the result would have been if we had coded the data into 4 possible outcomes as follows:

outcome $1 = -6, -5, -4$
outcome $2 = -3, -2, -1$
outcome $3 = 0, 1, 2$
outcome $4 = 3, 4, 5$

That is, for instance, an individual whose death day occurred 3 months before his or her birthday would be placed in outcome 2. With this classification, the data would be as follows:

Outcome	Number of Times Occurring
1	277
2	283
3	358
4	333

$n = 1251$.
$n/4 = 312.75$.

The test statistic for testing $H_0 = p_i = 1/4, i = 1, 2, 3, 4$ is

$$T = \frac{(277)^2 + (283)^2 + (358)^2 + (333)^2}{312.75} - 1.251$$
$$= 14.775$$

Hence, as $\chi^2_{.01,3} = 11.345$, the null hypothesis would be rejected even at the 1 percent level of significance. Indeed, R yields

$$p\text{-value} = P\{\chi^2_3 \geq 14.775\} = 1 - \text{pchisq}(14.775, 3) = 0.00201938$$

The foregoing analysis is, however, subject to the criticism that the null hypothesis was chosen after the data were observed. Indeed, while there is nothing incorrect about using a set of data to determine the "correct way" of phrasing a null hypothesis, the additional use of those data to test that very hypothesis is certainly questionable. Therefore, to be quite certain of the conclusion to be drawn from this example, it seems prudent to choose a second random sample — coding the values as before — and again test $H_0 : p_i = 1/4, i = 1, 2, 3, 4$ (see Problem 3). ■

R can be used to both compute the value of the test statistic and determine the resulting p-value. Suppose $x = c(x_1, \ldots, x_k)$ is the vector giving the number of times that each of the k possible values occurred, and suppose we want to test the null hypothesis that the probability vector for outcomes is $p = c(p_1, \ldots, p_k)$. To do so, just use the R commands

$$\begin{aligned} &> \quad x = c(x_1, \ldots, x_k) \\ &> \quad \text{chisq.test}(x,\ p = c(p_1, \ldots, p_k)) \end{aligned}$$

For instance, for the example just considered

$$\begin{aligned} &> \qquad x = c(277, 283, 358, 333) \\ &> \qquad \text{chisq.test}(x,\ p = c(1/4, 1/4, 1/4, 1/4)) \\ &\qquad\quad \text{Chi-squared test for given probabilities} \\ &\text{data :} \qquad x \\ &X - \text{squared} = 14.775, \quad \text{df} = 3, \quad p\text{-value} = 0.00202 \end{aligned}$$

Example 11.2.b. A contractor who purchases a large number of fluorescent lightbulbs has been told by the manufacturer that these bulbs are not of uniform quality but rather have been produced in such a way that each bulb produced will, independently, either be of quality level A, B, C, D, or E, with respective probabilities .15, .25, .35, .20, .05. However, the contractor feels that he is receiving too many type E (the lowest quality) bulbs, and so he decides to test the producer's claim by taking the time and expense to ascertain the quality of 30 such bulbs. Suppose that he discovers that of the 30 bulbs, 3 are of quality level A, 6 are of quality level B, 9 are of quality level C, 7 are of quality level D, and 5 are of quality level E. Do these data, at the 5 percent level of significance, enable the contractor to reject the producer's claim?

Solution. We use R as follows:

$$\begin{aligned} &> \qquad x = c(3, 6, 9, 7, 5) \\ &> \qquad \text{chisq.test}(x,\ p = c(.15, .25, .35, .20, .05)) \\ &\qquad\quad \text{Chi-squared test for given probabilities} \\ &\text{data : } x \\ &X - \text{squared} = 9.3476, \quad \text{df} = 4, \quad \text{p-value} = 0.05297 \end{aligned}$$

$$\begin{aligned} &\text{Warning message:} \\ &\text{In chi-sq.test}(x,\ p = c(.15, .25, .35, .20, .05)) \\ &\text{Chi-squared approximation may be incorrect} \end{aligned}$$

Note that R gives the warning that, because of the relatively small numbers falling in each category, the chi-squared approximation for the distribution of the test statistic may not be totally accurate. Assuming that its accuracy is

not an issue, the hypothesis would not be rejected at the 5 percent level of significance (but since it would be rejected at all significance levels above .053, the contractor should certainly remain skeptical). ■

11.2.1 Determining the critical region by simulation[1]

From 1900 when Karl Pearson first showed that T has approximately (becoming exact as n approaches infinity) a chi-square distribution with $k-1$ degrees of freedom, until relatively recently, this approximation was the only means available for determining the p-value of the goodness of fit test. However, with the advent of inexpensive, fast, and easily available computational power a second, potentially more accurate, approach has become available: namely, the use of simulation to obtain to a high level of accuracy the p-value of the test statistic.

The simulation approach is as follows. First, the value of T is determined — say, $T = t$. Now to determine whether or not to accept H_0, at a given significance level α, we need to know the probability that T would be at least as large as t when H_0 is true. To determine this probability, we simulate n independent random variables $Y_1^{(1)}, \ldots, Y_n^{(1)}$ each having the probability mass function $\{p_i, i = 1, \ldots, k\}$ — that is,

$$P\{Y_j^{(1)} = i\} = p_i, \qquad i = 1, \ldots, k, \qquad j = 1, \ldots, n$$

Now let

$$X_i^{(1)} = \text{number } j : Y_j^{(1)} = i$$

and set

$$T^{(1)} = \sum_{i=1}^{k} \frac{(X_i^{(1)} - np_i)^2}{np_i}$$

Now repeat this procedure by simulating a second set, independent of the first set, of n independent random variables $Y_1^{(2)}, \ldots, Y_n^{(2)}$ each having the probability mass function $\{p_i, i = 1, \ldots, k\}$ and then, as for the first set, determining $T^{(2)}$. Repeating this a large number, say, r, of times yields r independent random variables $T^{(1)}, T^{(2)}, \ldots, T^{(r)}$, each of which has the same distribution as does the test statistic T when H_0 is true. Hence, by the law of large numbers, the proportion of the T_i that are as large as t will be very nearly equal to the probability that T is as large as t when H_0 is true—that is,

$$\frac{\text{number } l : T^{(l)} \geq t}{r} \approx P_{H_0}\{T \geq t\}$$

[1] Optional section.

In fact, by letting r be large, the foregoing can be considered to be, with high probability, almost an equality. Hence, if that proportion is less than or equal to α, then the p-value, equal to the probability of observing a T as large as t when H_0 is true, is less than α and so H_0 should be rejected.

Remarks

(a) To utilize the foregoing simulation approach to determine whether or not to accept H_0 when T is observed, we need to specify how one can simulate, or generate, a random variable Y such that $P\{Y = i\} = p_i, i = 1, \ldots, k$. One way is as follows:

Step 1: Generate a random number U.

Step 2: If

$$p_1 + \cdots + p_{i-1} \le U < p_1 + \cdots + p_i$$

set $Y = i$ (where $p_1 + \cdots + p_{i-1} \equiv 0$ when $i = 1$). That is,

$$\begin{aligned} U < p_1 &\Rightarrow Y = 1 \\ p_1 \le U < p_1 + p_2 &\Rightarrow Y = 2 \\ &\vdots \\ p_1 + \cdots + p_{i-1} \le U < p_1 + \cdots + p_i &\Rightarrow Y = i \\ &\vdots \\ p_1 + \cdots + p_{n-1} < U &\Rightarrow Y = n \end{aligned}$$

Since a random number is equivalent to a uniform (0, 1) random variable, we have that

$$P\{a < U < b\} = b - a, \qquad 0 < a < b < 1$$

and so

$$P\{Y = i\} = P\{p_1 + \cdots + p_{i-1} < U < p_1 + \cdots + p_i\} = p_i$$

(b) A significant question that remains is how many simulation runs are necessary. It has been shown that the value $r = 100$ is usually sufficient at the conventional 5 percent level of significance.[2]

Example 11.2.c. Let us reconsider the problem presented in Example 11.2.b. A simulation study yielded the result

$$P_{H_0}\{T \le 9.52381\} = .95$$

[2]See Hope, A., "A Simplified Monte Carlo Significance Test Procedure," *J. of Royal Statist. Soc.*, B. 30, 582–598, 1968.